



US012385474B2

(12) **United States Patent**
Bodishbaugh et al.

(10) **Patent No.:** **US 12,385,474 B2**
(45) **Date of Patent:** ***Aug. 12, 2025**

(54) **SYSTEMS FOR GENERATING
GEOTHERMAL POWER IN AN ORGANIC
RANKINE CYCLE OPERATION DURING
HYDROCARBON PRODUCTION BASED ON
WORKING FLUID TEMPERATURE**

(58) **Field of Classification Search**
CPC F03G 7/04; F03G 7/045; F03G 4/00-074
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

1,982,745 A 12/1934 Koenemann
3,517,208 A 6/1970 Williams et al.
(Continued)

FOREIGN PATENT DOCUMENTS

AU 2007204830 A1 7/2007
AU 2009238733 B2 8/2013
(Continued)

OTHER PUBLICATIONS

Richter, Alexander, GreenFire Energy and Mitsui Oil Exploration Co. are partnering on a closed-loop geothermal pilot project in Japan, Think GeoEnergy, Apr. 6, 2021.
(Continued)

Primary Examiner — Laert Dounis

(74) *Attorney, Agent, or Firm* — Womble Bond Dickinson (US) LLP

(71) Applicant: **ICE Thermal Harvesting, LLC,**
Houston, TX (US)

(72) Inventors: **Adrian Benjamin Bodishbaugh,**
Houston, TX (US); **Carrie Jeanne Murtland,** Houston, TX (US)

(73) Assignee: **ICE Thermal Harvesting, LLC,**
Houston, TX (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

This patent is subject to a terminal disclaimer.

(21) Appl. No.: **18/671,018**

(22) Filed: **May 22, 2024**

(65) **Prior Publication Data**

US 2024/0309856 A1 Sep. 19, 2024

Related U.S. Application Data

(63) Continuation of application No. 17/305,297, filed on Jul. 2, 2021, now Pat. No. 12,060,867.
(Continued)

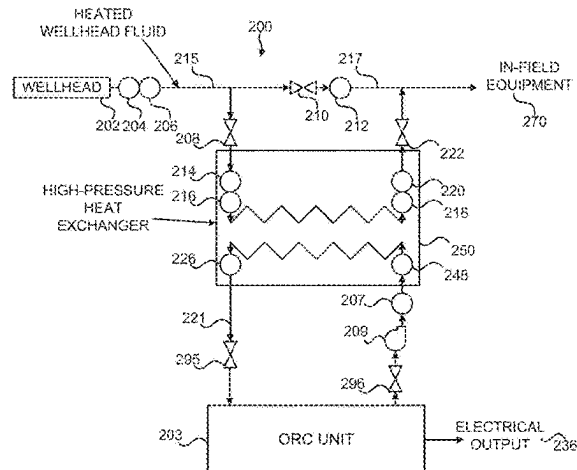
(51) **Int. Cl.**
F03G 7/04 (2006.01)
E21B 34/02 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC **F03G 7/045** (2021.08); **E21B 34/025** (2020.05); **E21B 34/066** (2013.01); **E21B 36/00** (2013.01);
(Continued)

ABSTRACT

Systems and methods for generating and a controller for controlling generation of geothermal power in an organic Rankine cycle (ORC) operation in the vicinity of a wellhead during hydrocarbon production to thereby supply electrical power to one or more of in-field operational equipment, a grid power structure, and an energy storage device. In an embodiment, during hydrocarbon production, a temperature of a flow of wellhead fluid from the wellhead or working fluid may be determined. If the temperature is above a vaporous phase change threshold of the working fluid, heat exchanger valves may be opened to divert flow of wellhead fluid to heat exchangers to facilitate heat transfer from the flow of wellhead fluid to working fluid through the heat exchangers, thereby to cause the working fluid to change

(Continued)



from a liquid to vapor, the vapor to cause a generator to generate electrical power via rotation of an expander.

12 Claims, 26 Drawing Sheets

Related U.S. Application Data

(60) Provisional application No. 63/200,908, filed on Apr. 2, 2021.

(51) Int. Cl.

E21B 34/06 (2006.01)
E21B 36/00 (2006.01)
E21B 41/00 (2006.01)
E21B 47/07 (2012.01)
F01K 11/02 (2006.01)
F01K 17/02 (2006.01)
F01K 25/08 (2006.01)
F03G 4/00 (2006.01)
F03G 4/06 (2006.01)
F03G 7/00 (2006.01)
F24T 50/00 (2018.01)
H02K 7/18 (2006.01)

(52) U.S. Cl.

CPC *E21B 41/0085* (2013.01); *E21B 47/07* (2020.05); *F01K 11/02* (2013.01); *F01K 17/02* (2013.01); *F01K 25/08* (2013.01); *F03G 4/023* (2021.08); *F03G 4/035* (2021.08); *F03G 4/06* (2021.08); *F03G 4/072* (2021.08); *F03G 7/027* (2021.08); *F24T 50/00* (2018.05); *H02K 7/1823* (2013.01); *F02G 2243/08* (2013.01); *F05D 2220/76* (2013.01)

(56) References Cited

U.S. PATENT DOCUMENTS

3,757,516 A 9/1973 McCabe
 3,808,794 A 5/1974 Wood
 3,875,749 A 4/1975 Baciu
 3,908,381 A 9/1975 Barber et al.
 3,988,895 A 11/1976 Sheinbaum
 4,063,417 A 12/1977 Shields
 4,079,590 A 3/1978 Sheinbaum
 4,112,687 A 9/1978 Dixon
 4,112,745 A 9/1978 McCabe
 4,149,385 A 4/1979 Sheinbaum
 4,157,730 A 6/1979 Despois et al.
 4,191,021 A 3/1980 Nakamura
 4,224,796 A 9/1980 Stiel et al.
 4,228,657 A 10/1980 Leo
 4,275,563 A 6/1981 Kuroda
 4,292,808 A 10/1981 Lohmiller
 4,356,401 A 10/1982 Santi
 4,369,373 A 1/1983 Wiseman
 4,484,446 A 11/1984 Goldsberry
 4,542,625 A 9/1985 Bronicki
 4,558,568 A 12/1985 Hoshino et al.
 4,576,005 A 3/1986 Force
 4,590,384 A 5/1986 Bronicki
 4,982,568 A 1/1991 Kalina
 4,996,846 A 3/1991 Bronicki
 5,038,567 A 8/1991 Mortiz
 5,117,908 A 6/1992 Hofmann
 5,131,231 A 7/1992 Trimble
 5,199,507 A 4/1993 Westmoreland
 5,311,741 A 5/1994 Blaize
 5,421,157 A 6/1995 Rosenblatt
 5,440,882 A 8/1995 Kalina
 5,483,797 A 1/1996 Rigal et al.

5,497,624 A 3/1996 Amir et al.
 5,517,822 A 5/1996 Haws et al.
 5,526,646 A 6/1996 Bronicki et al.
 5,555,731 A 9/1996 Rosenblatt
 5,570,579 A 11/1996 Larjola
 5,595,059 A 1/1997 Huber et al.
 5,598,706 A 2/1997 Bronicki et al.
 5,660,042 A 8/1997 Bronicki et al.
 5,661,977 A 9/1997 Shnell
 5,671,601 A 9/1997 Bronicki et al.
 5,685,362 A 11/1997 Brown
 5,816,048 A 10/1998 Bronicki et al.
 5,839,282 A 11/1998 Bronicki et al.
 5,860,279 A 1/1999 Bronicki et al.
 RE36,282 E 8/1999 Nitschke
 5,970,714 A 10/1999 Bronicki et al.
 5,974,804 A 11/1999 Sterling
 6,073,448 A 6/2000 Lozada
 6,212,890 B1 4/2001 Amir
 6,536,360 B2 3/2003 O'Connor
 6,585,047 B2 7/2003 McClung
 6,691,531 B1 2/2004 Martinez
 6,695,061 B2 2/2004 Fripp et al.
 6,724,687 B1 4/2004 Stephenson et al.
 6,853,798 B1 2/2005 Weiss
 6,857,268 B2 2/2005 Stinger et al.
 6,857,486 B2 2/2005 Chitwood et al.
 6,989,989 B2 1/2006 Brasz et al.
 7,096,665 B2 8/2006 Stinger et al.
 7,174,716 B2 2/2007 Brasz et al.
 7,224,080 B2 5/2007 Smedstad
 7,225,621 B2 6/2007 Zimron et al.
 7,234,314 B1 6/2007 Wiggs
 7,237,383 B2 7/2007 Ahrens-Botzong et al.
 7,254,949 B2 8/2007 Brasz et al.
 7,281,379 B2 10/2007 Brasz
 7,287,381 B1 10/2007 Pierson et al.
 7,289,325 B2 10/2007 Brasz et al.
 7,313,926 B2 1/2008 Gurin
 7,320,221 B2 1/2008 Bronicki
 7,334,410 B2 2/2008 Creighton et al.
 7,337,842 B2 3/2008 Roddy et al.
 7,353,653 B2 4/2008 Bronicki
 7,428,816 B2 9/2008 Singh et al.
 7,472,548 B2 1/2009 Meksvanh
 7,493,768 B2 2/2009 Klaus et al.
 7,753,122 B2 7/2010 Curlett
 7,823,386 B2 11/2010 Zimron et al.
 7,891,187 B2 2/2011 Mohr
 7,891,189 B2 2/2011 Bottger et al.
 7,900,450 B2 3/2011 Gurin
 7,926,276 B1 4/2011 Dunn
 7,934,383 B2 5/2011 Gutierrez et al.
 7,942,001 B2 5/2011 Radcliff et al.
 7,950,230 B2 5/2011 Nishikawa et al.
 7,987,676 B2 8/2011 Ast et al.
 8,046,999 B2 11/2011 Doty
 8,096,128 B2 1/2012 Held et al.
 8,099,198 B2 1/2012 Gurin
 8,146,360 B2 4/2012 Myers et al.
 8,166,761 B2 5/2012 Moghtaderi et al.
 8,193,659 B2 6/2012 Bronicki et al.
 8,272,217 B2 9/2012 Lengert
 8,309,498 B2 11/2012 Funkhouser et al.
 8,371,099 B2 2/2013 Gutierrez et al.
 8,381,523 B2 2/2013 Zadok
 8,430,166 B2 4/2013 Danko
 8,438,849 B2 5/2013 Kaplan et al.
 8,459,029 B2 6/2013 Lehar
 8,511,085 B2 8/2013 Frey et al.
 8,528,333 B2 9/2013 Juchymenko
 8,534,069 B2 9/2013 Parrella
 8,555,643 B2 10/2013 Kalina
 8,555,912 B2 10/2013 Woolley et al.
 8,572,970 B2 11/2013 Matteson et al.
 8,578,714 B2 11/2013 Nagurny et al.
 8,596,066 B2 12/2013 Zimron et al.
 8,616,000 B2 12/2013 Parrella
 8,616,001 B2 12/2013 Held et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

8,616,323	B1	12/2013	Gurin	9,784,140	B2	10/2017	Huntington et al.
8,656,720	B1	2/2014	Hardgrave	9,784,248	B2	10/2017	Batscha et al.
8,667,797	B2	3/2014	Woodland	9,797,273	B2	10/2017	Nishiguchi et al.
8,667,799	B2	3/2014	Batscha	9,803,803	B1	10/2017	Adams
8,674,525	B2	3/2014	Van den Bossche et al.	9,816,402	B2	11/2017	Kauffman et al.
8,680,704	B1	3/2014	Rooney	9,816,443	B2	11/2017	Sheridan et al.
8,707,697	B2	4/2014	Nitschke	9,829,194	B2	11/2017	Aumann et al.
8,707,698	B2	4/2014	Conry	9,840,662	B2	12/2017	Pascarella et al.
8,708,046	B2	4/2014	Montgomery et al.	9,845,423	B2	12/2017	Frantz et al.
8,720,563	B2	5/2014	Joseph et al.	9,863,282	B2	1/2018	Hart et al.
8,752,382	B2	6/2014	Lehar	9,874,112	B2	1/2018	Giegel
8,756,908	B2	6/2014	Sheridan et al.	9,932,861	B2	4/2018	Preuss et al.
8,771,603	B2	7/2014	Harless et al.	9,932,970	B1	4/2018	Jeter
8,783,034	B2	7/2014	Held	9,957,432	B2	5/2018	Galindo et al.
8,791,054	B2	7/2014	Deville	9,994,751	B2	6/2018	Hulse et al.
8,820,075	B2	9/2014	Kaminsky	10,005,950	B2	6/2018	Smith et al.
8,820,079	B2	9/2014	Zyhowski et al.	10,024,198	B2	7/2018	Held et al.
8,839,857	B2	9/2014	Schultz et al.	10,059,870	B2	8/2018	Joseph et al.
8,841,041	B2	9/2014	Biederman et al.	10,060,283	B2	8/2018	Tomigashi et al.
8,850,814	B2	10/2014	Kaplan et al.	10,060,302	B2	8/2018	Weng et al.
8,857,186	B2	10/2014	Held	10,060,652	B2	8/2018	Tahara
8,869,531	B2	10/2014	Held	10,077,683	B2	9/2018	Close
8,881,805	B2	11/2014	Klemencic	10,082,030	B2	9/2018	Genrup et al.
8,919,123	B2	12/2014	Gibble et al.	10,113,389	B2	10/2018	Pandey et al.
8,959,914	B2	2/2015	Kasuya et al.	10,113,535	B2	10/2018	Conlon
8,984,883	B2	3/2015	Riley	10,138,405	B2	11/2018	Kulkarni et al.
8,984,884	B2	3/2015	Xu et al.	10,138,560	B2	11/2018	Reyes et al.
9,003,798	B2	4/2015	Yanagi	10,221,770	B2	3/2019	Sheridan
9,014,791	B2	4/2015	Held	10,227,893	B2	3/2019	McCune et al.
9,016,063	B2	4/2015	Gaia et al.	10,234,183	B2	3/2019	Hashimoto
9,062,898	B2	6/2015	Held	10,247,044	B2	4/2019	Barmeier et al.
9,077,220	B2	7/2015	Kyle et al.	10,247,046	B2	4/2019	Schuster et al.
9,080,789	B2	7/2015	Hamstra et al.	10,267,184	B2	4/2019	Bowan et al.
9,091,278	B2	7/2015	Vermeersch	10,323,545	B2	6/2019	Johnson
9,109,398	B2	8/2015	Harris et al.	10,352,197	B2	7/2019	Grill et al.
9,115,603	B2	8/2015	Leibowitz et al.	10,357,726	B2	7/2019	Qin et al.
9,115,604	B2	8/2015	Bronicki	10,400,635	B2	9/2019	Johnson et al.
9,118,226	B2	8/2015	Kacludis et al.	10,435,604	B2	10/2019	Kontomaris et al.
9,121,259	B2	9/2015	Bryant et al.	10,436,075	B2	10/2019	Freund et al.
9,150,774	B2	10/2015	Reddy et al.	10,458,206	B2	10/2019	Al-Dossary et al.
9,181,930	B2	11/2015	Klemencic	10,465,104	B2	11/2019	Ravi et al.
9,217,370	B2	12/2015	Wang et al.	10,465,491	B2	11/2019	Moore
9,234,522	B2	1/2016	Jonsson et al.	10,472,994	B2	11/2019	Avadhanula et al.
9,243,616	B2	1/2016	Lee et al.	10,494,897	B2	12/2019	Pandey et al.
9,297,367	B2	3/2016	Ramaswamy et al.	10,495,098	B2	12/2019	Preuss et al.
9,316,404	B2	4/2016	Gurin	10,519,814	B2	12/2019	Quoilin
9,322,300	B2	4/2016	Mirmobin et al.	10,527,026	B2	1/2020	Muir et al.
9,331,547	B2	5/2016	Bronicki	10,563,927	B2	2/2020	Papadopoulos et al.
9,341,084	B2	5/2016	Xie et al.	10,570,777	B2	2/2020	Bowan
9,341,086	B2	5/2016	Batscha et al.	10,570,782	B2	2/2020	Lintl et al.
9,359,919	B1	6/2016	Berry	10,584,660	B2	3/2020	Sheridan et al.
9,376,937	B2	6/2016	Goswami et al.	10,590,324	B2	3/2020	Kulkarni et al.
9,394,764	B2	7/2016	Favilli et al.	10,590,802	B2	3/2020	McCune et al.
9,394,771	B2	7/2016	Wiggs	10,598,160	B2	3/2020	Sumrall
9,403,102	B2	8/2016	Wu et al.	10,619,520	B2	4/2020	Juchymenko
9,441,504	B2	9/2016	Held	10,626,709	B2	4/2020	Al-Dossary
9,458,738	B2	10/2016	Held	10,670,340	B2	6/2020	Batscha et al.
9,488,160	B2	11/2016	Fisher et al.	10,724,805	B2	7/2020	Barmeier et al.
9,499,732	B2	11/2016	Reddy et al.	10,767,904	B2	9/2020	von Düring
9,512,348	B2	12/2016	Reyes et al.	10,788,267	B2	9/2020	Dokic
9,512,741	B2	12/2016	Myogan et al.	10,794,292	B2	10/2020	Kupratis et al.
9,574,551	B2	2/2017	Parrella, Sr. et al.	10,883,388	B2	1/2021	Held
9,587,161	B2	3/2017	Fisk, Jr.	10,934,895	B2	3/2021	Held et al.
9,587,162	B2	3/2017	Fisk, Jr.	10,947,626	B2	3/2021	Pinder et al.
9,638,065	B2	5/2017	Vermeersch et al.	10,947,839	B2	3/2021	Cuthbert et al.
9,649,582	B2	5/2017	Shnell	10,975,279	B2	4/2021	Kontomaris et al.
9,671,138	B2	6/2017	Batscha et al.	11,022,070	B2	6/2021	Aumann et al.
9,683,463	B2	6/2017	Juchymenko	11,137,169	B2	10/2021	Buscheck et al.
9,726,157	B2	8/2017	Sweatman et al.	11,168,673	B2	11/2021	Younes et al.
9,726,441	B2	8/2017	Reissner et al.	11,174,715	B2	11/2021	Atisele
9,732,634	B2	8/2017	Hikichi et al.	11,187,112	B2	11/2021	Held
9,745,870	B2	8/2017	Johnson et al.	11,187,212	B1	11/2021	Bodishbaugh et al.
9,759,096	B2	9/2017	Vermeersch	11,220,932	B2	1/2022	Kontomaris et al.
9,762,460	B2	9/2017	Pawlowski et al.	11,236,735	B1	2/2022	Bodishbaugh et al.
9,777,602	B2	10/2017	Juchymenko	11,255,315	B1	2/2022	Bodishbaugh et al.
				11,255,576	B2	2/2022	Higgins et al.
				11,274,660	B2	3/2022	Radke
				11,274,663	B1	3/2022	Bodishbaugh et al.
				11,280,322	B1	3/2022	Bodishbaugh et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

11,293,414	B1	4/2022	Bodishbaugh et al.
11,326,479	B2	5/2022	Radke
11,326,550	B1	5/2022	Bodishbaugh et al.
11,359,576	B1	6/2022	Bodishbaugh et al.
11,359,612	B1	6/2022	Bodishbaugh et al.
11,365,652	B2	6/2022	Gaia et al.
11,396,828	B2	7/2022	Chase
11,421,625	B1	8/2022	Bodishbaugh et al.
11,421,663	B1	8/2022	Bodishbaugh et al.
11,480,074	B1	10/2022	Bodishbaugh et al.
11,486,330	B2	11/2022	Bodishbaugh et al.
11,486,370	B2	11/2022	Bodishbaugh et al.
11,493,029	B2	11/2022	Bodishbaugh et al.
11,542,888	B2	1/2023	Bodishbaugh et al.
11,549,402	B2	1/2023	Bodishbaugh et al.
11,572,849	B1	2/2023	Bodishbaugh et al.
11,578,706	B2	2/2023	Bodishbaugh et al.
11,592,009	B2	2/2023	Bodishbaugh et al.
11,598,320	B2	3/2023	Bodishbaugh et al.
11,624,355	B2	4/2023	Bodishbaugh et al.
11,644,014	B2	5/2023	Bodishbaugh et al.
11,644,015	B2	5/2023	Bodishbaugh et al.
11,668,209	B2	6/2023	Bodishbaugh et al.
11,680,541	B2	6/2023	Bodishbaugh et al.
11,732,697	B2	8/2023	Bodishbaugh et al.
11,761,353	B2	9/2023	Bodishbaugh et al.
11,761,433	B2	9/2023	Bodishbaugh et al.
11,773,805	B2	10/2023	Bodishbaugh et al.
11,879,409	B2	1/2024	Bodishbaugh et al.
11,905,934	B2	2/2024	Bodishbaugh et al.
11,933,279	B2	3/2024	Bodishbaugh et al.
11,933,280	B2	3/2024	Bodishbaugh et al.
11,946,459	B2	4/2024	Bodishbaugh et al.
11,959,466	B2	4/2024	Bodishbaugh et al.
11,971,019	B2	4/2024	Bodishbaugh et al.
12,049,875	B2	7/2024	Bodishbaugh
12,060,867	B2	8/2024	Bodishbaugh
12,104,553	B2	10/2024	Bodishbaugh
12,110,878	B2	10/2024	Bodishbaugh
12,135,016	B2	11/2024	Bodishbaugh
12,140,124	B2	11/2024	Bodishbaugh
12,146,475	B2	11/2024	Bodishbaugh
12,163,485	B2	12/2024	Bodishbaugh
12,180,861	B1	12/2024	Bodishbaugh
12,305,624	B2	5/2025	Bodishbaugh et al.
12,312,981	B2	5/2025	Bodishbaugh et al.
2002/0178723	A1	12/2002	Bronicki et al.
2003/0010652	A1	1/2003	Hunt
2003/0029169	A1	2/2003	Hanna et al.
2004/0237890	A1	12/2004	Bour
2005/0034467	A1	2/2005	Varney
2005/0109495	A1	5/2005	Cheng et al.
2005/0247056	A1	11/2005	Cogswell et al.
2005/0247059	A1	11/2005	Cogswell et al.
2006/0026961	A1	2/2006	Bronicki
2006/0130480	A1	6/2006	Lovelace
2007/0025854	A1	2/2007	Moore et al.
2008/0095611	A1	4/2008	Storage
2008/0168772	A1	7/2008	Radcliff et al.
2008/0217523	A1	9/2008	O'Sullivan
2009/0071155	A1	3/2009	Boyapati
2009/0211253	A1	8/2009	Radcliff et al.
2009/0217664	A1	9/2009	Rapp et al.
2009/0313999	A1	12/2009	Hunter
2009/0320477	A1	12/2009	Juchymenko
2010/0018207	A1	1/2010	Juchymenko
2010/0034684	A1	2/2010	Ast
2010/0045042	A1	2/2010	Hinders et al.
2010/0071366	A1	3/2010	Klemencic
2010/0071409	A1	3/2010	Kaart
2010/0077752	A1	4/2010	Papile
2010/0077792	A1	4/2010	Gurin
2010/0187319	A1	7/2010	Isom et al.
2010/0192573	A1	8/2010	Hamilton
2010/0194111	A1	8/2010	Van den Bossche et al.
2010/0218930	A1	9/2010	Proeschel
2010/0300093	A1	12/2010	Doty
2010/0319354	A1	12/2010	Guidati et al.
2011/0000210	A1	1/2011	Miles
2011/0000227	A1	1/2011	Kamiya
2011/0030404	A1	2/2011	Gurin et al.
2011/0041502	A1	2/2011	Zimron et al.
2011/0041505	A1	2/2011	Kasuya et al.
2011/0083620	A1	4/2011	Yoon
2011/0100003	A1	5/2011	McLeod et al.
2011/0126539	A1	6/2011	Ramaswamy et al.
2011/0138809	A1	6/2011	Ramaswamy et al.
2011/0175358	A1	7/2011	Langson
2011/0272166	A1	11/2011	Hunt
2011/0314818	A1	12/2011	Breen et al.
2012/0001429	A1	1/2012	Saar et al.
2012/0042650	A1	2/2012	Ernst et al.
2012/0111004	A1	5/2012	Conry
2012/0131918	A1	5/2012	Held
2012/0145397	A1	6/2012	Schultz et al.
2012/0174581	A1	7/2012	Vaughan et al.
2012/0174622	A1	7/2012	Granier
2012/0192560	A1	8/2012	Ernst et al.
2012/0198844	A1	8/2012	Kaminsky
2012/0261092	A1	10/2012	Heath et al.
2012/0291433	A1	11/2012	Meng et al.
2012/0292112	A1	11/2012	Lakic
2012/0292909	A1	11/2012	Erikson
2012/0315158	A1	12/2012	Klaus
2013/0041068	A1	2/2013	Reddy et al.
2013/0067910	A1	3/2013	Ishiguro et al.
2013/0091843	A1	4/2013	Zyhowski et al.
2013/0129496	A1	5/2013	Eckert
2013/0139509	A1	6/2013	Berti
2013/0168089	A1	7/2013	Berg et al.
2013/0168964	A1	7/2013	Xu et al.
2013/0186089	A1	7/2013	Bruckner
2013/0217604	A1	8/2013	Fisk, Jr.
2013/0227947	A1	9/2013	Bronicki et al.
2013/0247569	A1	9/2013	Suter
2013/0298568	A1	11/2013	Pierson et al.
2013/0299123	A1	11/2013	Matula
2013/0299170	A1	11/2013	Joseph et al.
2013/0341010	A1	12/2013	Nevison
2014/0011908	A1	1/2014	Reddy et al.
2014/0026574	A1	1/2014	Leibowitz et al.
2014/0033713	A1	2/2014	Juchymenko
2014/0057810	A1	2/2014	Fisk, Jr.
2014/0087978	A1	3/2014	Deville
2014/0102098	A1	4/2014	Bowan et al.
2014/0102103	A1	4/2014	Yamamoto
2014/0123643	A1	5/2014	Ming
2014/0130498	A1	5/2014	Randolph
2014/0158429	A1	6/2014	Kader et al.
2014/0178180	A1	6/2014	Sheridan
2014/0206912	A1	7/2014	Iglesias
2014/0224469	A1	8/2014	Mirmobin et al.
2014/0296113	A1	10/2014	Reyes et al.
2014/0305125	A1	10/2014	Wang et al.
2014/0360702	A1	12/2014	Cook
2014/0366540	A1	12/2014	Zyhowski et al.
2015/0021924	A1	1/2015	Parella
2015/0047351	A1	2/2015	Ishikawa et al.
2015/0135708	A1	5/2015	Lutz et al.
2015/0192038	A1	7/2015	Sharp et al.
2015/0226500	A1	8/2015	Reissner et al.
2015/0252653	A1	9/2015	Shelton, Jr.
2015/0300327	A1	10/2015	Sweatman et al.
2015/0330261	A1	11/2015	Held
2015/0345341	A1	12/2015	Kacludis et al.
2015/0345482	A1	12/2015	Klitzing et al.
2015/0361831	A1	12/2015	Myers
2016/0003108	A1	1/2016	Held et al.
2016/0010512	A1	1/2016	Close
2016/0017758	A1	1/2016	Vermeersch et al.
2016/0017759	A1	1/2016	Gayawal et al.
2016/0040557	A1	2/2016	Vermeersch et al.
2016/0047540	A1	2/2016	Aumann et al.
2016/0061055	A1	3/2016	Bowan

(56)

References Cited

U.S. PATENT DOCUMENTS

2016/0076405 A1 3/2016 Hashimoto et al.
 2016/0084115 A1 3/2016 Ludewig et al.
 2016/0130985 A1 5/2016 O'Connor et al.
 2016/0160111 A1 6/2016 Smith et al.
 2016/0177887 A1 6/2016 Fischer
 2016/0201521 A1 7/2016 Karthausier
 2016/0222275 A1 8/2016 Galindo et al.
 2016/0257869 A1 9/2016 Kulkarni et al.
 2016/0312646 A1 10/2016 Juano
 2016/0340572 A1 11/2016 Pascarella et al.
 2016/0369408 A1 12/2016 Reyes et al.
 2017/0058181 A1 3/2017 Frantz et al.
 2017/0058722 A1 3/2017 Noureldin et al.
 2017/0130614 A1 5/2017 Held et al.
 2017/0145815 A1 5/2017 Cuthbert et al.
 2017/0159504 A1 6/2017 Ostrom et al.
 2017/0175582 A1 6/2017 McCune et al.
 2017/0175583 A1 6/2017 McCune et al.
 2017/0211829 A1 7/2017 Slack
 2017/0226402 A1 8/2017 Patil et al.
 2017/0233635 A1 8/2017 Pandey et al.
 2017/0240794 A1 8/2017 Iverson et al.
 2017/0254223 A1 9/2017 Goethals et al.
 2017/0254226 A1 9/2017 Heber et al.
 2017/0261268 A1 9/2017 Barmeier et al.
 2017/0276026 A1 9/2017 Barmeier et al.
 2017/0276435 A1 9/2017 Papadopoulos et al.
 2017/0284230 A1 10/2017 Juchymenko
 2017/0314420 A1 11/2017 Bowan et al.
 2017/0321104 A1 11/2017 Ravi et al.
 2017/0321107 A1 11/2017 Joseph et al.
 2017/0362963 A1 12/2017 Hostler et al.
 2017/0370251 A1 12/2017 McGrail
 2018/0094548 A1 4/2018 Jeter
 2018/0128131 A1 5/2018 Zyhowski et al.
 2018/0135468 A1 5/2018 Kayahara
 2018/0179960 A1 6/2018 Apte et al.
 2018/0224164 A1 8/2018 Lakic
 2018/0274524 A1 9/2018 Moncarz et al.
 2018/0313232 A1 11/2018 Auciello
 2018/0313340 A1 11/2018 Spadacini et al.
 2018/0328138 A1 11/2018 Pandey et al.
 2018/0340450 A1 11/2018 Avadhanula et al.
 2018/0355703 A1 12/2018 Al-Dossary
 2018/0356044 A1 12/2018 Monti et al.
 2019/0048759 A1 2/2019 Noureldin et al.
 2019/0055445 A1 2/2019 Kulkarni et al.
 2019/0128147 A1 5/2019 Liu et al.
 2019/0128567 A1 5/2019 Redfern
 2019/0390660 A1 12/2019 McBay
 2020/0011426 A1 1/2020 Juchymenko
 2020/0025032 A1 1/2020 McCune et al.
 2020/0041071 A1 2/2020 Werlen
 2020/0095899 A1 3/2020 Merswolke et al.
 2020/0200123 A1 6/2020 Aumann et al.
 2020/0200483 A1 6/2020 Ahlbom
 2020/0217304 A1 7/2020 Sumrall
 2020/0232342 A1 7/2020 McCune et al.
 2020/0248063 A1 8/2020 Stone
 2020/0292240 A1 9/2020 Chase
 2020/0308992 A1 10/2020 Juchymenko
 2020/0309101 A1 10/2020 Muir et al.
 2020/0354839 A1 11/2020 Pinder et al.
 2020/0386212 A1 12/2020 Atisele
 2020/0399524 A1 12/2020 Pisklak et al.
 2021/0017439 A1 1/2021 Ramirez Angulo et al.
 2021/0047963 A1 2/2021 Conde
 2021/0062682 A1 3/2021 Higgins et al.
 2021/0071063 A1 3/2021 Stone
 2021/0140684 A1 5/2021 Younes et al.
 2021/0172344 A1 6/2021 Juchymenko
 2021/0205738 A1 7/2021 Blomqvist et al.
 2021/0215440 A1 7/2021 Morris
 2021/0285693 A1 9/2021 Gerson
 2021/0372668 A1 12/2021 Buscheck et al.

2021/0396116 A1 12/2021 Yeung
 2021/0404309 A1 12/2021 Yeung
 2022/0090521 A1 3/2022 Kontomaris et al.
 2022/0186984 A1 6/2022 Gaia et al.
 2022/0389844 A1 12/2022 Zemlak
 2024/0026838 A1 1/2024 Bodishbaugh et al.
 2024/0093660 A1 3/2024 Bodishbaugh et al.
 2024/0159224 A1 5/2024 Bodishbaugh
 2024/0159225 A1 5/2024 Bodishbaugh
 2024/0191703 A1 6/2024 Bodishbaugh
 2024/0209844 A1 6/2024 Bodishbaugh
 2024/0218772 A1 7/2024 Bodishbaugh
 2024/0254973 A1 8/2024 Bodishbaugh

FOREIGN PATENT DOCUMENTS

AU 2011336831 B2 12/2016
 AU 2012306439 B2 3/2017
 AU 2014225990 B2 7/2018
 CA 2692629 1/2009
 CA 2698334 4/2009
 CA 2676502 A1 2/2011
 CA 2679612 5/2018
 CA 2676502 12/2018
 CA 2952379 C 4/2019
 CN 204661610 U 9/2015
 CN 103174473 10/2015
 CN 102812212 4/2016
 CN 103174475 8/2016
 CN 106517718 3/2017
 CN 107246550 10/2017
 CN 107387176 A 11/2017
 CN 107461221 A 12/2017
 CN 108302946 A 7/2018
 CN 108457609 A 8/2018
 CN 207761721 U 8/2018
 CN 209457990 U 10/2019
 CN 209704638 U 11/2019
 CN 111837006 10/2020
 CN 111911255 11/2020
 CN 113137286 A 7/2021
 CN 113266815 A 8/2021
 CN 113983844 A 1/2022
 CN 114370309 A 4/2022
 CN 216922244 U 7/2022
 DE 10337240 A1 3/2005
 DE 102011006066 A1 9/2012
 DE 102012214907 A1 10/2013
 DE 102012014443 1/2014
 DE 102013009351 1/2014
 DE 102018201172 A1 7/2019
 EP 0652368 5/1995
 EP 1507069 A1 2/2005
 EP 2530255 A2 12/2012
 EP 2201666 3/2013
 EP 1573173 4/2013
 EP 1713877 5/2013
 EP 1869293 5/2013
 EP 2222939 11/2013
 EP 1706667 B1 10/2014
 EP 2167872 2/2016
 EP 2446122 B1 8/2017
 EP 2478201 B1 8/2017
 EP 3102796 B1 1/2018
 EP 3514339 A1 7/2019
 EP 2550436 B1 8/2019
 EP 3464836 4/2020
 EP 3631173 A1 4/2020
 EP 2948649 B1 12/2020
 EP 3540331 B1 12/2020
 FR 2738872 3/1997
 GB 2336943 6/2003
 IN 247090 12/2003
 IN 256000 1/2005
 IN 202111000822 10/2021
 JP 08192150 7/1996
 JP 2001183030 7/2001
 JP 2009127627 A 6/2009
 JP 2010166805 A 7/2010

(56)	References Cited			WO	2013/082575	6/2013
	FOREIGN PATENT DOCUMENTS			WO	2013103592	7/2013
				WO	2013110375 A2	8/2013
				WO	2013115668 A1	8/2013
JP	2010249501 A	11/2010		WO	2013/136131	9/2013
JP	2010249502 A	11/2010		WO	2014019755 A1	2/2014
JP	2011064451 A	3/2011		WO	2014042580	3/2014
JP	4668189 B2	4/2011		WO	2014053292 A2	4/2014
JP	2011069370 A	4/2011		WO	2014059235 A1	4/2014
JP	2011106459 A	6/2011		WO	2014065977	5/2014
JP	2011137449 A	7/2011		WO	2014/124061	8/2014
JP	2013151931 A	8/2013		WO	2014/167795	10/2014
JP	2013238228 A	11/2013		WO	2014154405 A1	10/2014
JP	2014016124 A	1/2014		WO	2014159520 A1	10/2014
JP	2014080975 A	5/2014		WO	2014159587 A1	10/2014
JP	2014109279 A	6/2014		WO	2014160257 A1	10/2014
JP	2015149885 A	8/2015		WO	2014164620	10/2014
JP	2016006323 A	1/2016		WO	2014164620 A1	10/2014
JP	2016105687 A	6/2016		WO	2014165053	10/2014
JP	2016188640 A	11/2016		WO	2014165053 A1	10/2014
JP	2021167601 A	10/2021		WO	2014165144 A1	10/2014
KR	101126833	3/2012		WO	2014191157 A2	12/2014
KR	20120067710	6/2012		WO	2015/040279	3/2015
KR	20130023578 A	3/2013		WO	2015034987 A1	3/2015
KR	1691908	1/2017		WO	2015/059069	4/2015
KR	2075550	2/2020		WO	2015078829 A1	6/2015
KR	2185002	12/2020		WO	2015/135796	9/2015
NZ	581457	11/2011		WO	2015131940 A1	9/2015
RU	2006142350	6/2008		WO	2015/158600	10/2015
SG	191467	7/2013		WO	2015192005 A1	12/2015
SG	191468	7/2013		WO	2016039655 A1	3/2016
SG	192327	8/2013		WO	2016049712	4/2016
WO	1993/001397	1/1993		WO	2016050365 A1	4/2016
WO	1994028298	12/1994		WO	2016050366 A1	4/2016
WO	2005014981 A1	2/2005		WO	2016050367 A1	4/2016
WO	2005019606 A1	3/2005		WO	2016050368 A1	4/2016
WO	2005049975	6/2005		WO	2016050369 A1	4/2016
WO	2005100755 A1	10/2005		WO	2016/069242	5/2016
WO	2006/014609	2/2006		WO	2013103631	5/2016
WO	2006/027770	3/2006		WO	2016073245 A1	5/2016
WO	2006060253	6/2006		WO	2016/087920	6/2016
WO	2006/092786	9/2006		WO	2016099975 A1	6/2016
WO	2006138459	12/2006		WO	2016147419	9/2016
WO	2007048999	5/2007		WO	2016/196144	12/2016
WO	20070079245	7/2007		WO	2016204287	12/2016
WO	2007137373	12/2007		WO	2017041147	3/2017
WO	2008052809	5/2008		WO	2017065683	4/2017
WO	2008106774 A1	9/2008		WO	2017123132	7/2017
WO	2008125827	10/2008		WO	2017/146712	8/2017
WO	2009017471	2/2009		WO	2017/147400	8/2017
WO	2009017474	2/2009		WO	2017203447	11/2017
WO	2009027302 A2	3/2009		WO	2018/044690	3/2018
WO	2009030283 A2	3/2009		WO	2018/107279	6/2018
WO	2009/058112	5/2009		WO	2018106528	6/2018
WO	2009095127 A2	8/2009		WO	2018/210528	11/2018
WO	2009/142608	11/2009		WO	2018217969 A1	11/2018
WO	2010021618	2/2010		WO	2018227068 A1	12/2018
WO	2010/039448	4/2010		WO	2019004910	1/2019
WO	2010/065895	6/2010		WO	2019060844	3/2019
WO	2009017473	8/2010		WO	2019067618	4/2019
WO	2010106089 A2	9/2010		WO	2019086960 A1	5/2019
WO	2010127932 A2	11/2010		WO	2019/157341	8/2019
WO	2010/143046	12/2010		WO	2019155240	8/2019
WO	2010/143049	12/2010		WO	2019/178447	9/2019
WO	2011012047 A1	2/2011		WO	2020152485	7/2020
WO	2011/066032	6/2011		WO	2020153896	7/2020
WO	2011073469	6/2011		WO	2020/186044	9/2020
WO	2011/061601	8/2011		WO	2020201843	10/2020
WO	2011/103560	8/2011		WO	2020/229901	11/2020
WO	2011093854	8/2011		WO	2020/097714	12/2020
WO	2011/137980	11/2011		WO	2020239067	12/2020
WO	2012060510	5/2012		WO	2020239068	12/2020
WO	2012/079694	6/2012		WO	2020239069	12/2020
WO	2012074940 A2	6/2012		WO	2020251980 A1	12/2020
WO	2012/112889	8/2012		WO	2021004882	1/2021
WO	2012/142765	10/2012		WO	2021013465 A1	1/2021
WO	2012/151447	11/2012		WO	2021096696 A1	5/2021
WO	2013/014509	1/2013		WO	2021107834	6/2021
WO	2013059695	4/2013		WO	2022/061320	3/2022

(56)

References Cited

FOREIGN PATENT DOCUMENTS

OTHER PUBLICATIONS

- Edwards, Alex, Dallas Innovates, Hunt Energy Network's New Venture Will Put 50 Batteries Across Texas, Giving ERCOT a Portfolio of Energy Generation, Apr. 1, 2021.
- Guo, Boyun, Petroleum Engineering, A Computer-Assisted Approach, Dec. 21, 2006.
- Li, Tailu et al., Cascade utilization of low temperature geothermal water in oilfield combined power generation, gathering heat tracing and oil recovery, *Applied Thermal Engineering* 40 (2012).
- Sherven, Bob, Automation Maximizes performance for shale wells, *Oil&Gas Journal*, 2013.
- Hu, Kaiyong et al., A case study of an ORC geothermal power demonstration system under partial load conditions in Huabei Oilfield, China, *ScienceDirect*, 2017.
- Liu, Xiaolei et al., A systematic study of harnessing low-temperature geothermal energy from oil and gas reservoirs, *Elsevier, ScienceDirect, Energy*, 2017.
- Wang, Kai, et al., A comprehensive review of geothermal energy extraction and utilization in oilfields, *Elsevier, ScienceDirect, Journal of Petroleum Science and Engineering*, 2017.
- Cutright, Bruce L., The Transformation of Tight Shale Gas Reservoirs to Geothermal Energy Production, Bureau of Economic Geology University of Texas, Austin Texas, Jun. 14, 2011.
- Khennich, Mohammed et al., Optimal Design of ORC Systems with a Low-Temperature Heat Source, *Entropy* 2012, 14, 370-389; doi:10.3390/e14020370.
- Dambly, Benjamin W., et al., The Organic Rankine Cycle for Geothermal Power Generation, *Geothermal Energy*, 2007.
- Obi, John Besong, State of art on ORC applications for waste heat recovery and micro-cogeneration for installations up to 100kW, *Elsevier, Energy Procedia* 82 (2015) 994-1001.
- Obafunmi, Jaiyejeje Sunday, Thermodynamic Analysis of Organic Rankine Cycles, Eastern Mediterranean University Jul. 2014, Gazimağusa, North Cyprus.
- Dong, Benshi et al., Potential of low temperature organic Rankine cycle with zeotropic mixtures as working fluid, *Elsevier, ScienceDirect, Energy Procedia* 105 (2017) 1489-1494.
- Iqbal, MdArbab et al., Trilateral Flash Cycle (TFC): a promising thermodynamic cycle for low grade heat to power generation, *Elsevier, ScienceDirect, Energy Procedia* 160 (2019) 208-214.
- Bao, Junjiang et al., A review of working fluid and expander selections for organic Rankine cycle, *Elsevier, ScienceDirect, Renewable and Sustainable Energy Reviews* 24 (2013) 325-342.
- Ajimoto, Kan, Habeeb A. et al., Trilateral Flash Cycle for Recovery of Power from a Finite Low-Grade Heat Source, *Proceedings of the 24th European Symposium on Computer Aided Process Engineering—ESCAPE Jun. 15-18, 24, 2014, Budapest, Hungary. Copyright © 2014 Elsevier B.V.*
- Hung Tzu-Chen, et al., The Development and Application of a Small-Scale Organic Rankine Cycle for Waste Heat Recovery, *IntechOpen*, 2019.
- Kong, Rithy et al., Thermodynamic performance analysis of a R245fa organic Rankine cycle (ORC) with different kinds of heat sources at evaporator, *Elsevier, ScienceDirect, Case Studies in Thermal Engineering* 13 (2019) 100385.
- Lukawski, Maciej Z. et al., Impact of molecular structure of working fluids on performance of organic Rankine cycles (ORCs), *Sustainable Energy Fuels*, 2017, 1, 1098.
- Saleh, Bahaa et al., Working fluids for low-temperature organic Rankine cycles, *Elsevier, ScienceDirect, Energy* 32 (2007) 1210-1221.
- Brasz, Lars J. et al., Ranking of Working Fluids for Organic Rankine Cycle Applications, *Purdue University, Purdue e-Pubs*, (2004).
- International Refrigeration and Air Conditioning Conference. Paper 722.
- Miller, Patrick C., Research uses landfill gas tech for Bakken flaring solution, *The Bakken magazine*, Sep. 16, 2015.
- Electratherm, Inc., Power+ Generator 4400B & 4400B+, Nov. 24, 2020.
- Electratherm, Inc., Heat To Power Generation Base Load Renewable Energy, Mar. 2020.
- Electratherm, Inc., Power+ Generator, Nov. 25, 2020.
- Electratherm, Inc., Generating Clean Power From Waste Heat, Nov. 2020.
- Electratherm, Inc., Power+ Generator, May 19, 2020.
- Sneary, Loy et al., Gulf Coast Green Energy, Flare Gas Reduction Trial Using an Organic Rankine Cycle Generator, Jan. 11, 2016.
- Enertime, ORC for Industrial Waste Heat Recovery, Aug. 2017.
- Enogia, Generate power from your waste heat thanks to our ORC, 2019.
- UTC Power, PureCycle, 200 Heat-to-Electricity Power System, 2004.
- Rank, MT3 machine, Dec. 17, 2018.
- Heat Recovery Solutions, Clean Cycle Containerized Solution, 2009.
- Triogen BV, Specification: E-Box Engine Application, Feb. 15, 2019.
- "From Waste Heat To High Performance", *PBOG (Permian Basin Oil and Gas Magazine)*, Apr. 26, 2013.
- "Turning Waste Heat Into Clean Power; GNP's Expander System", *Great Northern Power Corporation*, 2022.
- Ng et al., "Thermo-Economic Performance of an Organic Rankine Cycle System Recovering Waste Heat Onboard an Offshore Service Vessel", *Journal of Marine Science and Engineering*, May 14, 2020.
- "First Flare Elimination Demonstration", *ElectraTherm*, 2022.
- International Search Report and Written Opinion for PCT/US2022/071472, May 9, 2022.
- International Search Report and Written Opinion for PCT/US2022/071475, May 17, 2022.
- International Search Report and Written Opinion for PCT/US2022/071474, Jun. 10, 2022.
- International Search Report and Written Opinion for PCT/US2022/071486, Jun. 14, 2022.
- International Search Report and Written Opinion for PCT/US2022/071323, Jun. 28, 2022.
- International Search Report and Written Opinion for PCT/US2022/071325, Jun. 28, 2022.
- Invitation to Pay Additional Fees and Communication Relating to Results of Partial International Search for PCT/US2022/071329, mailed Jun. 27, 2022.
- Invitation to Pay Additional Fees and Communication Relating to Results of Partial International Search for PCT/US2022/071327, mailed Jul. 4, 2022.
- International Search Report and Written Opinion for PCT/US2022/071313, Jul. 5, 2022.
- International Search Report and Written Opinion for PCT/US2022/071517, Jun. 27, 2022.
- International Search Report and Written Opinion for PCT/US2022/071484, Jun. 27, 2022.
- International Search Report and Written Opinion for PCT/US2022/071319, Jul. 12, 2022.
- International Search Report and Written Opinion for PCT/US2022/071480, Aug. 3, 2022.
- International Search Report and Written Opinion for PCT/US2022/071482, Aug. 2, 2022.
- Li et al., "Thermodynamic analysis of subcritical/transcritical ORCs with metal-organic heat carriers for efficient power generation from low-grade thermal energy", *Energy* 255 (2022) 124519.
- International Search Report and Written Opinion for PCT/US2022/071327, Aug. 29, 2022.
- International Search Report and Written Opinion for PCT/US2022/071328, Sep. 9, 2022.
- International Search Report and Written Opinion for PCT/US2022/071329, Aug. 25, 2022.

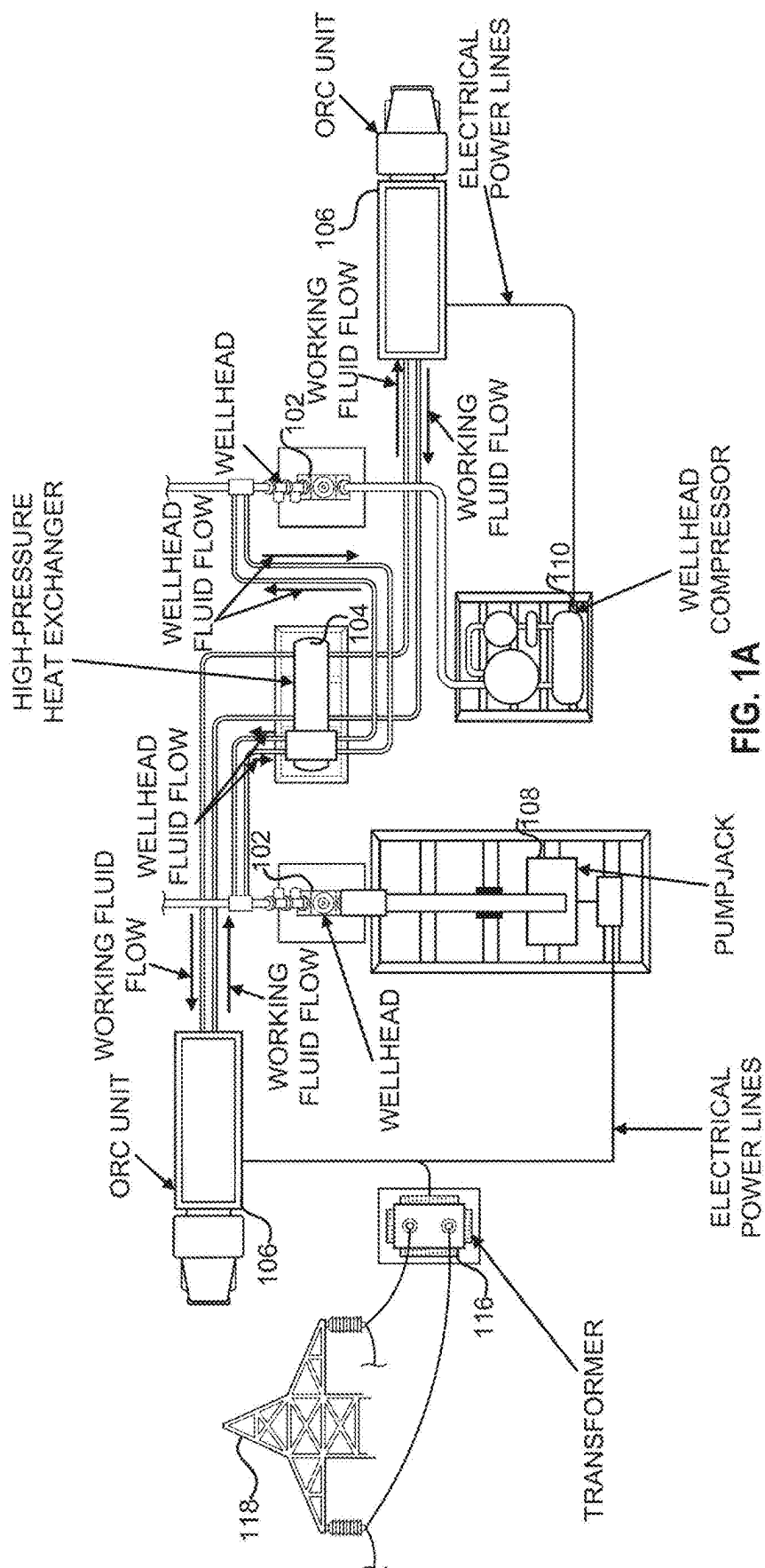
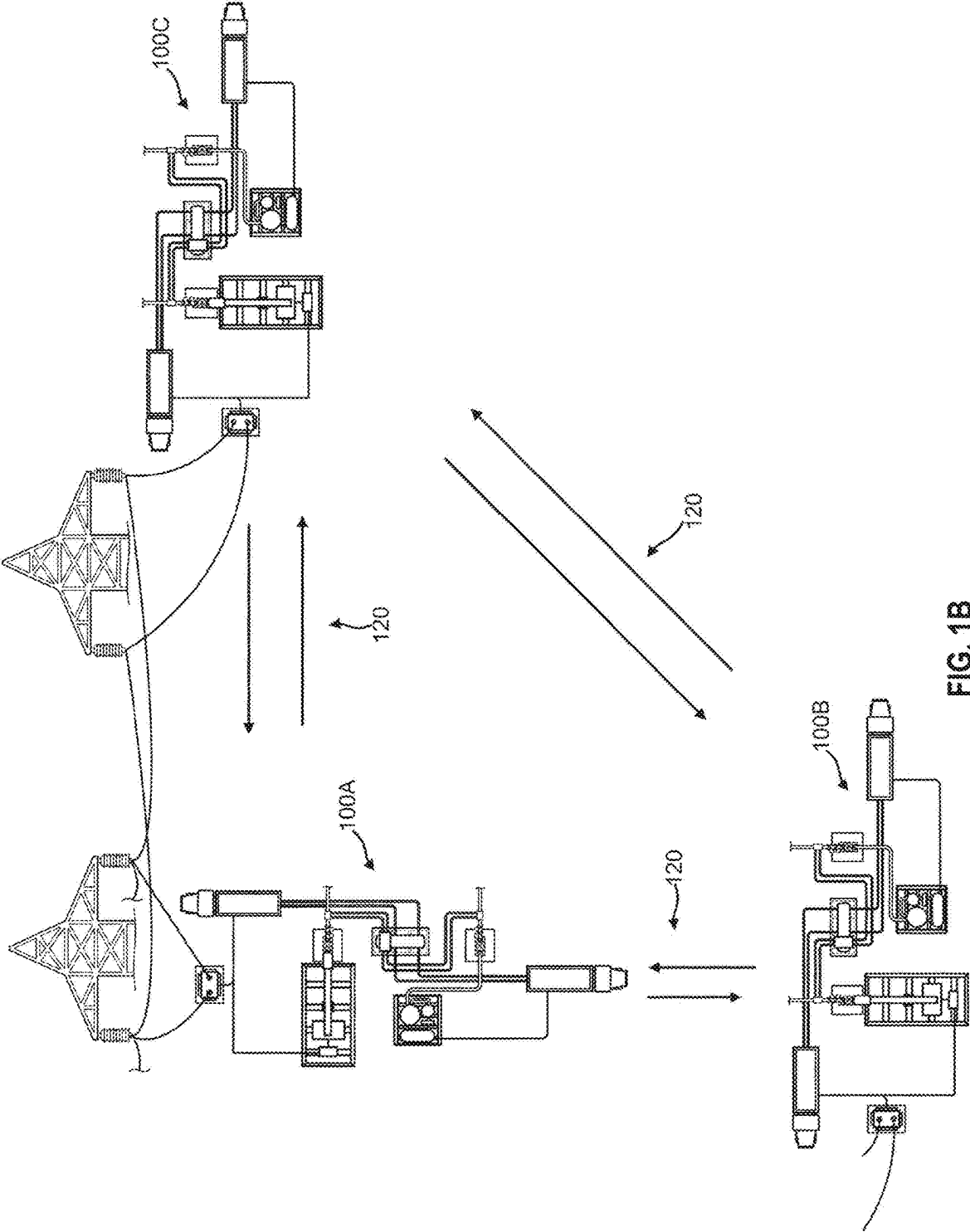


FIG. 1A



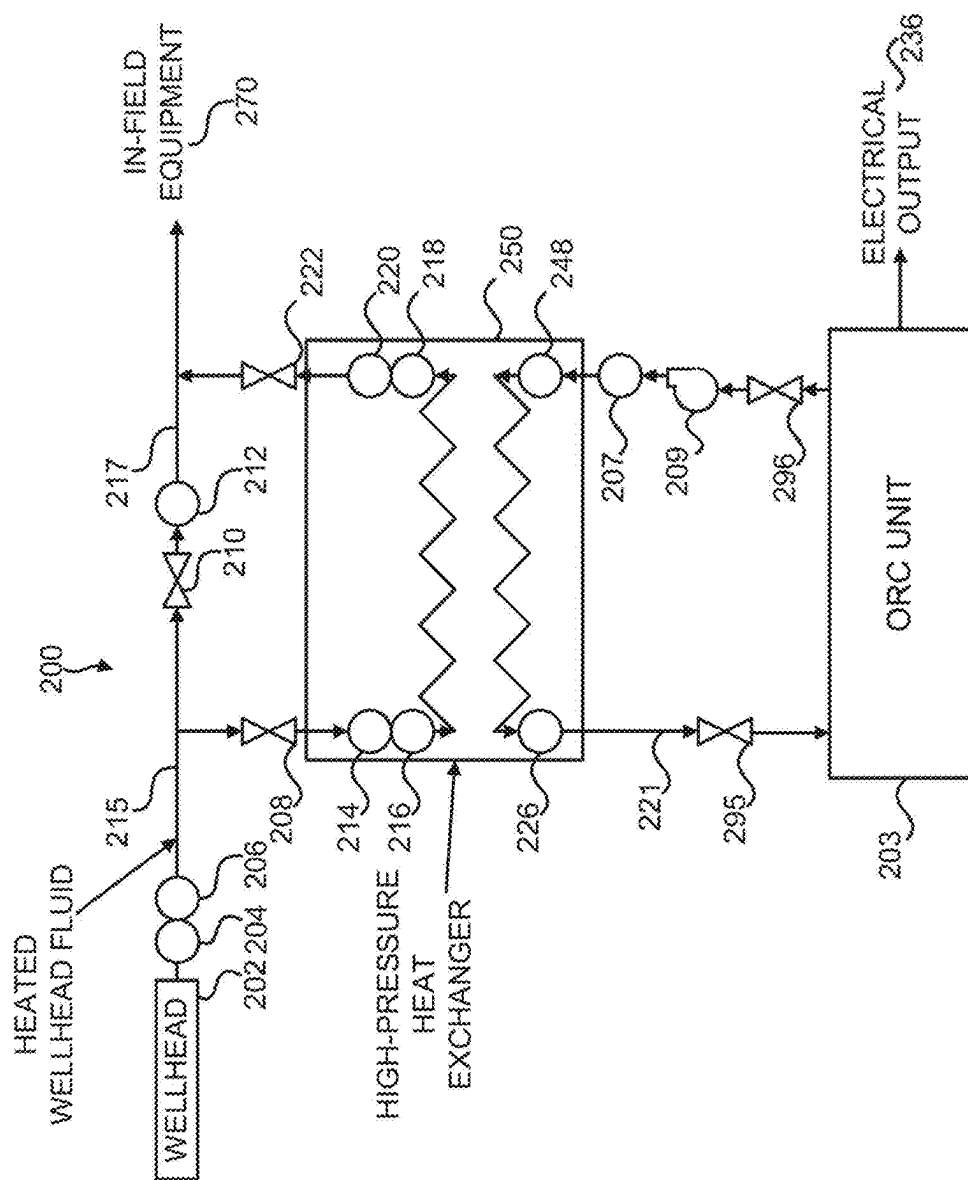


FIG. 2A

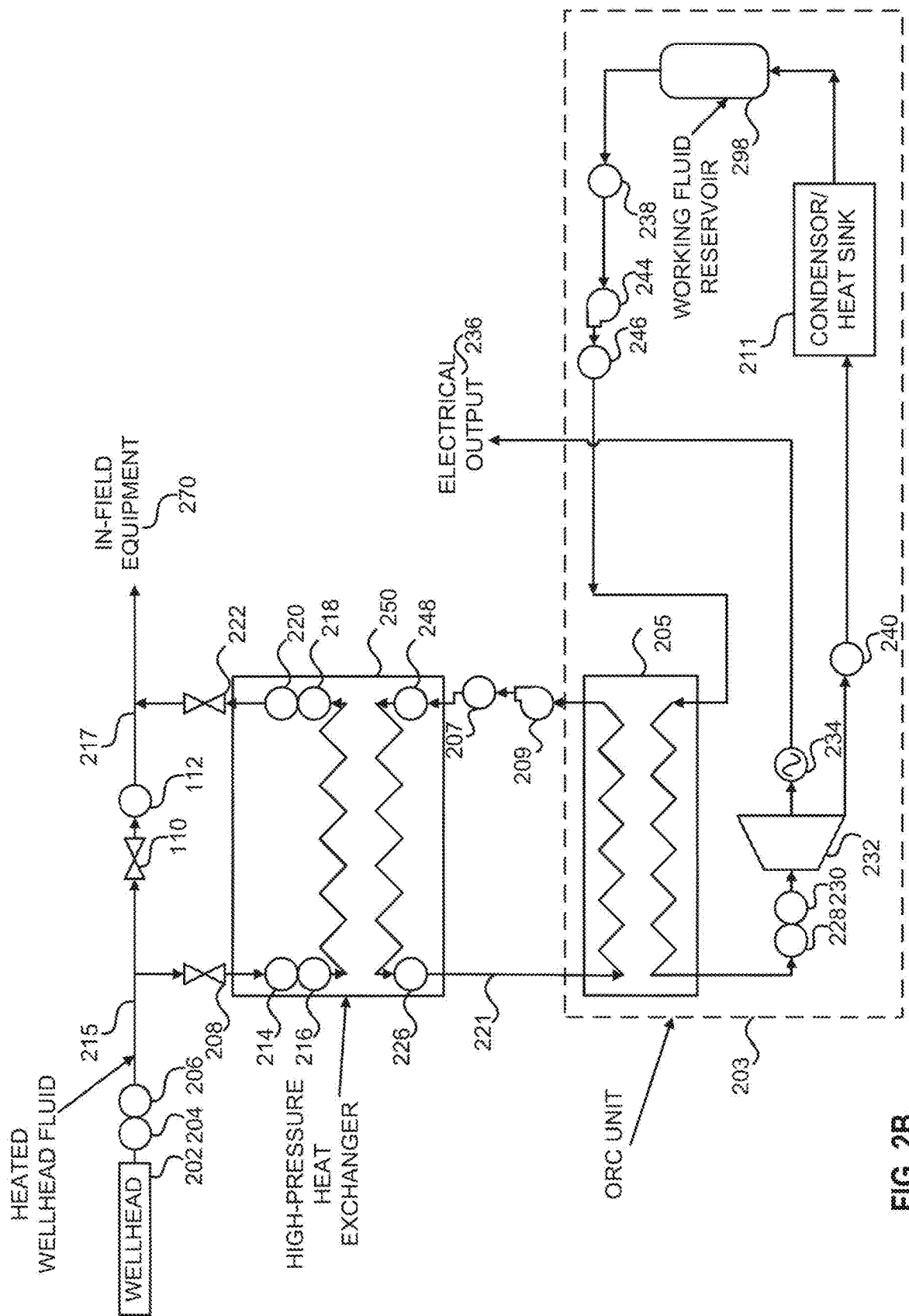
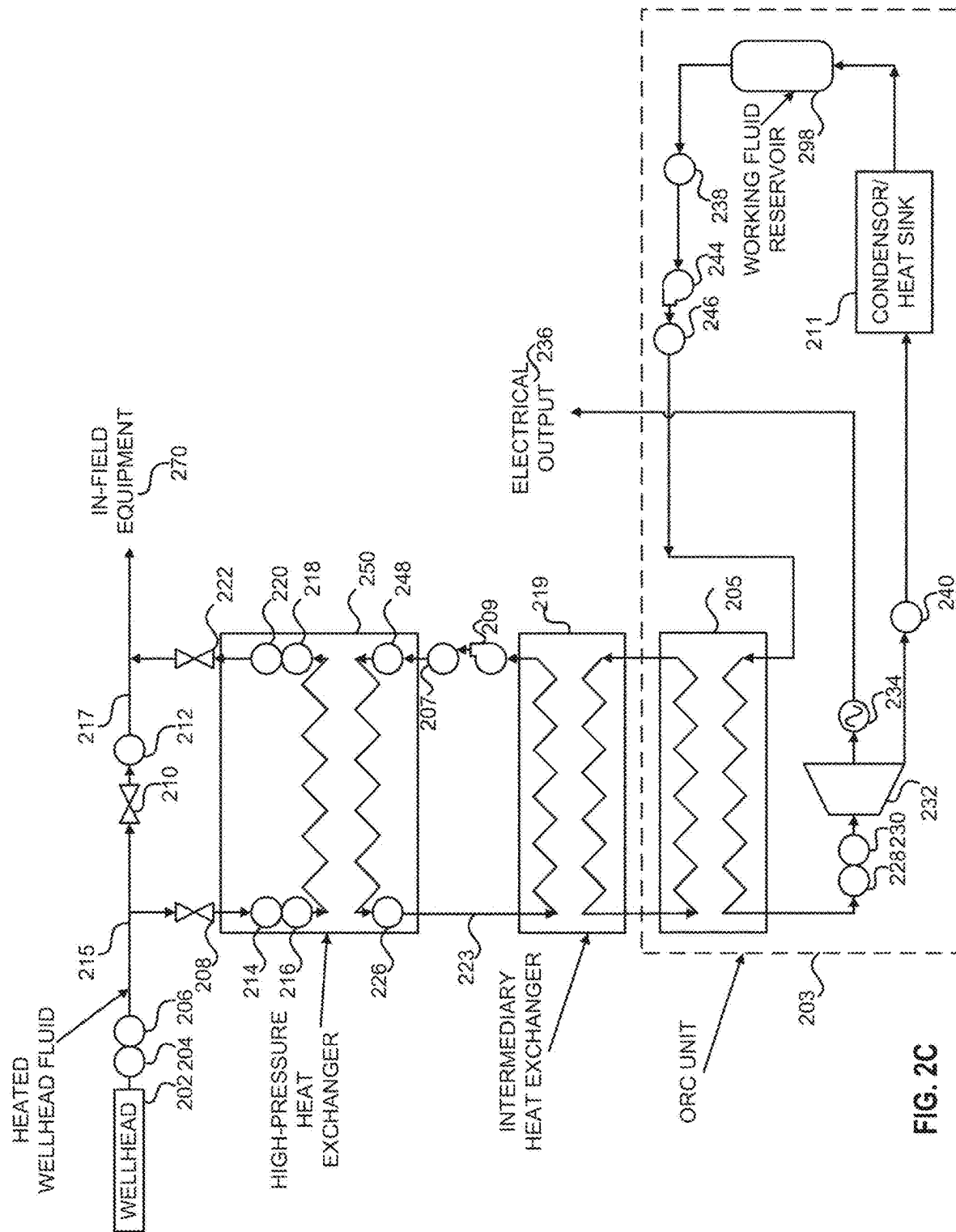


FIG. 2B



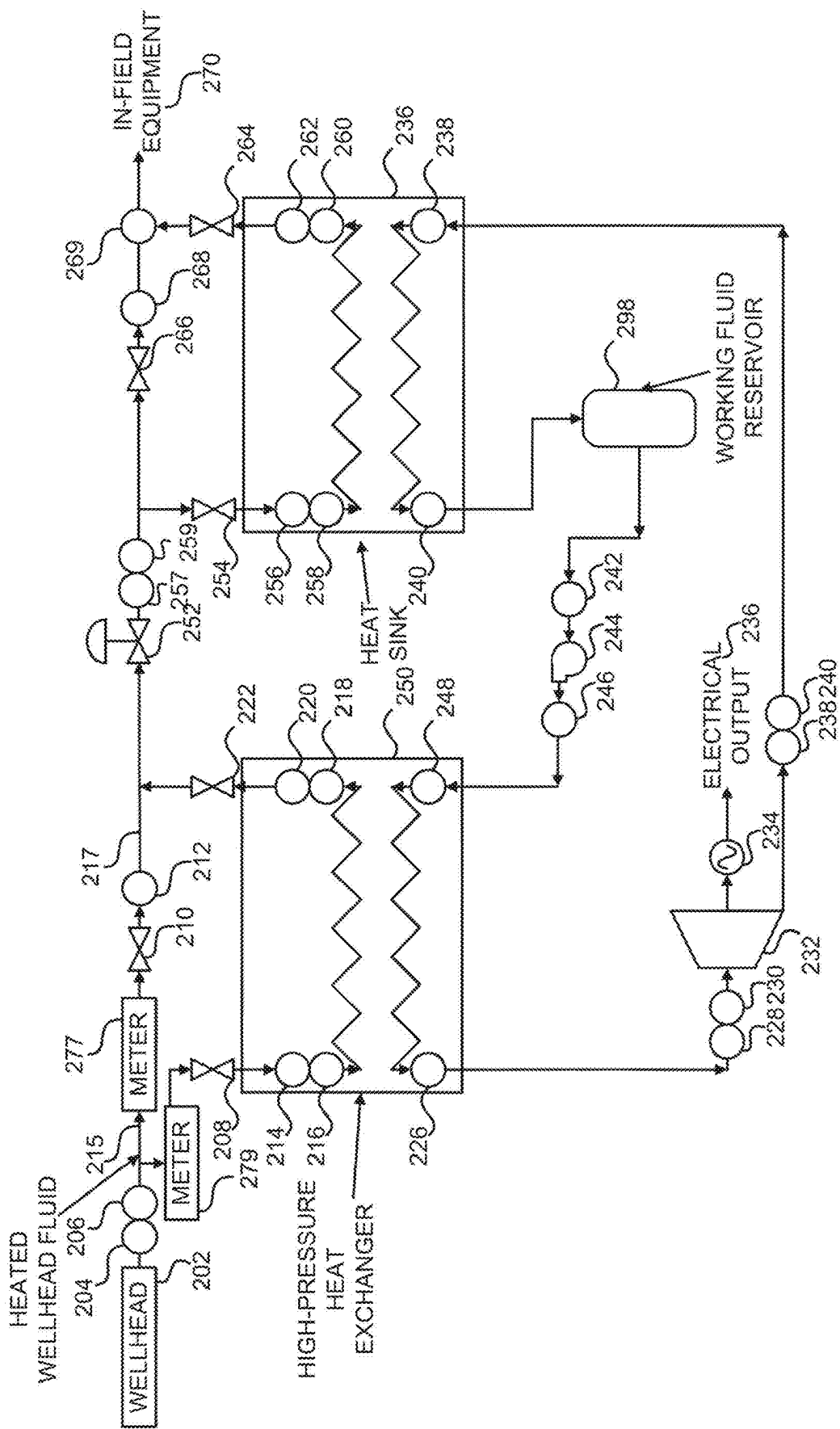
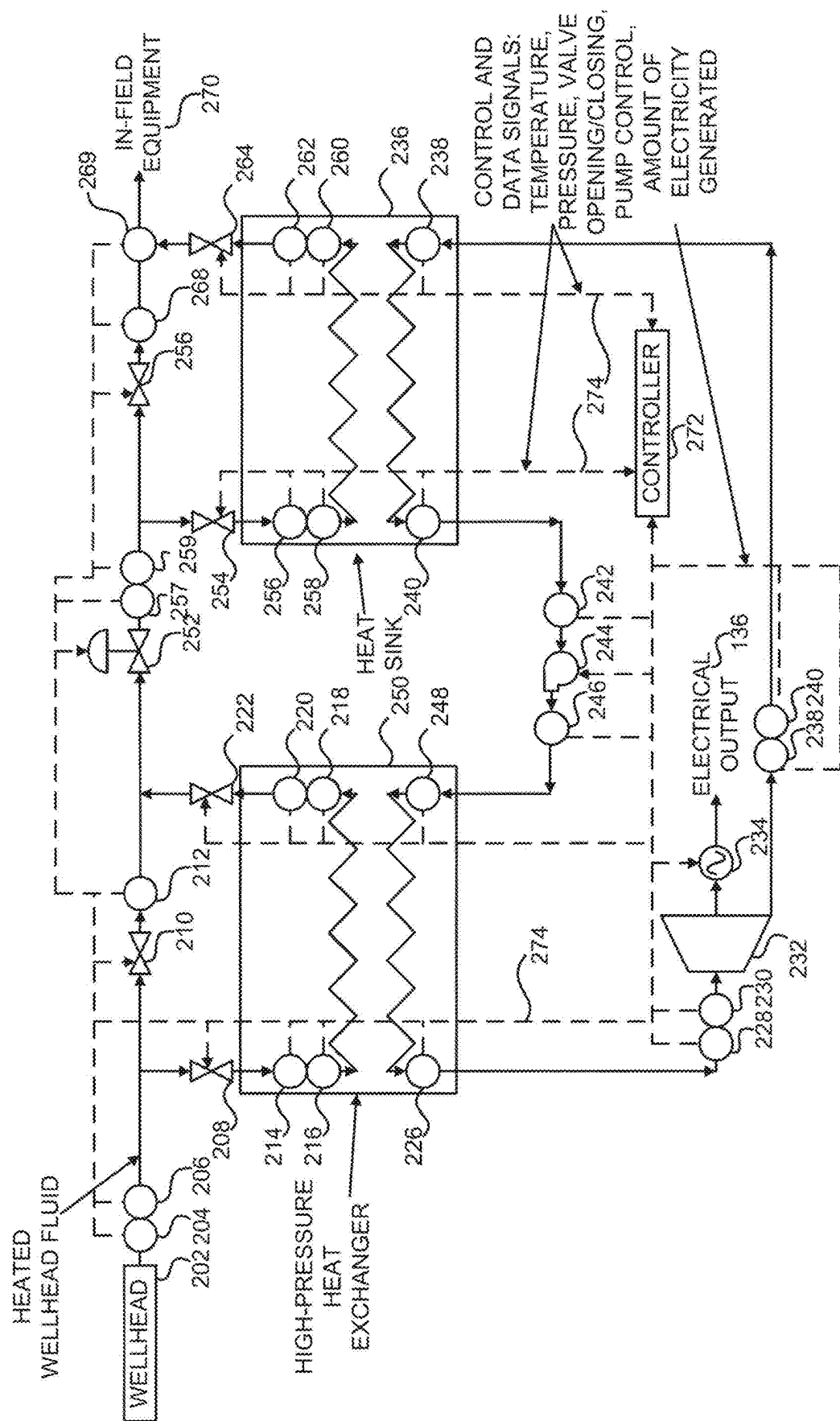


FIG. 2D



WZG
XXXXXX
LL

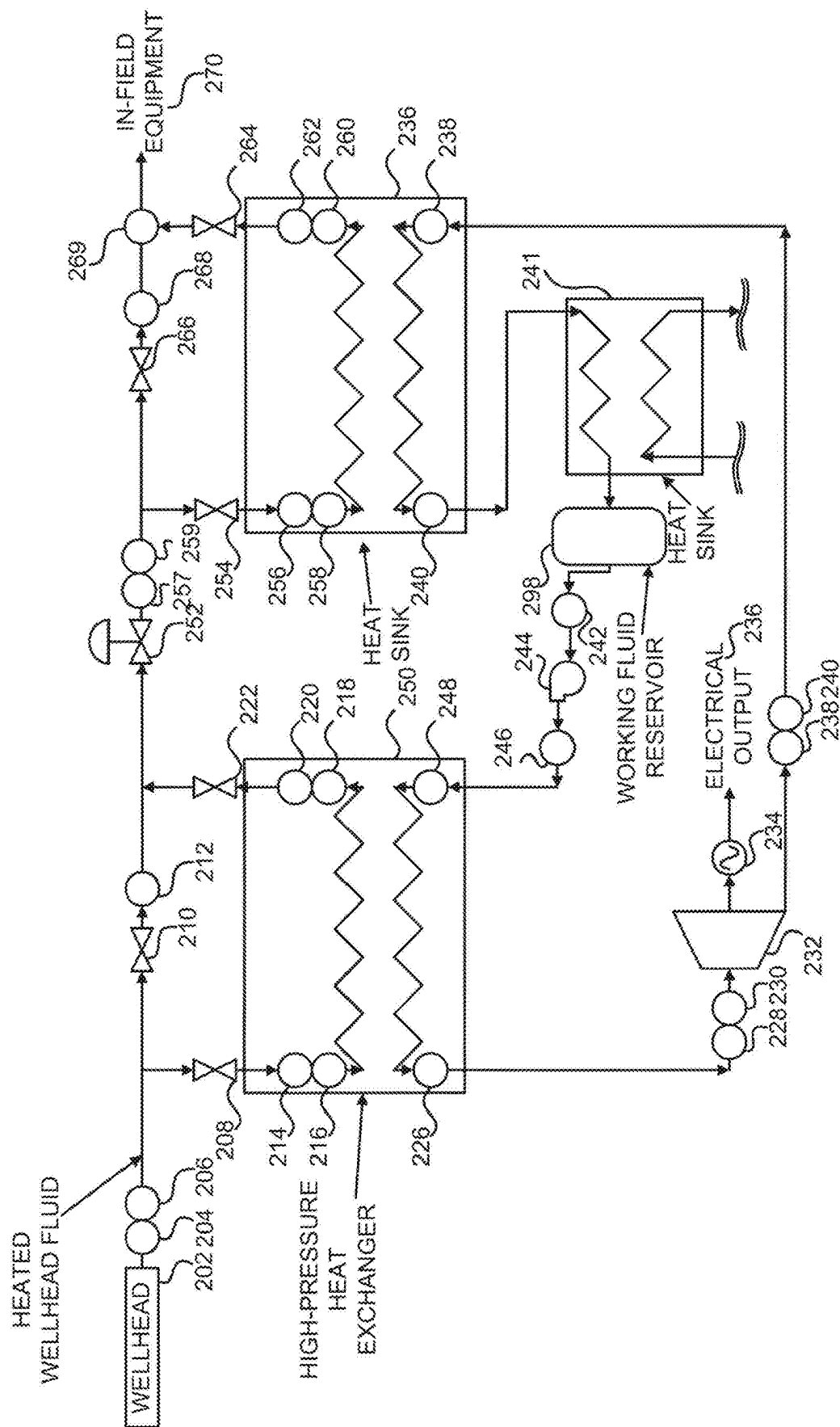


FIG. 2F

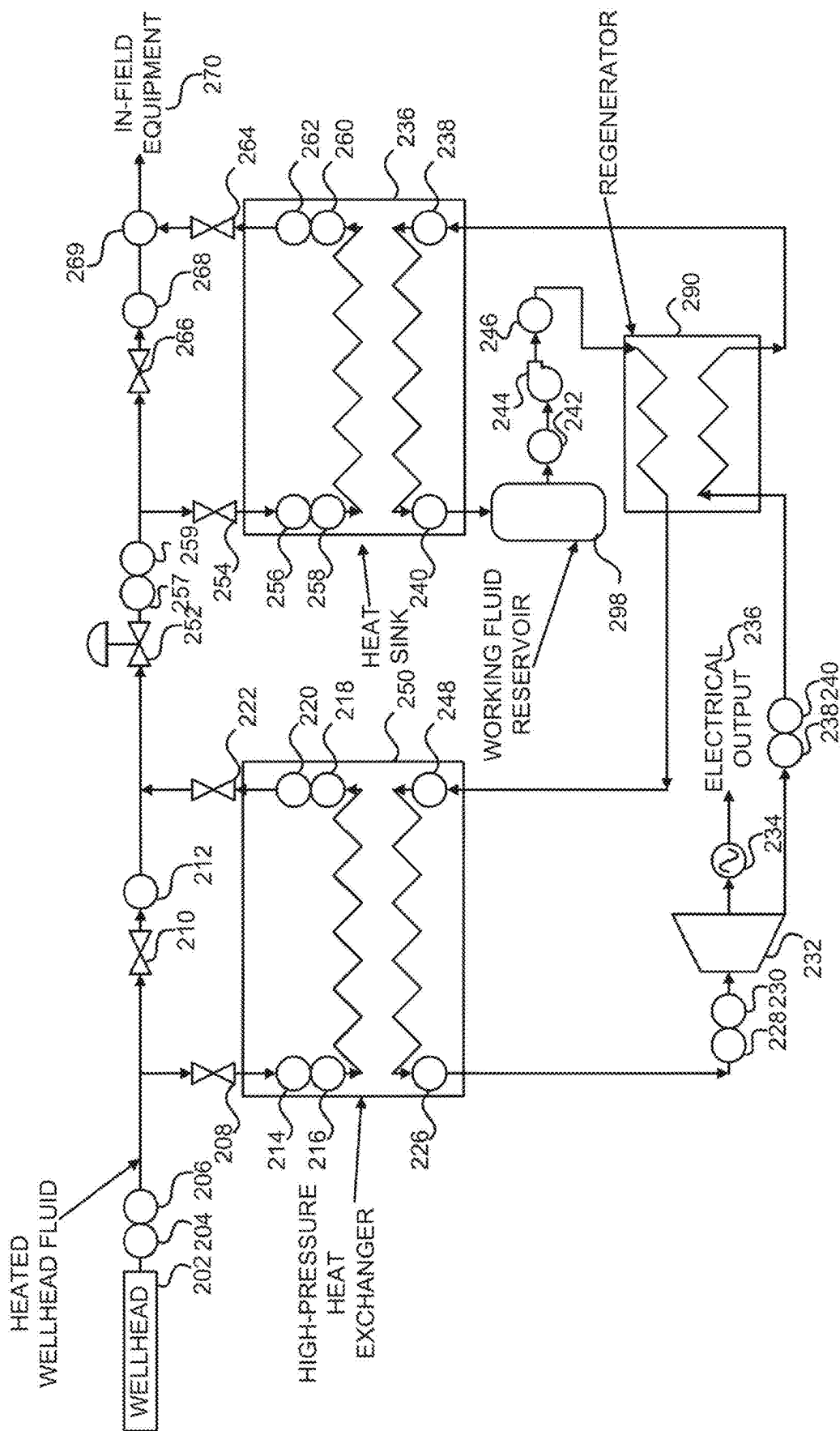


FIG. 2G

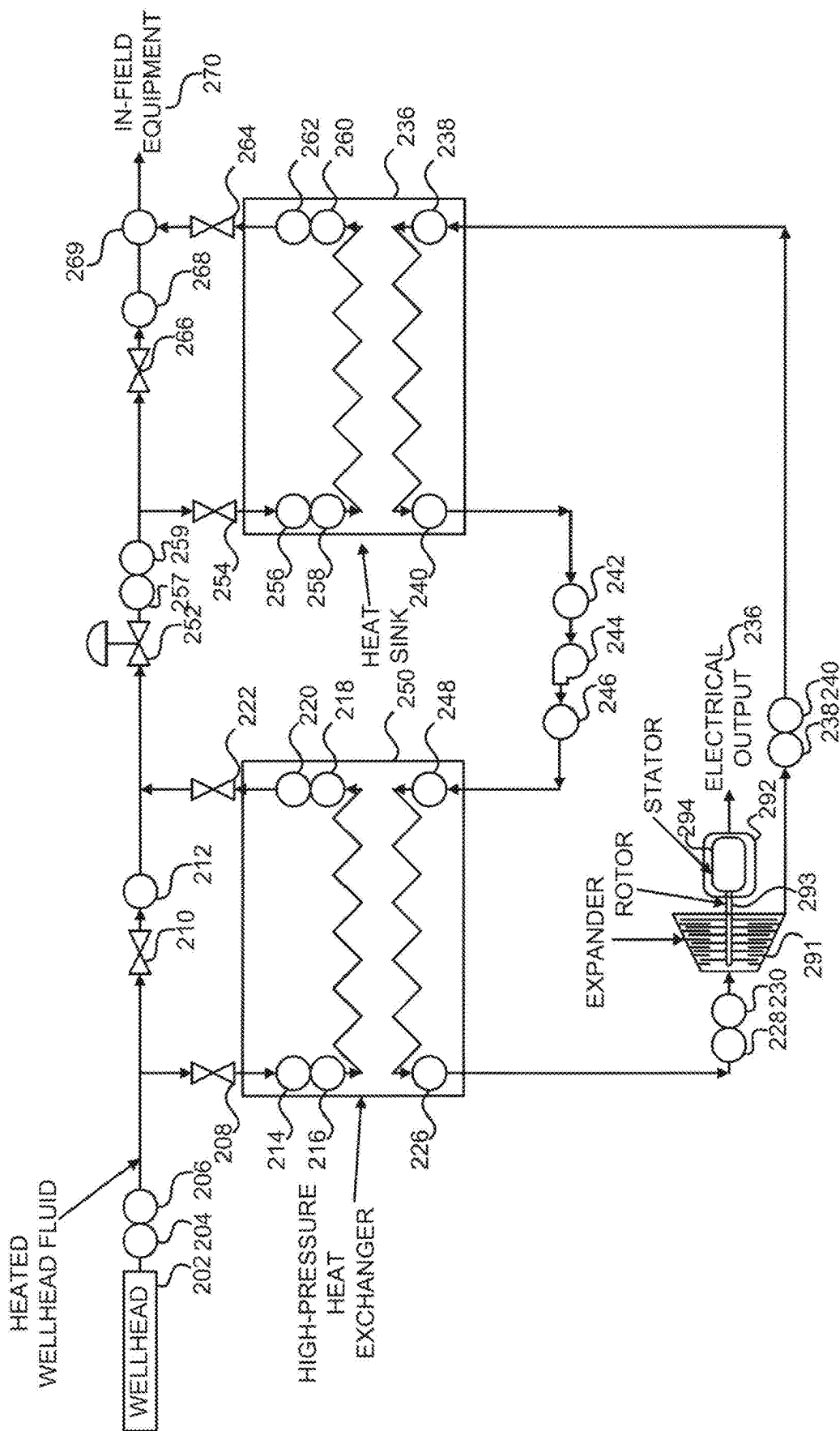


FIG. 2H

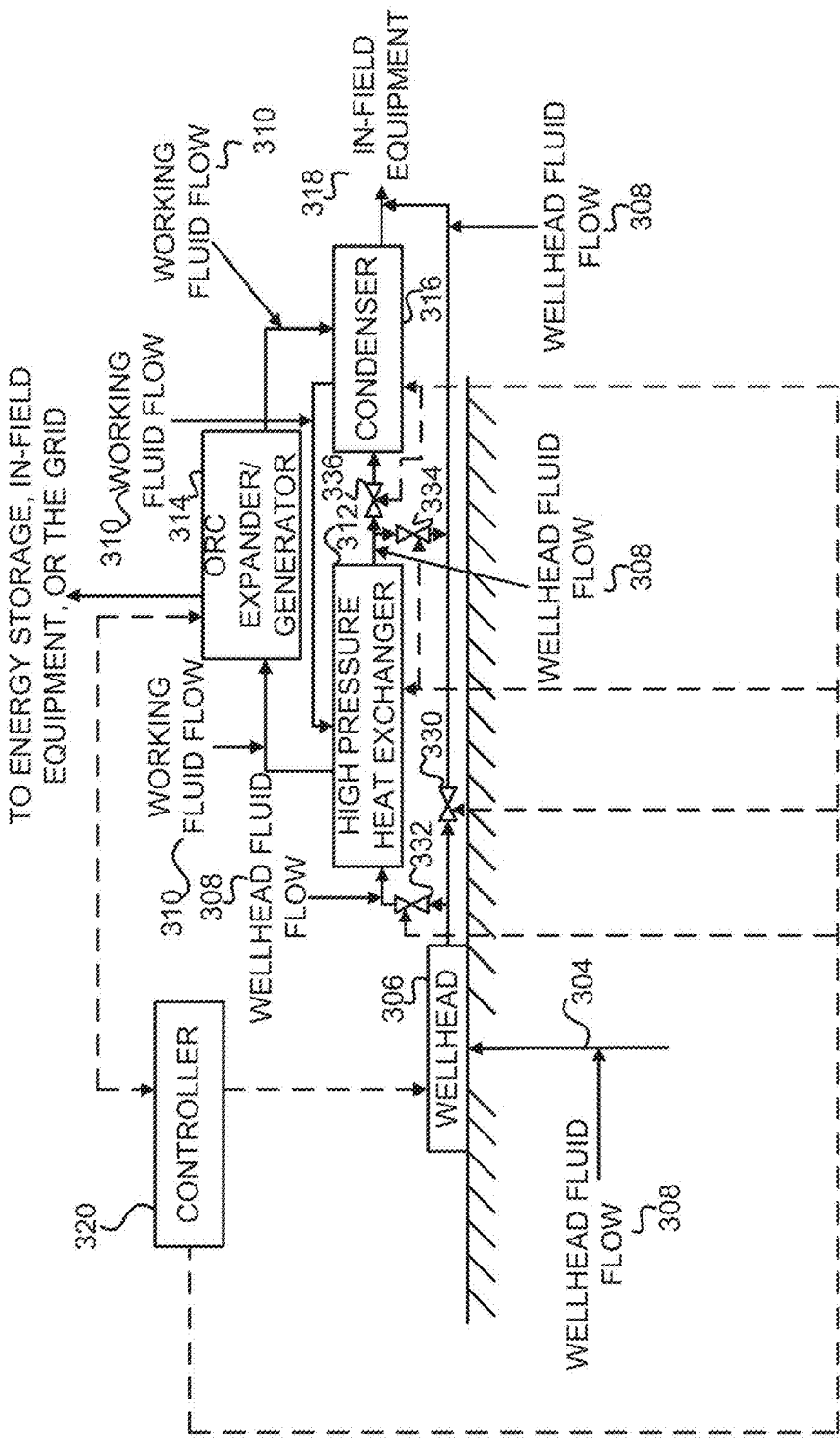


FIG. 3A

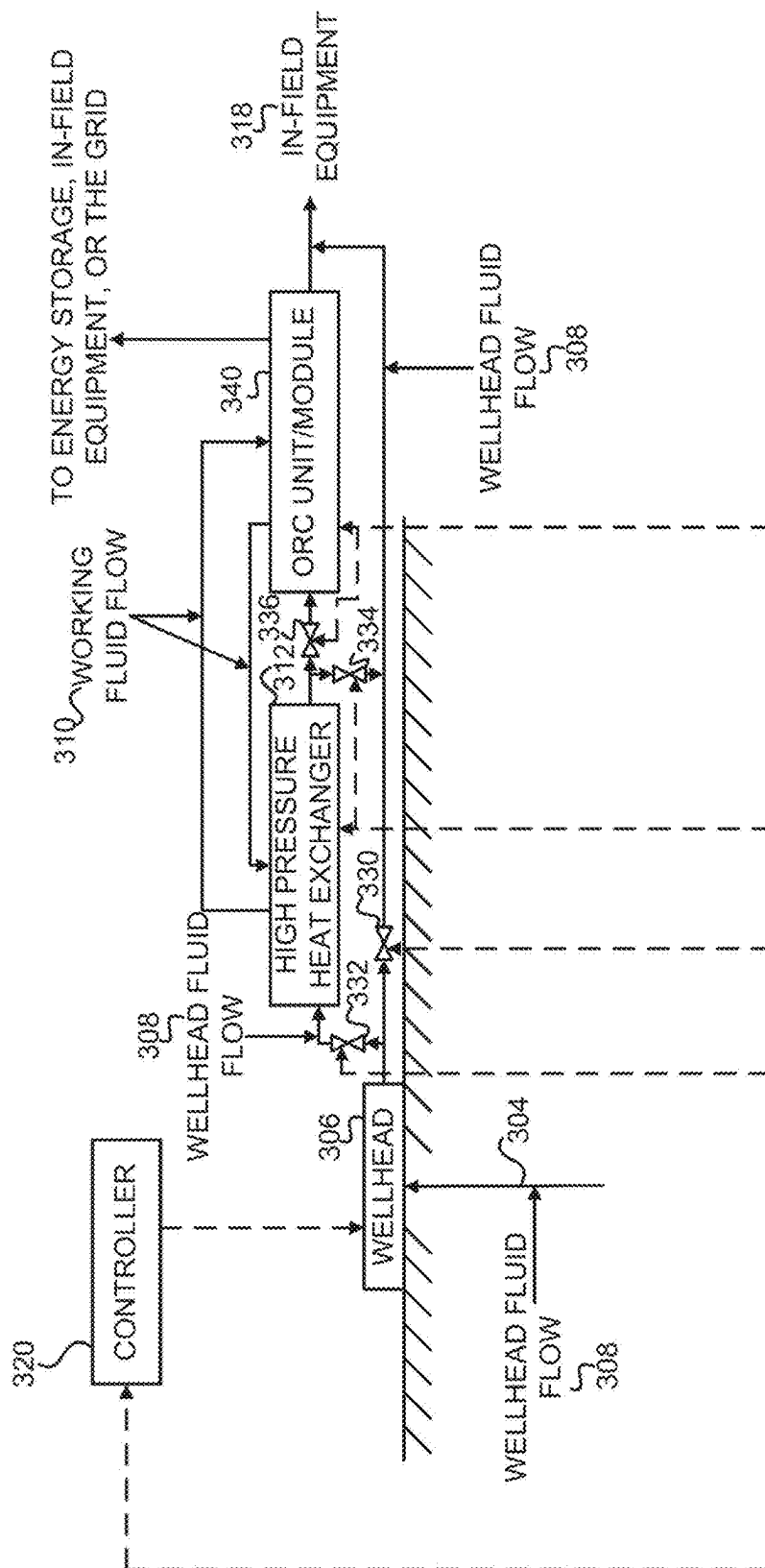


FIG. 3B

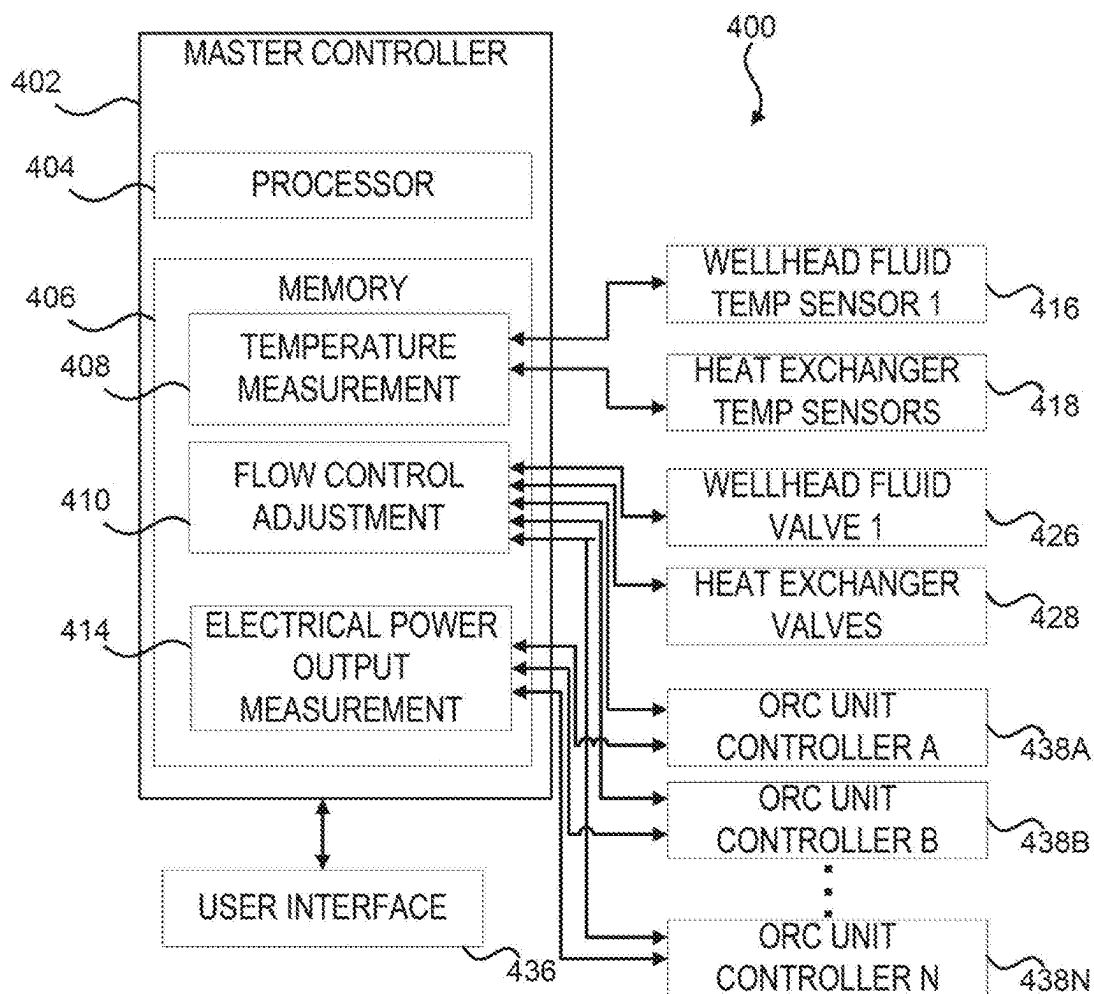
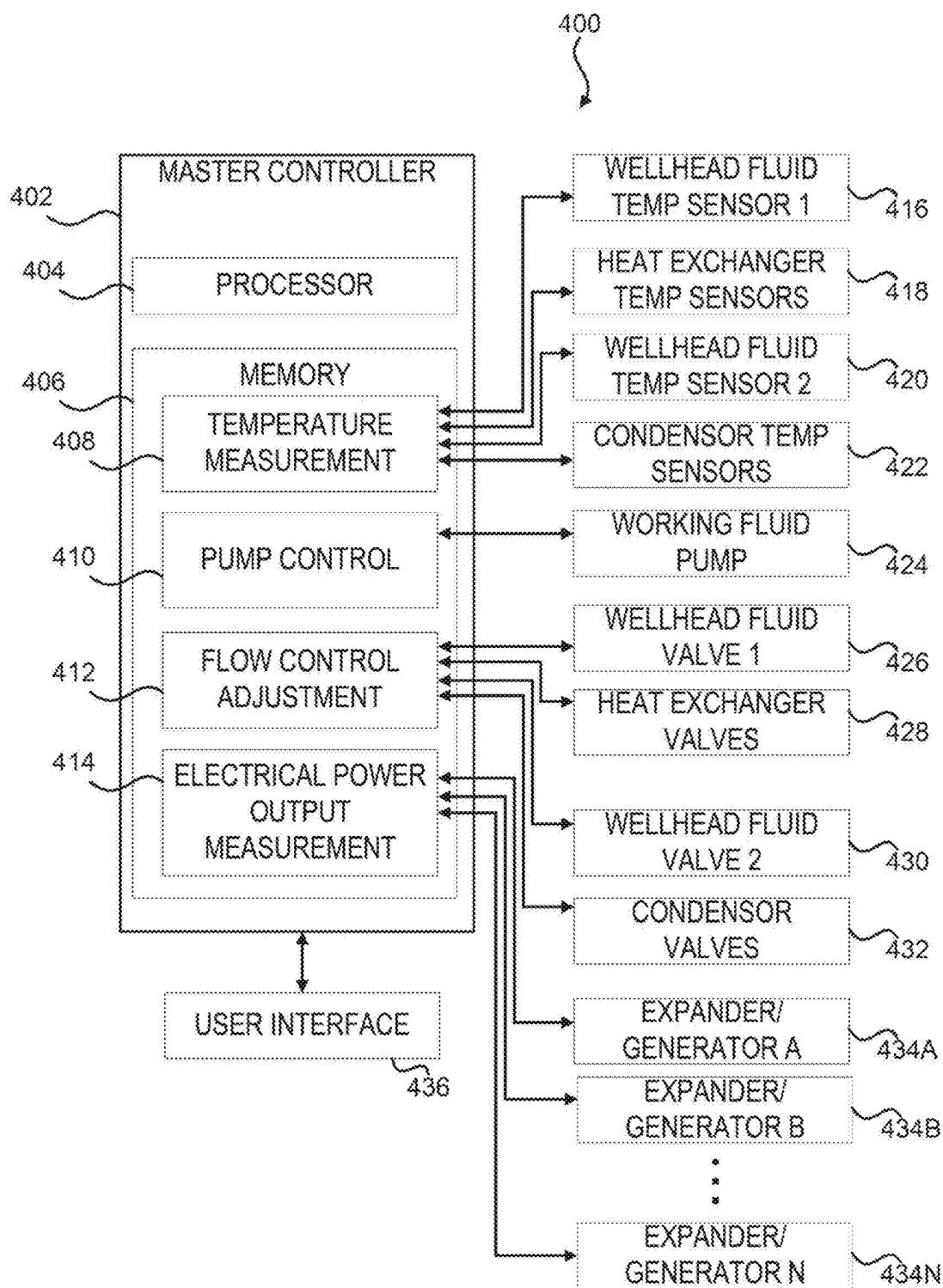


FIG. 4A



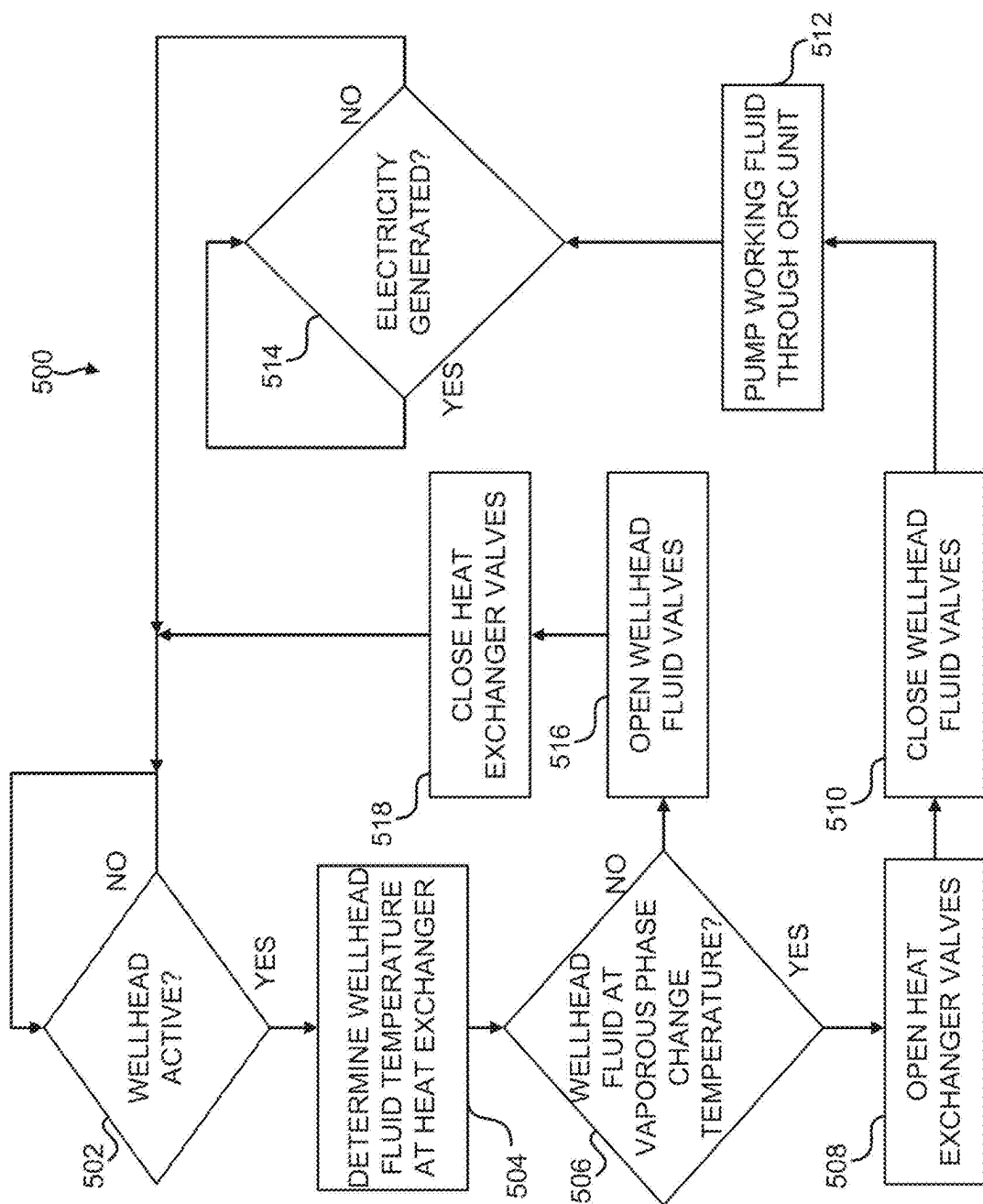


FIG. 5

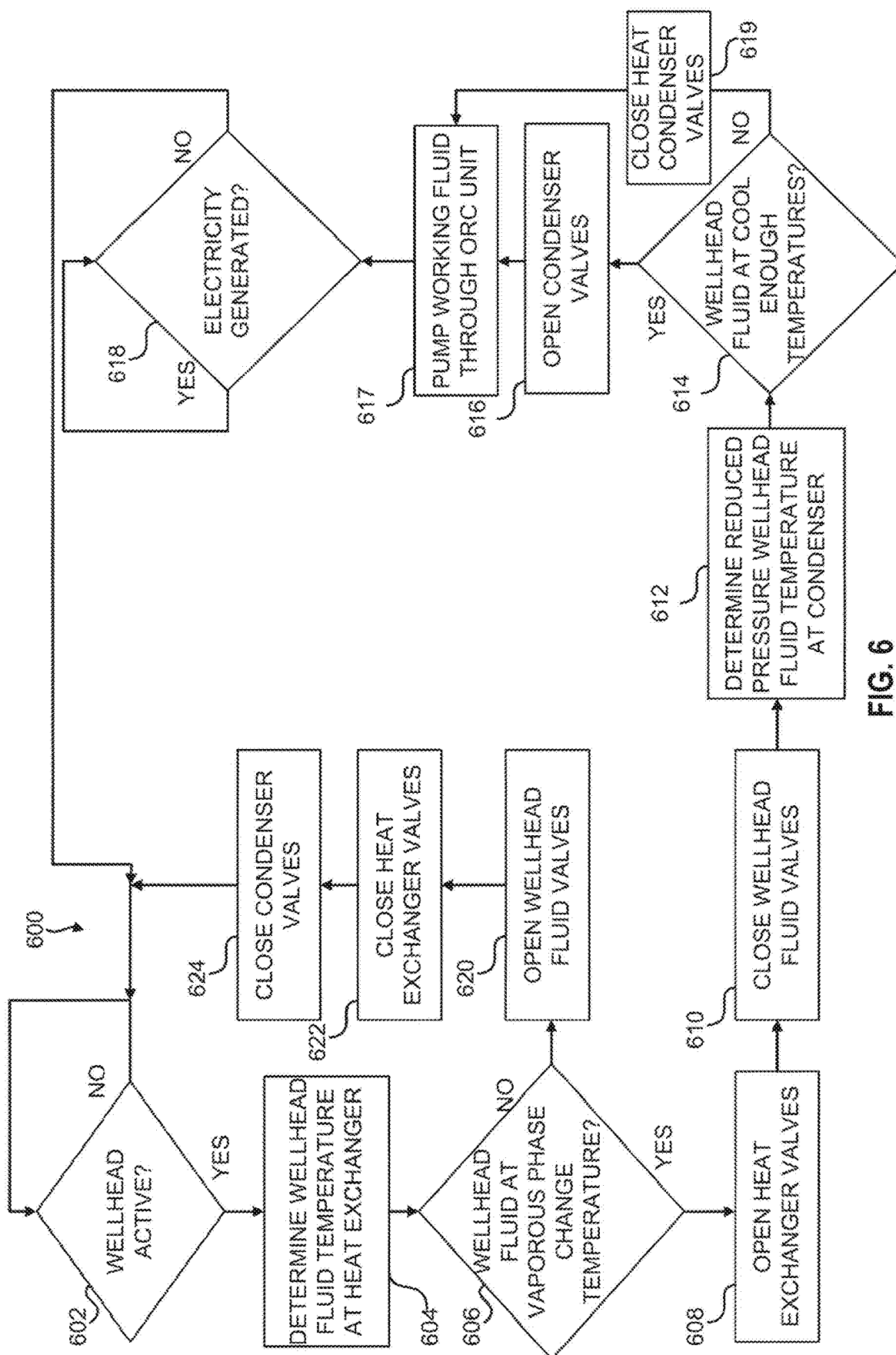


FIG. 6

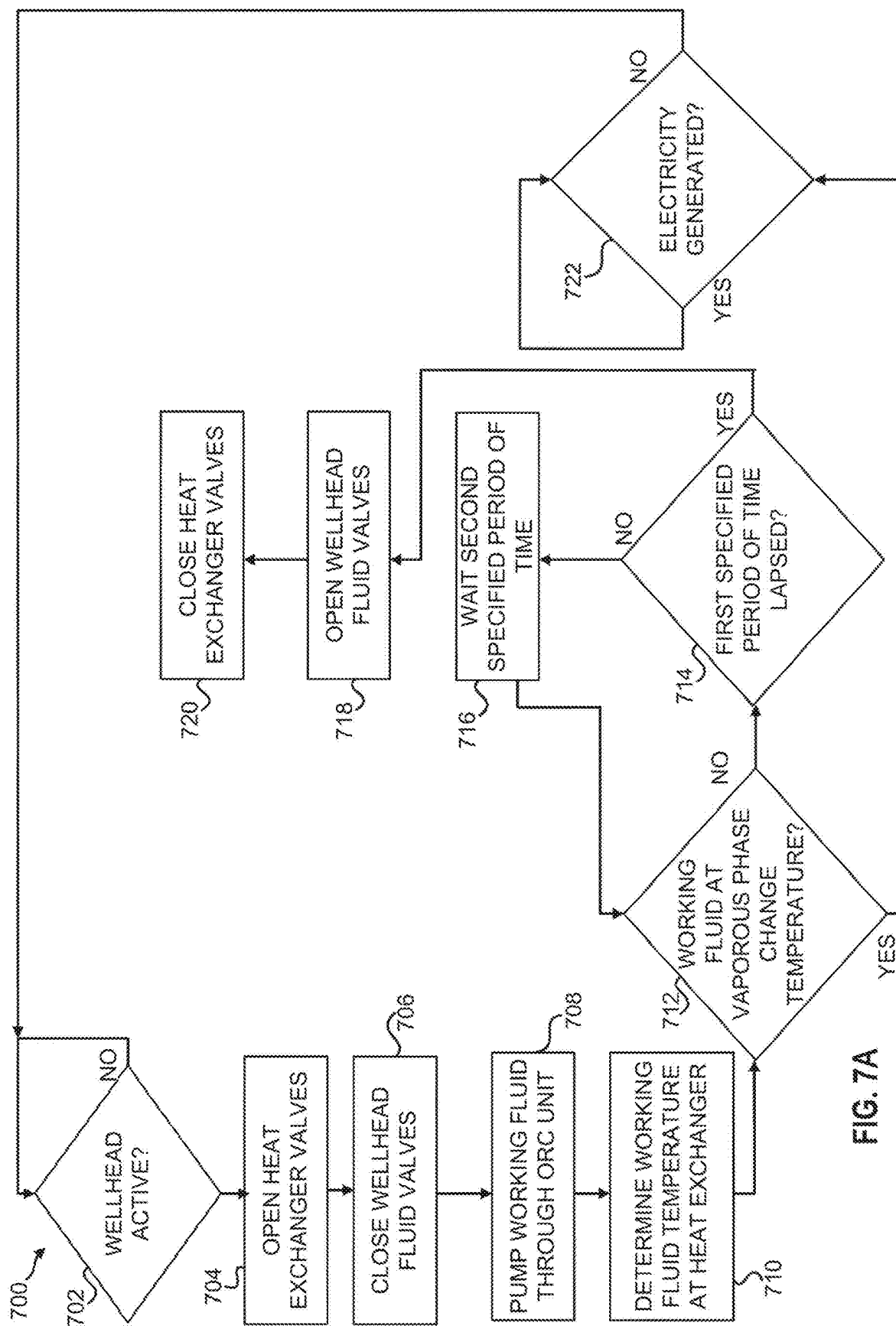


FIG. 7A

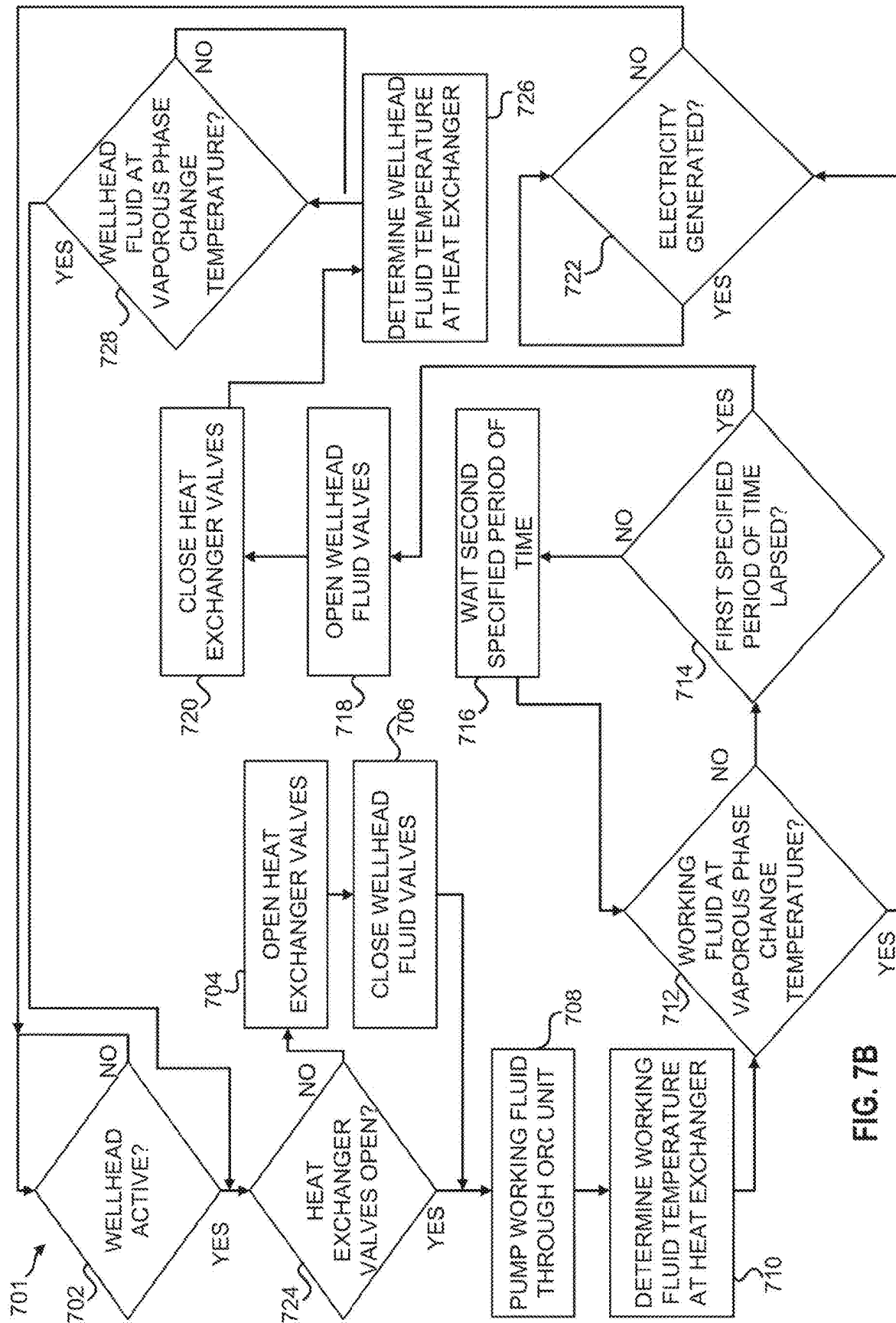


FIG. 7B

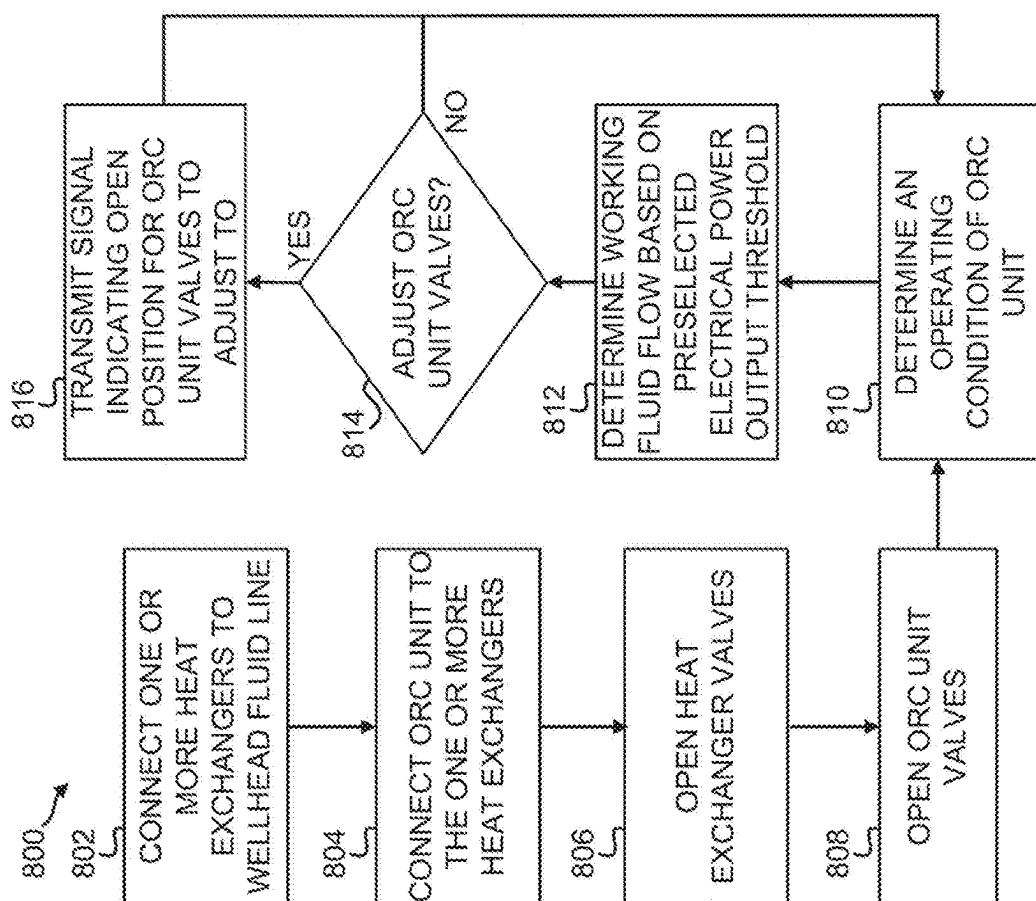
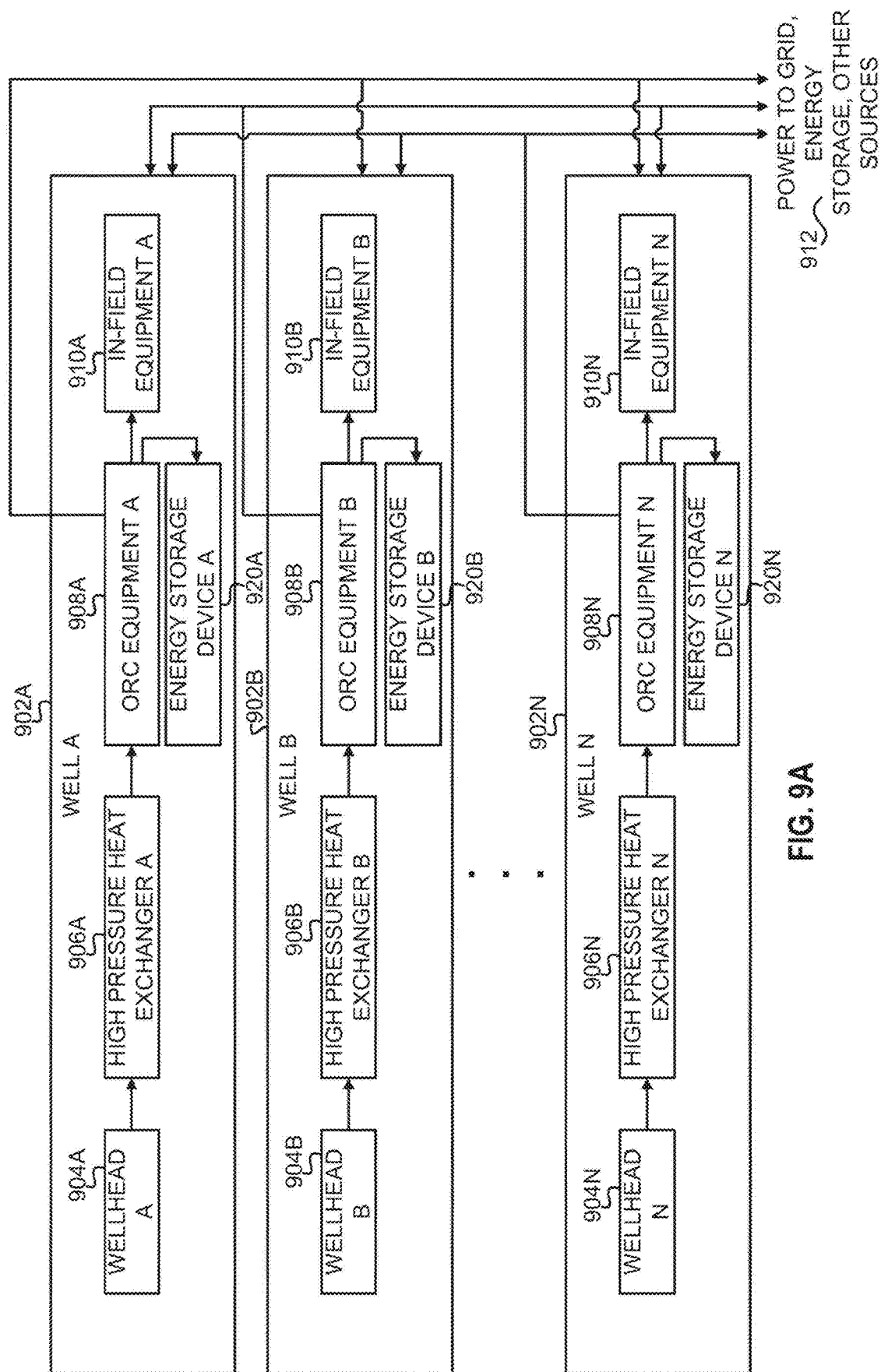


FIG. 8



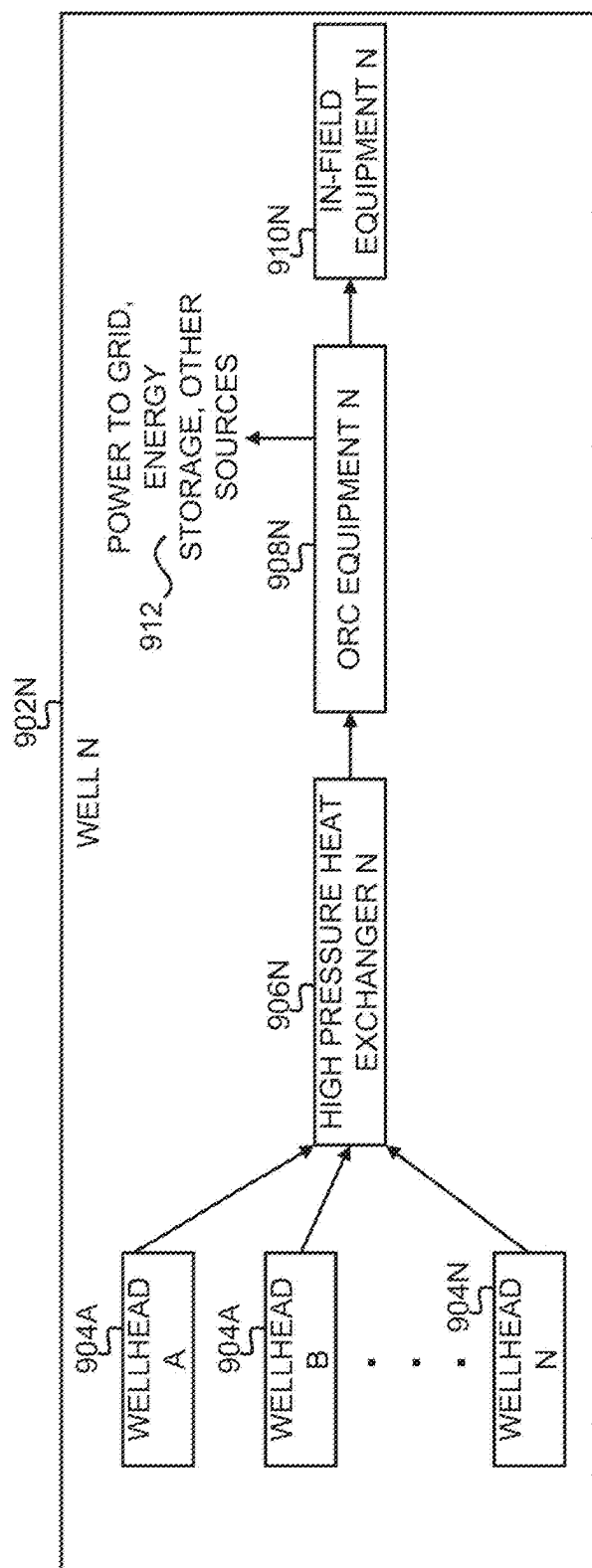
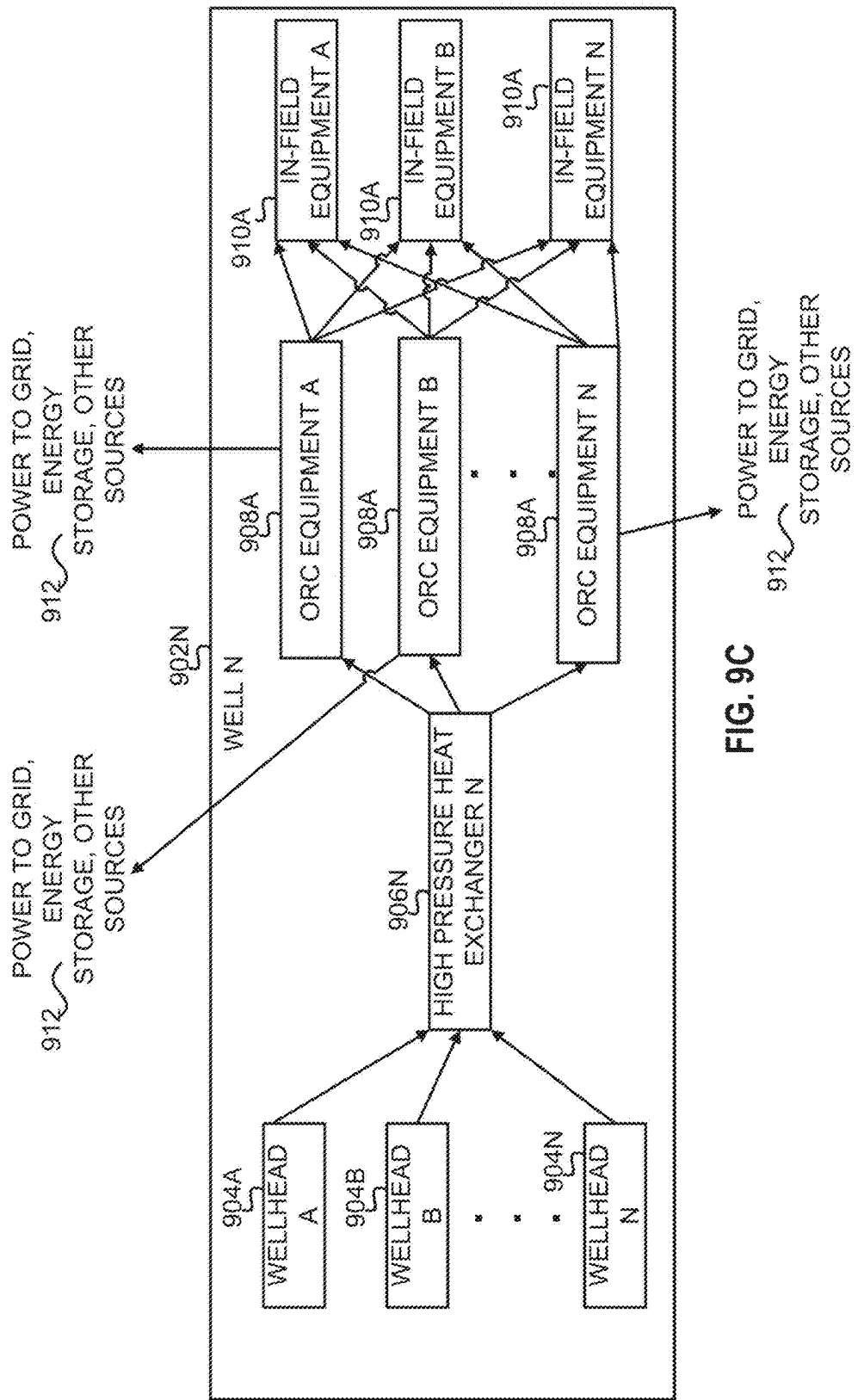


FIG. 9B



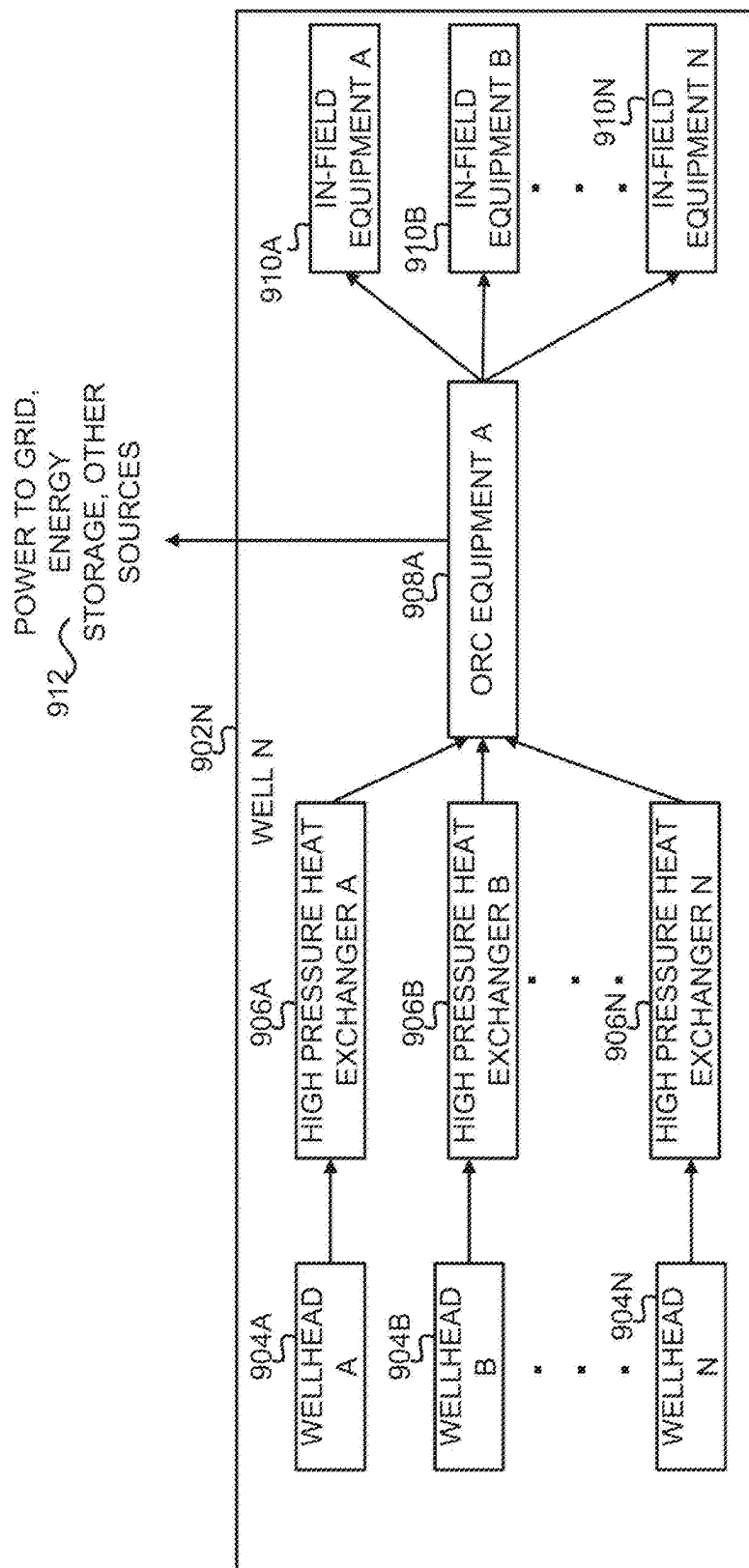
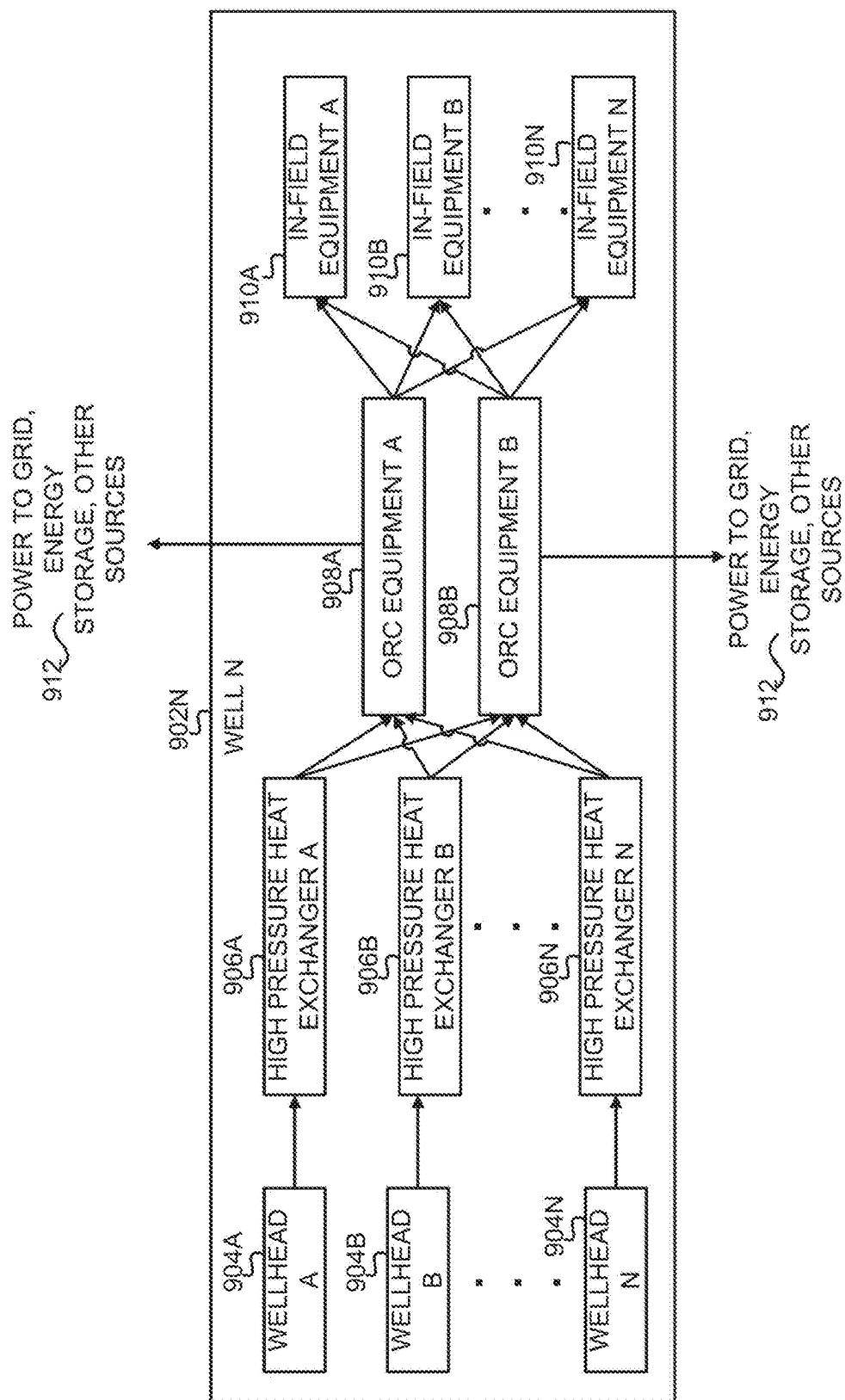


FIG. 9D



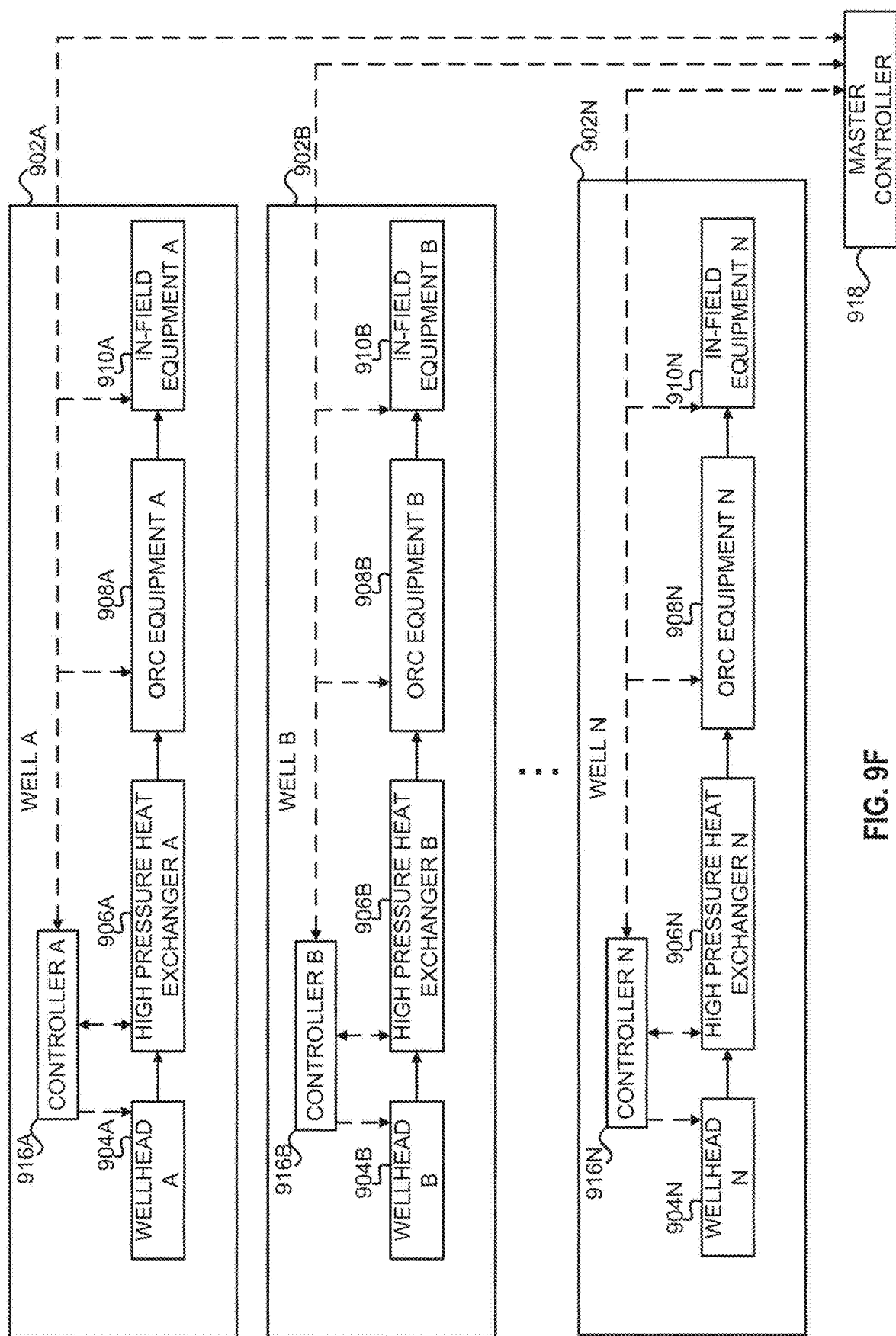


FIG. 9F

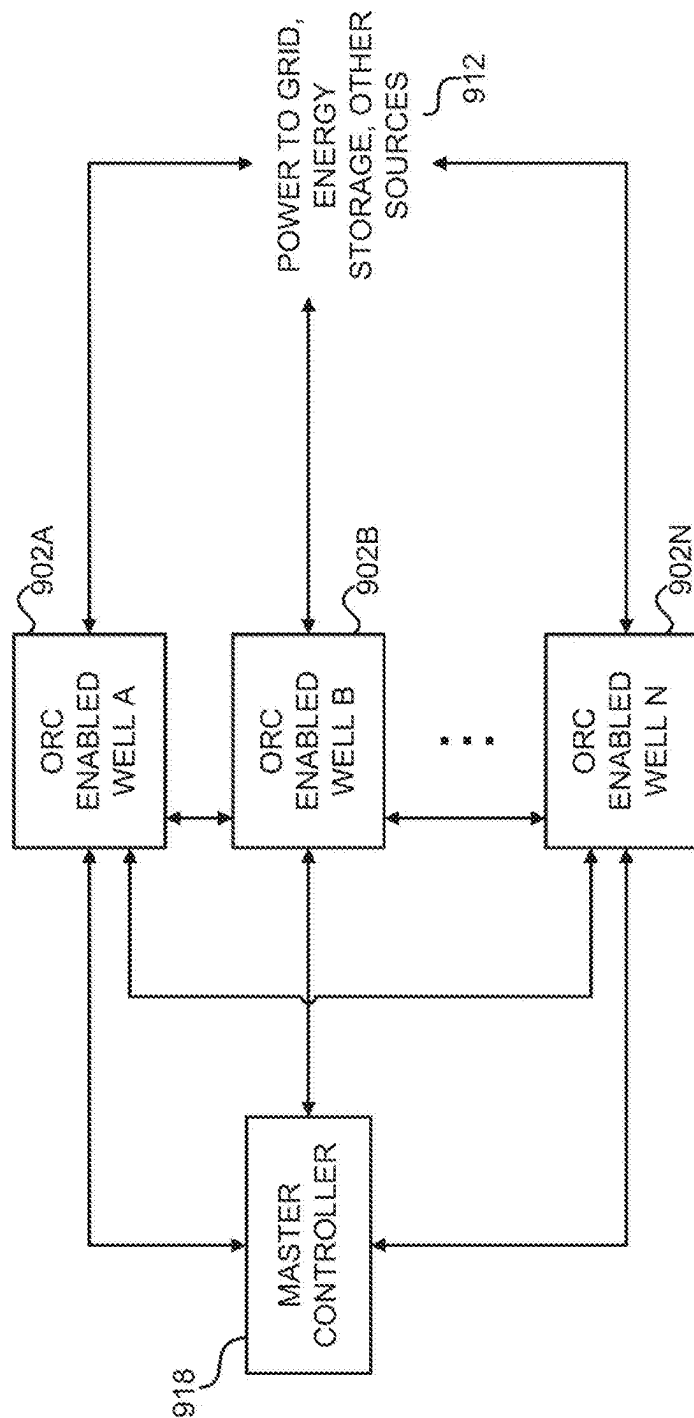


FIG. 9G

1

SYSTEMS FOR GENERATING GEOTHERMAL POWER IN AN ORGANIC RANKINE CYCLE OPERATION DURING HYDROCARBON PRODUCTION BASED ON WORKING FLUID TEMPERATURE

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. application Ser. No. 17/305,297, filed Jul. 2, 2021, titled "SYSTEMS FOR GENERATING GEOTHERMAL POWER IN AN ORGANIC RANKINE CYCLE OPERATION DURING HYDROCARBON PRODUCTION BASED ON WORKING FLUID TEMPERATURE", which claims priority to and the benefit of U.S. Provisional Application No. 63/200,908, filed Apr. 2, 2021, titled "SYSTEMS AND METHODS FOR GENERATING GEOTHERMAL POWER DURING HYDROCARBON PRODUCTION," the disclosures of which are incorporated herein by reference in their entirety.

FIELD OF DISCLOSURE

Embodiments of this disclosure relate to generating geothermal power during hydrocarbon production, and more particularly, to systems and methods for generating and controllers for controlling generation of geothermal power in an organic Rankine cycle (ORC) operation in the vicinity of a wellhead during hydrocarbon production to thereby supply electrical power to one or more of in-field operational equipment, a grid power structure, and an energy storage device.

BACKGROUND

Typically, geothermal generators include a working fluid loop that flows to varying depths underground, such that underground heat causes the working fluid in the loop to change phases from a liquid to a vapor. The vaporous working fluid may then flow to a gas expander, causing the gas expander to rotate. The rotation of the gas expander may cause a generator to generate electrical power. The vaporous working fluid may then flow to a condenser or heat sink. The condenser or heat sink may cool the working fluid, causing the working fluid to change phase from the vapor to the liquid. The working fluid may circulate through the loop in such a continuous manner, thus the geothermal generator may generate electrical power.

Heat exchangers within geothermal generators are not built to withstand high pressures. Typically, the working fluid does not flow through the loop under high pressure. Further, rather than including heat exchangers, typically, geothermal generators utilize geothermal heat from varying depths underground, the heat being transferred to the working fluid. For example, a geothermal generator may simply include a conduit or pipe buried deep underground. As working fluid flows through the conduit or pipe, the heat at such depths causes the working fluid to change to a vapor and the vapor may flow up to the gas expander. As the vapor is cooled, the vapor flows back underground, the cycle repeating. While old or non-producing wells have been utilized for geothermal power generation, no such solution is known to exist for generating geothermal power at a well during hydrocarbon production.

Accordingly, Applicants have recognized a need for systems and methods to generate geothermal power in an organic Rankine cycle (ORC) operation in the vicinity of a

2

wellhead during hydrocarbon production to thereby supply electrical power to one or more of in-field operational equipment, a grid power structure, and an energy storage device. The present disclosure is directed to embodiments of such systems and methods

SUMMARY

The present disclosure is generally directed to systems and methods for generating and controlling generation of geothermal power in an organic Rankine cycle (ORC) operation in the vicinity of a wellhead during hydrocarbon production to thereby supply electrical power to one or more of in-field operational equipment, a grid power structure, and an energy storage device. The wellhead may produce a wellhead fluid. The wellhead fluid, as it exits the wellhead, may be under high-pressure and at a high temperature. The wellhead fluid's temperature may be determined, as well as the pressure, and based on such determinations, one or more heat exchanger valves may be actuated to partially open or completely open, thereby diverting a portion or all of the flow of wellhead fluid to the heat exchanger. The heat exchanger may be a high-pressure heat exchanger configured to withstand the high-pressure of the wellhead fluid from a wellhead. The heat exchanger may indirectly transfer heat from the flow of the wellhead fluid to the flow of a working fluid. As heat is transferred from the flow of the wellhead fluid to the flow of a working fluid, such a heat transfer may cause the working fluid to change phases from a liquid to a vapor. The vaporous working fluid may then flow through an ORC unit to cause a generator to generate electrical power via rotation of a gas expander of the ORC unit. Such an operation may be defined as or may be an ORC operation or process. The ORC unit may be an off-the-shelf unit, while the high-pressure heat exchanger may be a stand-alone component or device. In another embodiment, the ORC unit may be a high-pressure ORC unit and may include the high-pressure heat exchanger apparatus.

Accordingly, an embodiment of the disclosure is directed to a system for generating geothermal power in an organic Rankine cycle (ORC) operation in the vicinity of a wellhead during hydrocarbon production to thereby supply electrical power to one or more of in-field equipment, a grid power structure, and energy storage devices. The system may include a first temperature sensor to provide a first temperature, the first temperature defined by a temperature of a flow of organic working fluid. The system may include a heat exchanger valve to divert flow of wellhead fluid from one or more wellheads based on the first temperature. The system may include a high-pressure heat exchanger including a first fluid path to accept and output the flow of wellhead fluid from the heat exchanger valve and a second fluid path to accept and output the flow of the organic working fluid, the high-pressure heat exchanger to indirectly transfer heat from the flow of wellhead fluid to the flow of the organic working fluid to cause the organic working fluid to change phases from a liquid to a vapor. The system may include an ORC unit. The ORC unit may include a generator, a gas expander, and a partial loop for the flow of the organic working fluid, the partial loop defined by a fluid path through the condenser, generator, and pump, the partial loop forming a complete loop when connected to the fluid path of the high-pressure heat exchanger, the flow of the organic working fluid, as a vapor, so as to cause the generator to generate electrical power via rotation of a gas expander as defined by an ORC operation, the condenser to cool the flow of the organic working fluid, the cooling to cause the organic

3

working fluid to change phases from the vapor to the liquid, the pump to transport the liquid state organic working fluid from the condenser for heating.

In another embodiment, the system may include a pump and/or a condenser. The system may include a first wellhead fluid valve to adjust flow of wellhead fluid from the one or more wellheads based on the diversion of the flow of wellhead fluid to the heat exchanger valve. The system may include another one or more ORC units connected to the high-pressure heat exchanger. The ORC unit may include a working fluid reservoir to store organic working fluid flowing from the condenser.

In another embodiment, the first fluid path of the high-pressure heat exchanger may be configured to withstand corrosion caused by the wellhead fluid. The high-pressure heat exchanger may include a vibration induction device to reduce scaling caused by the flow of the wellhead fluid.

Another embodiment is directed to a system for generating geothermal power in the vicinity of a wellhead during hydrocarbon production to thereby supply electrical power to one or more of in-field equipment, a grid power structure, and energy storage devices. The system may include a first pipe connected to and in fluid communication with the wellhead. The first pipe may be configured to transport wellhead fluid under high-pressure. The system may include a first wellhead fluid valve having a first end and second end, the first end of the first wellhead fluid valve connected to and in fluid communication with the first pipe, the first wellhead fluid valve to control flow of wellhead fluid based on an organic working fluid temperature. The system may include a heat exchanger valve connected to and in fluid communication with the first pipe, the heat exchanger valve to control flow of wellhead fluid on an organic working fluid temperature. The system may include a high-pressure heat exchanger to accept the flow of wellhead fluid when the heat exchanger valve is open. The high-pressure heat exchanger may include a first opening and a second opening connected via a first fluidic path and a third opening and a fourth opening connected via a second fluidic path, the first fluidic path and the second fluidic path to facilitate heat transfer from the flow of wellhead fluid to an organic working fluid, the transfer of heat from the wellhead fluid to the organic working fluid to cause the organic working fluid to change phases from a liquid to a vapor, the flow of wellhead fluid flowing into the first opening of the high-pressure heat exchanger from the heat exchanger valve through the first fluidic path and to the second opening of the high-pressure heat exchanger, and a flow of the organic working fluid flowing into the third opening through the second fluidic path and out of the fourth opening. The system may include a first temperature sensor connected to the fourth opening, the first temperature sensor to provide the organic working fluid temperature, the organic working fluid temperature defined by a temperature of organic working fluid flowing through the second fluidic path. The system may include a second pipe connected to and in fluid communication with the second end of the first wellhead fluid valve and connected to and in fluid communication with the second opening of the high-pressure heat exchanger. The system may include a generator connected to and in fluid communication with the fourth opening of the high-pressure heat exchanger, the organic working fluid flowing from the fourth opening to the generator and causing the generator to generate electrical power via rotation of a vapor expander as defined by an ORC operation. The system may include a condenser to facilitate heat transfer from the organic working fluid to a coolant. The system may include a pump

4

connected to the condenser to pump the organic working fluid from the condenser to the third opening of the high-pressure heat exchanger.

In an embodiment, the organic working fluid may include one of pentafluoropropane, carbon dioxide, ammonia and water mixtures, tetrafluoroethane, isobutene, propane, pentane, perfluorocarbons, and other hydrocarbons. The generator may include a rotation mechanism, a stator, and rotor, the rotor connected to the rotation mechanism, the rotor to rotate as the rotation mechanism spins via the flow of organic working fluid. The rotation mechanism may include one of a turbine expander, a positive displacement expander, or a twin-screw expander. The rotation mechanism may connect to the rotor via one of a transmission and gearbox.

In another embodiment, when the heat exchanger valve is opened, a portion of the flow of wellhead fluid is diverted from the first wellhead fluid valve. The organic working fluid may change phase from liquid to vapor when the wellhead fluid is at a temperature from greater than or equal to about 50 degrees Celsius.

In another embodiment, after the organic working fluid flows through the expander, the organic working fluid flows through a regenerator and, after the working fluid flows through the condenser, the organic working fluid flows through the regenerator, the regenerator pre-heating the organic working fluid from the condenser via heat of the organic working fluid from the generator.

In another embodiment, the high-pressure heat exchanger may be configured to withstand pressures up to about 15,000 pounds per square inch (PSI). The high-pressure heat exchanger may include pressure relief valves, in an event that the wellhead fluid flowing to the high-pressure heat exchanger exceeds a pressure rating of the high-pressure heat exchanger.

In another embodiment, the wellhead fluid includes one of a liquid and a gas. The wellhead fluid may include hydrocarbons. The wellhead fluid may further include a mixture of the hydrocarbons and one or more of water and other chemical residuals. The working fluid may include a mixture of two or more fluids, each of the two or more fluids including different vaporous phase change points and different condensation points.

In another embodiment, the generator, condenser, and pump define an ORC unit. The system may include another one or more high-pressure heat exchangers. The ORC unit may connect to each of the high-pressure heat exchanger and the another one or more high-pressure heat exchangers. Each of one or more ORC unit valves may connect and be positioned between the ORC unit and each of the high-pressure heat exchanger and the another one or more high-pressure heat exchangers. Each of the one or more ORC unit valves may be adjustingly opened to a position based on an efficient working fluid flow to the ORC unit. The efficient working fluid flow may be based on one or more of a temperature of working fluid flowing from each of the high-pressure heat exchanger and the another one or more high-pressure heat exchangers, an electrical output from the ORC unit, and a current open position of each of the one or more ORC unit valves.

Still other aspects and advantages of these embodiments and other embodiments, are discussed in detail herein. Moreover, it is to be understood that both the foregoing information and the following detailed description provide merely illustrative examples of various aspects and embodiments, and are intended to provide an overview or framework for understanding the nature and character of the claimed aspects and embodiments. Accordingly, these and

5

other objects, along with advantages and features of the present invention herein disclosed, will become apparent through reference to the following description and the accompanying drawings. Furthermore, it is to be understood that the features of the various embodiments described herein are not mutually exclusive and may exist in various combinations and permutations.

BRIEF DESCRIPTION OF DRAWINGS

These and other features, aspects, and advantages of the disclosure will become better understood with regard to the following descriptions, claims, and accompanying drawings. It is to be noted, however, that the drawings illustrate only several embodiments of the disclosure and, therefore, are not to be considered limiting of the scope of the disclosure.

FIG. 1A and FIG. 1B are schematic top-down perspectives of novel implementations of a geothermal power generation enabled well, according to one or more embodiment of the disclosure.

FIG. 2A, FIG. 2B, FIG. 2C, FIG. 2D, FIG. 2E, FIG. 2F, FIG. 2G, and FIG. 2H are block diagrams illustrating novel implementations of a geothermal power generation enabled well to provide electrical power to one or more of in-field equipment, equipment at other wells, energy storage devices, and the grid power structure, according to one or more embodiment of the disclosure.

FIG. 3A and FIG. 3B are block diagrams illustrating other novel implementations of a geothermal power generation enabled well to provide electrical power to one or more of in-field equipment, equipment at other wells, energy storage devices, and the grid power structure, according to one or more embodiment of the disclosure.

FIG. 4A and FIG. 4B are simplified diagrams illustrating a control system for managing geothermal power production at a well, according to one or more embodiment of the disclosure.

FIG. 5 is a flow diagram of geothermal power generation in which, when a wellhead fluid is at or above a vaporous phase change temperature, heat exchanger valves may be opened to allow wellhead fluid to flow therethrough, thereby facilitating heating of a working fluid for use in an organic Rankine cycle (ORC) unit, according to one or more embodiment of the disclosure.

FIG. 6 is another flow diagram of geothermal power generation in which, when a wellhead fluid is at or above a vaporous phase change temperature, heat exchanger valves may be opened to allow wellhead fluid to flow therethrough, thereby facilitating heating of a working fluid for use in an ORC unit, according to one or more embodiment of the disclosure.

FIG. 7A is a flow diagram of geothermal power generation in which, when an ORC fluid is at or above a vaporous phase change temperature, heat exchanger valves may remain open to allow wellhead fluid to flow therethrough, thereby facilitating heating of a working fluid for use in an ORC unit, according to one or more embodiment of the disclosure.

FIG. 7B is another flow diagram of geothermal power generation in which, when an ORC fluid is at or above a vaporous phase change temperature, heat exchanger valves may remain open to allow wellhead fluid to flow therethrough, thereby facilitating heating of a working fluid for use in an ORC unit, according to one or more embodiment of the disclosure.

6

FIG. 8 is a flow diagram of geothermal power generation in which a working fluid flow is determined based on a preselected electrical power output threshold, according to one or more embodiment of the disclosure.

FIG. 9A, FIG. 9B, FIG. 9C, FIG. 9D, FIG. 9E, FIG. 9F, and FIG. 9G are block diagrams illustrating novel implementations of one or more geothermal power generation enabled wells to provide electrical power to one or more of in-field equipment, equipment at one of the other wells, energy storage devices, and the grid power structure, according to one or more embodiment of the disclosure.

DETAILED DESCRIPTION

So that the manner in which the features and advantages of the embodiments of the systems and methods disclosed herein, as well as others that will become apparent, may be understood in more detail, a more particular description of embodiments of systems and methods briefly summarized above may be had by reference to the following detailed description of embodiments thereof, in which one or more are further illustrated in the appended drawings, which form a part of this specification. It is to be noted, however, that the drawings illustrate only various embodiments of the systems and methods disclosed herein and are therefore not to be considered limiting of the scope of the systems and methods disclosed herein as it may include other effective embodiments as well.

The present disclosure is directed to systems and methods for generating geothermal power in an organic Rankine cycle (ORC) operation in the vicinity of a wellhead during hydrocarbon production to thereby supply electrical power to one or more of in-field equipment or operational equipment, a grid power structure, other equipment, and an energy storage device. Wellhead fluids flowing from a wellhead at a well are typically under high-pressure. In-field equipment at the well is not rated for such high pressures. Prior to further processing or transport, the pressure of the flow of the wellhead fluid may be reduced, e.g., from 15,000 PSI to 200 PSI, from 10,000 PSI to 200 PSI, from 2,000 PSI to 200 PSI, or any other range from 20,000 PSI to 100 PSI, based on the pressure rating of the in-field equipment at the well. As the wellhead fluid flows from the wellhead, the temperature of the flow of the wellhead fluid may be at a high temperature, at least partially due to the high pressure of the flow of the wellhead fluid. As the pressure is reduced, the wellhead fluid temperature may also be reduced, as result of the pressure drop. Typically, the heat of the flow of wellhead fluid from the wellhead is not utilized and may be considered heat waste.

Geothermal power generators typically use a looping pipe or pipeline buried at depths with sufficient temperature to allow a working fluid to change phase from liquid to vapor. As the working fluid changes phase from a liquid to a vaporous state, the vaporous state working fluid may flow up the pipe or pipeline to a gas expander. The vaporous state working fluid may flow through and cause the gas expander to rotate. The rotation of the gas expander may cause a generator to generate electrical power, as will be described below. The vaporous state working fluid may flow through the gas expander to a heat sink, condenser, or other cooling apparatus. The heat sink, condenser, or other cooling apparatus may cool the working fluid thereby causing the working fluid to change phases from a vapor to a liquid. Heat exchangers of typical geothermal generators are not rated for high-pressure operations and usually geothermal generators obtain heat from varying underground depths.

In the present disclosure, a high-pressure heat exchanger may be placed or disposed at the well and/or in the vicinity of one or more wellheads. The high-pressure heat exchanger may be connected to the wellhead and may accept a high temperature or heated flow of wellhead fluid. A working fluid may flow through the heat exchanger. As the wellhead fluid and working fluid flows through the high-pressure heat exchanger, the high-pressure heat exchanger may facilitate transfer of heat from the wellhead fluid to the working fluid. A heat exchanger may include two fluidic paths, one for a heated fluid and another for a cool fluid. The fluidic paths may be in close proximity, allowing heat to transfer from the heated fluid to the cool fluid. The fluidic paths may be loops, coils, densely packed piping, tubes, chambers, some other type of path to allow for fluid to flow therethrough, and/or a combination thereof, as will be understood by those skilled in the art. As fluids flow through the heat exchanger, the cool liquid's temperature may increase, while the heated liquid's temperature may decrease.

Additionally, a geothermal generator unit or ORC unit may be disposed, positioned, or placed at the wellhead. The geothermal generator unit or ORC unit may directly connect to the high-pressure heat exchanger, include the high-pressure heat exchanger, or may connect to the high-pressure heat exchanger via an intermediary heat exchanger. As the hot wellhead fluid heats the working fluid, either via direct connection or through an intermediary heat exchanger, the working fluid may change phases from a liquid to a vapor. In such examples, the working fluid utilized may be chosen based on a low boiling point and/or high condensing point. The vaporous state working fluid may flow through the geothermal generator unit or ORC unit to a generator, e.g., a gas expander and generator. The vaporous state working fluid may then flow to a condenser or heat sink, thereby changing state from the vapor to the liquid. Finally, the liquid may be pumped back to the high-pressure heat exchanger. Such a cycle, process, or operation may be considered a Rankine cycle or ORC.

Such systems may include various components, devices, or apparatuses, such as temperature sensors, pressure sensors or transducers, flow meters, control valves, smart valves, valves actuated via control signal, controllers, a master or supervisory controller, other computing devices, computing systems, user interfaces, in-field equipment, and/or other equipment. The controller may monitor and adjust various aspects of the system to ensure that hydrocarbon production continues at a specified rate, that downtime is limited or negligible, and that electrical power is generated efficiently, optimally, economically, and/or to meet or exceed a preselected electrical power output threshold.

FIGS. 1A and 1B are schematic top-down perspectives of novel implementations of a geothermal power generation enabled well, according to one or more embodiment of the disclosure. As illustrated in FIG. 1A, a well 100 may include various components or equipment, also referred to as in-field equipment. Such in-field equipment may include fracturing equipment, field compressors, pump stations, artificial lift equipment, drilling rigs, data vans, and/or any other equipment utilized or used at a well 100. For example, the well 100 may include one or more pumpjacks 108, one or more wellhead compressors 110, various other pumps, various valves, and/or other equipment that may use electrical power or other type of power to operate. To generate power from otherwise wasted heat, the well 100 may additionally include a high-pressure heat exchanger 104, one or more geothermal generators, one or more ORC units 106, a high-pressure geothermal generator, a high-pressure ORC

unit, and/or some combination thereof. As wellhead fluid flows from one of the one or more wellheads 102, a portion of the flow of wellhead fluid or all of the flow of wellhead fluid may flow through the high-pressure heat exchanger 104, high-pressure geothermal generator, a high-pressure ORC unit, or some combination thereof. As the hot and high-pressure wellhead fluid flows through, for example, the high-pressure heat exchanger 104, the high-pressure heat exchanger 104 may facilitate a transfer of heat from the wellhead fluid to a working fluid flowing through the high-pressure heat exchanger 104. In other words, the wellhead fluid may heat the working fluid. Such a heat transfer may cause the working fluid to change phases from a liquid to a vapor. The vaporous state working fluid may flow from the high-pressure heat exchanger to the one or more ORC units 106. The one or more ORC units 106 may then generate power using the vaporous state working fluid. The electrical power may be transferred to the in-field equipment at the well 100, to an energy storage device (e.g., if excess power is available), to equipment at other wells, to the grid or grid power structure (e.g., via a transformer 116 through power lines 118), or some combination thereof.

As illustrated in FIG. 1B, one or more wells 100A, 100B, 100C may be nearby or in close proximity to each of the other one or more wells 100A, 100B, 100C. Further, each of the one or more wells 100A, 100B, 100C may utilize different amounts of electrical power, in addition to generating different amounts of electrical power. As such, one well (e.g., well 100A, well 100B, and/or well 100C) of the one or more wells 100A, 100B, 100C may generate a surplus of electrical power or utilize electrical power from other sources. In an example, a controller may determine if a well (e.g., well 100A, well 100B, and/or well 100C) of the one or more wells 100A, 100B, 100C generates a surplus. If a surplus is generated, the controller may determine which, if any, of the other one or more wells 100A, 100B, 100C may have a deficit of electrical power. The controller may then transmit signals to equipment at the one or more wells 100A, 100B, 100C to enable electrical power transfer between the one or more wells 100A, 100B, 100C with excess and deficits, e.g., a well with a deficit may receive electrical power from a well with a surplus (see 120). In another example, the one or more wells 100A, 100B, 100C may include energy storage devices e.g., batteries, battery banks, or other solutions to store energy for short or long term time periods. The energy storage devices may be placed, disposed, or installed at one or more of the one or more wells 100A, 100B, 100C or at points in between the one or more wells 100A, 100B, 100C. As surplus electrical power is generated, that surplus electrical power may be transmitted and stored in the energy storage devices. The energy storage devices may be accessible by the in-field equipment of each of the one or more wells 100.

As illustrated in FIGS. 1A and 1B, the one or more wells 100A, 100B, 100C may include a high-pressure heat exchanger 104 and ORC units 106. In an example, the ORC units 106 may be modular and/or mobile. The ORC units 106 may be mounted to a vehicle, such as a truck or other vehicle type, or skid and transported to the well 100. Further, the high-pressure heat exchanger 104 may be modular and/or mobile. The high-pressure heat exchanger 104 may be mounted to a vehicle and/or skid. Upon arrival at one of the one or more wells 100A, 100B, 100C, the high-pressure heat exchanger 104 may be removed from the vehicle or the vehicle may be left on-site or at least while the one or more wells 100A, 100B, 100C are producing hydrocarbons. In an example, during hydrocarbon production, operation of the

high-pressure heat exchanger **104** and ORC unit **106** may occur. After hydrocarbon production has ceased, none, some, or all of the equipment may be removed from the well **100**. For example, the well **100** may be re-used for generating geothermal energy via a different method. In such examples the ORC units **106** and high-pressure heat exchanger **104** may remain on-site.

FIG. 2A, FIG. 2B, FIG. 2C, FIG. 2D, FIG. 2E, FIG. 2F, FIG. 2G, and FIG. 2H are block diagrams illustrating novel implementations of a geothermal power generation enabled well to provide electrical power to one or more of in-field equipment, equipment at other wells, energy storage devices, and the grid power structure, according to one or more embodiment of the disclosure. As illustrated in the FIGS. 2A through 2H, different embodiments may be utilized for geothermal power generation at the surface of a well during hydrocarbon production. As illustrated in FIG. 2A, a well **200** may include a wellhead **202**. The wellhead **202** may produce a stream or flow of wellhead fluid. The wellhead fluid may flow from a first pipe **215**. The first pipe **215** may include, at various positions along the length of the pipe **215**, sensors, meters, transducers, and/or other devices to determine the characteristics of the wellhead fluid flowing through pipe **215**. Further, sensors, meters, transducers and/or other devices may be included at various positions within the high-pressure heat exchanger **250** and/or ORC unit **203** to determine the characteristics of the working fluid or ORC fluid flowing through the high-pressure heat exchanger **250** and/or ORC unit **203**. Based on the determined characteristics of the wellhead fluid, working fluid, and/or ORC fluid and/or characteristics of other aspects of the well **200**, e.g., temperature, pressure, flow, power demand and/or power storage or transmission options and/or other factors, a first heat exchanger valve **208** connected to the first pipe **215** may be opened, e.g., partially opened or fully opened.

Further, when the first heat exchanger valve **208** is opened, a second heat exchanger valve **222** connected to a second pipe **217** may be opened, e.g., partially opened or fully opened. The second heat exchanger valve **222** may allow the flow of wellhead fluid to exit the high-pressure heat exchanger **250**. Based on the opening of the first heat exchanger valve **208** and second heat exchanger valve **222**, a first wellhead fluid valve **210** may be adjusted. Such adjustment may occur to ensure that the production of the wellhead **202** or the production of wellhead fluid from the wellhead **202** may not be impeded or slowed based on a diversion of the flow of the wellhead fluid to the high-pressure heat exchanger **250**. Downstream of the first wellhead fluid valve **210**, but prior to where the diverted portion of the flow of wellhead fluid is reintroduced to the primary or bypass wellhead fluid flow, a pressure sensor **212** may be disposed, e.g., the pressure sensor **212** may be disposed along pipe **217**. In another example, rather than a pressure sensor **212**, a flow meter may be disposed along pipe **217**, e.g., a wellhead fluid flow meter and/or a downstream flow meter. The wellhead fluid flow meter may measure the flow rate of a fluid exiting the wellhead **202**. A downstream flow meter may measure the flow rate of the wellhead fluid at a point downstream of the first wellhead fluid valve **210**. Such a pressure sensor **212** or flow meter may be utilized to determine whether the flow of wellhead fluid is at a pressure or flow that does not impede hydrocarbon production. In an example, where the flow is impeded to a degree that the production of hydrocarbons may be inhibited, the first wellhead fluid valve **210** may be opened further, if the first wellhead fluid valve **210** is not already fully opened. If the first wellhead fluid valve **210** is fully opened, the first heat

exchanger valve's **208** and second heat exchanger valve's **222** percent open may be adjusted. Other factors may be taken into account for such determinations, such as the pressure or flow from the high-pressure heat exchanger **250**, the pressure or flow at the point of a choke valve located further downstream along the second pipe **217**, a pressure rating for downstream equipment, and/or the temperature of the wellhead fluid within the high-pressure heat exchanger **250**. Once the flow of wellhead fluid passes through the first wellhead fluid valve **210** and/or the second heat exchanger valve **222**, the wellhead fluid may flow to in-field equipment **270** for further processing or transport.

Each of the valves described herein, e.g., first heat exchanger valve **208**, second heat exchanger valve **222**, first wellhead fluid valve **210**, and other valves illustrated in the FIG. 2B through FIGS. 9G, may be control valves, electrically actuated valves, pneumatic valves, or other valves suitable to receive signals and open and close under, potentially, high pressure. The valves may receive signals from a controller or other source and the signals may cause the valves to move to a partially or fully opened or closed position. The signal may indicate the position that the valve may be adjusted to, e.g., a position halfway open, a position a third of the way open, a position a particular degree open, or completely open, such positions to be understood as non-limiting examples. In such examples, as the valve receives the signal indicative of a position to adjust to, the valve may begin turning to the indicated position. Such an operation may take time, depending on the valve used. To ensure proper operation and prevent damage (e.g., damage to the high-pressure heat exchanger **250**, such as when pressure of the wellhead fluid exceeds a pressure rating of the high-pressure heat exchanger **250**), the valve may be configured to close in a specified period of time while high-pressure fluid flows therethrough. Such a configuration may be based on a torque value of the valve, e.g., a valve with a higher torque value may close faster than that of a control valve with a lower torque value. Such a specified period of time may be 5 seconds to 10 seconds, 5 seconds to 15 seconds, 5 seconds to 20 seconds, 10 seconds to 15 seconds, 10 seconds to 20 seconds, or 15 seconds to 20 seconds. For example, if a wellhead fluid flow exceeds a pressure of 15,000 PSI, the first heat exchanger valve **208** may close within 5 seconds of the pressure sensor **214** indicating the pressure exceeding 15,000 PSI. In other embodiments, the valves may either fully open or fully close, rather than open to positions in between. In yet other examples, the valves may be manually or physically opened by operators or technicians on-site.

As illustrated in FIG. 2A, pressure sensors **204**, **212**, **214**, **222**, **207** and/or temperature sensors **206**, **216**, **218**, **226**, **218** may be disposed at various points on or along different pipes, equipment, apparatuses, and/or devices at the well. Each sensor may provide information to adjust and/or control various aspects of the wellhead fluid flow. For example, pressure sensors **204**, **212** may provide pressure measurements to determine whether a flow of wellhead fluid is not impeded in relation to hydrocarbon production targets. Pressure sensors **214**, **220** may provide measurements to ensure that the flow of wellhead fluid through the high-pressure heat exchanger **250** is sufficient to facilitate heat transfer from the wellhead fluid to a working fluid in a ORC loop **221** and/or that the pressure within the high-pressure heat exchanger **250** does not exceed a pressure rating of the high-pressure heat exchanger **250**. In another example, temperature sensor **206** may provide data to control flow of the wellhead fluid. For example, if the flow of wellhead fluid

is at a temperature sufficient to cause the working fluid to exhibit a vaporous phase change, then the first heat exchanger valve **208** and second heat exchanger valve **222** may be opened. Further, temperature sensors **216**, **218**, **226**, **248** may provide measurements to ensure that the temperatures within the heat exchanger, of the flow of wellhead fluid, and of the flow of working fluid, are above the thresholds or within a range of thresholds, e.g., above a vaporous phase change threshold. For example, rather than or in addition to using temperature of the wellhead fluid as measured by temperature sensor **206**, the temperature of the working fluid as measured by temperature sensor **226** may be utilized to determine whether to maintain an open position of an already open first heat exchanger valve **208**. In such examples, the first heat exchanger valve **208** may initially be fully or partially open, e.g., prior to flow of the wellhead fluid.

In another embodiment, in addition to the first heat exchanger valve **208** and second heat exchanger valve **222**, the well **200** may include a first ORC unit valve **295** and a second ORC unit valve **296**. Such ORC unit valves **295**, **296** may be utilized to control flow of working fluid flowing into the ORC unit **203**. Further the ORC unit valves **295**, **296** may be utilized when more than one high-pressure heat exchanger **250** corresponding to one or more wellheads are connected to the ORC unit **203**. The ORC unit valves **295**, **296** may be utilized to optimize or to enable the ORC unit **203** to meet a preselected electrical power output threshold via the flow of working fluid from one or more high-pressure heat exchangers into the ORC unit **203**, as will be understood by a person skilled in the art. The flow of working fluid may be adjusted to ensure that the ORC unit **203** produces an amount of electrical power greater than or equal to a preselected electrical power output threshold, based on various factors or operating conditions (e.g., temperature of wellhead fluid flow, temperature of working fluid flow, electrical output **236** of the ORC unit **203**, electrical rating of the ORC unit **203**, flow and/or pressure of the wellhead fluid, and/or flow and/or pressure of the working fluid). In some examples, the ORC unit valves **295**, **296** may initially be fully open or at least partially open and as various factors or operating conditions are determined, then the ORC unit valves **295**, **296** for one or more heat exchangers may be adjusted to enable the ORC unit **203** to meet a preselected electrical power output threshold, as will be understood by a person skilled in the art. The preselected electrical power output threshold may be set by a user or may be a predefined value generated by a controller (e.g., controller **272**) based on various factors. The various factors may include an ORC unit electrical power rating or output rating or maximum potential temperature of the wellhead fluid and/or working fluid.

As shown, several pairs of sensors may be located adjacent to one another. In other examples, those positions, for example, the pressure sensor's **204** and the temperature sensor's **206** location, may be reversed. In yet another example, each one of the sensors may provide measurements for multiple aspects of the wellhead fluid, e.g., one sensor to provide a combination of flow, pressure, temperature, composition (e.g., amount of components in the wellhead fluid, such as water, hydrocarbons, other chemicals, proppant, etc.), density, or other aspect of the wellhead fluid or working fluid. Each sensor described above may be integrated in or within the pipes or conduits of each device or component, clamped on or over pipes or conduits, and or disposed in other ways, as will be understood by those skilled in the art. Further, the determinations, adjustments,

and/or other operations described above may occur or may be performed by or in a controller.

As noted, a high-pressure heat exchanger **250** may be disposed, placed, or installed at a well. The high-pressure heat exchanger **250** may be disposed nearby or at a distance from the wellhead **202**. The high-pressure heat exchanger **250** may be a modular and/or mobile apparatus. In such examples, the high-pressure heat exchanger **250** may be brought or moved to a well or site (e.g., via a vehicle, such as a truck), placed at the well or site during hydrocarbon production, and then moved to another well or site at the end of hydrocarbon production. The high-pressure heat exchanger **250** may be disposed on a skid, a trailer, a flatbed truck, inside a geothermal generator unit, or inside an ORC unit **203**. Once brought to a well or site, the high-pressure heat exchanger **250** may be secured to the surface at the well. The high-pressure heat exchanger **250** may be configured to withstand pressures in excess of about 5,000 PSI, about 10,000 PSI, about 15,000 PSI, and/or greater. In an example, the high-pressure heat exchanger **250** may be a high-pressure shell and tube heat exchanger, a spiral plate or coil heat exchanger, a heliflow heat exchanger, or other heat exchanger configured to withstand high pressures. In another example, portions of the high-pressure heat exchanger **250** may be configured to withstand high pressures. For example, if a shell and tube heat exchanger is utilized, the shell and/or tubes may be configured to withstand high pressures.

In another embodiment, at least one fluidic path of the high-pressure heat exchanger **250** may be coated or otherwise configured to reduce or prevent corrosion. In such examples, a wellhead fluid may be corrosive. To prevent damage to the high-pressure heat exchanger **250** over a period of time, the fluid path for the wellhead fluid may be configured to withstand such corrosion by including a permanent, semi-permanent, or temporary anti-corrosive coating, an injection point for anti-corrosive chemical additive injections, and/or some combination thereof. Further, at least one fluid path of the high-pressure heat exchanger **250** may be comprised of an anti-corrosive material, e.g., anti-corrosive metals or polymers. As noted, the wellhead fluid may flow into the high-pressure heat exchanger **250** at a high pressure. As the high-pressure heat exchanger **250** may operate at high pressure, the high-pressure heat exchange may include pressure relief valves to prevent failures if pressure within the high-pressure heat exchanger **250** were to exceed the pressure rating of the high-pressure heat exchanger **250**. Over time, wellhead fluid flowing through the high-pressure heat exchanger **250** may cause a buildup of deposits or scaling. To prevent scaling and/or other related issues, the high-pressure heat exchanger **250** may be injected with scaling inhibitors or other chemicals or may include vibration or radio frequency induction devices.

Once the high-pressure heat exchanger **250** facilitates heat transfer from the wellhead fluid to the working fluid, the working fluid may partially, substantially, or completely change phases from a liquid to a vapor, vaporous state, gas, or gaseous state. The vapor or gas may flow to the ORC unit **203** causing an expander to rotate. The rotation may cause a generator to generate electricity, as will be further described and as will be understood by those skilled in the art. The generated electricity may be provided as an electrical output **236**. The electricity generated may be provided to in-field equipment, energy storage devices, equipment at other wells, or to a grid power structure. The working fluid in the high-pressure heat exchanger may be a working fluid to carry heat. Further, the working fluid of the high-pressure

13

heat exchanger **250** may or may not exhibit a vaporous phase change. The working fluid may carry heat to another heat exchanger **205** of the ORC unit **203**. As such, heat may be transferred from the wellhead fluid to the working fluid of the high-pressure heat exchanger **250** and heat may be transferred from the working fluid of the high-pressure heat exchanger **250** to the working fluid of the ORC unit **203**.

In an example, the working fluid may be a fluid with a low boiling point and/or high condensation point. In other words, a working fluid may boil at lower than typical temperatures, while condensing at higher than typical temperatures. The working fluid may be an organic working fluid. The working fluid may be one or more of pentafluoropropane, carbon dioxide, ammonia and water mixtures, tetrafluoroethane, isobutene, propane, pentane, perfluorocarbons, other hydrocarbons, a zeotropic mixture of pentafluoropentane and cyclopentane, other zeotropic mixtures, and/or other fluids or fluid mixtures. The working fluid's boiling point and condensation point may be different depending on the pressure within the ORC loop **221**, e.g., the higher the pressure, the lower the boiling point.

FIG. **2B** illustrates an embodiment of the internal components of an ORC unit **203**. FIG. **2B** further illustrates several of the same components, equipment, or devices as FIG. **2A** illustrates. As such, the numbers used to label FIG. **2B** may be the same as those used in **2A**, as those number correspond to the same component. The ORC unit **203** as noted may be a modular or mobile unit. As power demand increases, additional ORC units **203** may be added, installed, disposed, or placed at the well. The ORC units **203** may stack, connect, or integrate with each other ORC unit. In an example, the ORC unit **203** may be a modular single-pass ORC unit.

An ORC unit **203** may include a heat exchanger **205** or heater. Connections to the heat exchanger **205** or heater may pass through the exterior of the ORC unit **203**. Thus, as an ORC unit **203** is brought or shipped to a well or other location, a user, technician, service person, or other person may connect pipes or hoses from a working fluid heat source (e.g., the high-pressure heat exchanger **250**) to the connections on the ORC unit **203**, allowing a heat source to facilitate phase change of a second working fluid in the ORC unit **203**. In such examples, the working fluid flowing through ORC loop **221** may include water or other organic fluid exhibiting a higher vaporous phase change threshold than the working fluid of the ORC unit **203**, to ensure proper heat transfer in heat exchanger **205**. Further, the heat exchanger **205** may not be a high-pressure heat exchanger. In such examples, the high-pressure heat exchanger **250** allows for utilization of waste heat from high-pressure wellhead fluids. In another embodiment and as will be described, a high-pressure heat exchanger **250** may be included in the ORC unit **203**.

In yet another embodiment, the high-pressure heat exchanger **250** may be considered an intermediary heat exchanger or another intermediary heat exchanger (e.g., intermediary heat exchanger **219**) may be disposed between the high-pressure heat exchanger **250** and the ORC unit **203** (as illustrated in FIG. **2C** and described below). The working fluid flowing through the high-pressure heat exchanger **250** may be sufficient to heat another working fluid of the ORC unit **203**. The working fluid of such an intermediary heat exchanger may not physically flow through any of the equipment in the ORC unit **203**, except for heat exchanger **205**, thereby transferring heat from the working fluid in ORC loop **221** to an ORC working fluid in a loop defined by

14

the fluid path through the heat exchanger **205**, condenser **211**, expander **232**, working fluid reservoir **298**, and/or pump **244**.

The ORC unit **203** may further include pressure sensors **228**, **238**, **246** and temperature sensors **230**, **240** to determine whether sufficient, efficient, and/or optimal heat transfer is occurring in the heat exchanger **205**. A sensor or meter may further monitor electrical power produced via the expander **232** and generator **234**. Further, the ORC unit **203** may include a condenser or heat sink **211** to transfer heat from the second working fluid or working fluid of the ORC unit **203**. In other words, the condenser or heat sink **211** may cool the second working fluid or working fluid of the ORC unit **203** causing the second working fluid or working fluid of the ORC unit **203** to condense or change phases from vapor to liquid. The ORC unit **203** may also include a working fluid reservoir **298** to store an amount of working fluid, e.g., in a liquid state, to ensure continuous operation of the ORC unit **203**. The liquid state working fluid, whether from the working fluid reservoir **298** or directly from the condenser/heat sink **211**, may be pumped, via pump **244**, back to the heat exchanger **205**. Further, the pressure prior to and after pumping, e.g., as measured by the pressure sensors **238**, **246**, may be monitored to ensure that the working fluid remains at a ORC unit or working fluid loop pressure rating.

As illustrated in FIG. **2C**, the well may include an intermediary heat exchanger **219**. In such examples, an ORC unit **203** may not be configured for high-pressure heat exchange. As such, an intermediary heat exchanger **219** may be disposed nearby the high-pressure heat exchanger **250**, nearby the ORC unit **203**, or disposed at some other point in between to alleviate such issues. The intermediary heat exchanger **219** may include a working fluid, also referred to as an intermediary working fluid, to flow through the intermediary loop **223**. The intermediary fluid may include water, a water and glycol mixture, or other organic fluid exhibiting a higher vaporous phase change threshold than the working fluid of the ORC unit **203**. In an embodiment, the intermediary heat exchanger **219** may include sensors, meters, transducers and/or other devices at various positions throughout to determine characteristic of fluids flowing therein, similar to that of the high-pressure heat exchanger **250**.

As illustrated in FIG. **2D**, rather than utilizing an ORC unit **203** that may not withstand high pressure, a high-pressure ORC unit or an ORC unit with integrated high-pressure heat exchanger **250** may be utilized for geothermal power generation. In such examples, the components, equipment, and devices may be similar to those described above. In another example, such a system, as illustrated in FIG. **2D**, may include a heat sink **236** utilizing a cooled flow of wellhead fluid to cool the flow of working fluid. In such examples, as the flow of wellhead fluid passes through a choke valve **252**, the pressure of the flow of wellhead fluid may be reduced, e.g., for example, from about 15,000 PSI to about 1,500 PSI, from about 15,000 PSI to about 200 PSI, from about 15,000 PSI to about 100 PSI, about 15,000 PSI to about 50 PSI or lower, from about 10,000 PSI to about 200 PSI or lower, from about 5,000 PSI to about 200 PSI or lower, or from 15,000 PSI to lower than 200 PSI. In such examples, the temperature from a point prior to the choke valve **252** and after the choke valve **252**, e.g., a temperature differential, may be about 100 degrees Celsius, about 75 degrees Celsius, about 50 degrees Celsius, about 40 degrees Celsius, about 30 degrees Celsius, and lower. For example, the temperature of the wellhead fluid prior to the choke valve **252** may be about 50 degrees and higher, while, after passing

15

through the choke valve **252**, the temperature, as measured by the temperature sensor **259**, may be about 30 degrees Celsius, about 25 degrees Celsius, about 20 degrees Celsius, to about 0 degrees Celsius.

The system may include, as noted, a temperature sensor **259** and pressure sensor **257** to determine the temperature of the wellhead fluid after the choke valve **252**. The system may include temperature sensor **240** to determine the temperature of the working fluid or ORC fluid exiting the heat sink **236** and temperature sensor **238** to determine the temperature of the working fluid or ORC fluid entering the heat sink **236**. The pressure and/or temperature of the wellhead fluid may be used to determine whether the heat sink **236** may be utilized based on pressure rating of the heat sink **236** and/or a liquid phase change threshold of the working fluid. In other words, if the flow of wellhead fluid is at a temperature sufficient to cool the working fluid and/or below a pressure rating of the heat sink **236**, the heat sink valve **254** may open to allow wellhead fluid to flow through the heat sink **236** to facilitate cooling of the working fluid. In another embodiment, the heat sink valve **254** may initially be fully or partially open. The temperature of the working fluid or ORC fluid may be measured as the fluid enters the heat sink **236** and exits the heat sink **236**. If the temperature differential indicates that there is no change or an increase in temperature, based on the temperature of the working fluid or ORC fluid entering the heat sink **236** and then leaving the heat sink **236**, then the heat sink valve **254** may be closed. Temperature sensors **238**, **240**, **256**, **262**, and pressure sensors **258**, **260** may be disposed within the heat sink **236** to ensure that the temperature of the wellhead fluid is suitable for cooling the working fluid and that the pressure of wellhead fluid does not exceed the pressure rating of the heat sink **236**.

FIG. 2D also illustrates two flow meters **277**, **279** disposed prior to the first wellhead fluid valve **210** and first heat exchanger valve **208**. Such flow meters may measure the flow of the wellhead fluid at the point where the meter is disposed. Utilizing flow measurements may allow for fine-tuning or adjustment of the open percentage or position of the valves included in the system. Such fine-tuning or adjustment may ensure that the production of hydrocarbons at the well is not impeded by the use of the high-pressure heat exchanger. Other flow meters may be disposed at various other points of the system, e.g., after the first wellhead fluid valve **210**, prior to or after the choke valve **252**, at a point after the heat sink **236**, and/or at various other points in the system. As stated, these flow meter may be utilized to ensure proper flow wellhead fluid throughout the system. In an example, the sensors and/or meters disposed throughout the system may be temperature sensors, densitometers, density measuring sensors, pressure transducers, pressure sensors, flow meters, mass flow meters, Coriolis meters, spectrometer, other measurement sensors to determine a temperature, pressure, flow, composition, density, or other variable as will be understood by those skilled in the art, or some combination thereof. Further, the sensors and/or meters may be in fluid communication with a liquid to measure the temperature, pressure, or flow or may indirectly measure flow (e.g., an ultrasonic sensor). In other words, the sensors or meters may be a clamp-on device to measure flow indirectly (such as via ultrasound passed through the pipe to the liquid).

As illustrated in FIG. 2E, a controller **272** may be included at the well. The controller **272** may be utilized in any of the previous or following drawings. The controller **272** may include one or more controllers, a supervisory

16

controller, and/or a master controller. The controller **272** may connect to all the equipment and devices shown, including additional equipment and devices not shown, and may transmit control signals, receive or request data or measurements, control pumps, monitor electricity generated, among other things (see **274**). The controller **272** may, in another example, control a subset of the components shown. In another example, a controller may be included in an ORC unit (see **203** in FIGS. 2A through 2C). The controller **272** may connect to and control the controller in the ORC unit **203**. The controller **272** may transmit signals to the various control valves to open and close the valves by determined amounts or percentages or fully open or close the valves. The controller **272** may further determine, via a meter or other device or sensor, an amount of electrical power generated or being generated by the generator **234** or ORC unit **203**.

As illustrated in FIG. 2F, the system may include another heat sink **241**, in the case that the cooling offered by the flow of wellhead fluid in heat sink **236** is not sufficient. In an example, the heat sink **241** may be a fin fan cooler, a heat exchanger, a condenser, any other type of heat sink, a sing-pass parallel flow heat exchanger, a 2-pass crossflow heat exchanger, a 2-pass countercurrent heat exchanger, or other type of apparatus. As illustrated in FIG. 2G, the system may include a regenerator **290**. In such examples, the working fluid may flow through a first fluid path of the regenerator **290**. After the working fluid is cooled by a primary heat sink (e.g., heat sink **236**), the working fluid may flow back through another fluid path of the regenerator **290**. As such, the heat from the first fluid path may pre-heat the working fluid, while the second fluid path may offer some level of cooling to the working fluid.

As illustrated in FIG. 2H, the system may include a gas expander **291**. In an example, the gas expander **291** may be a turbine expander, positive displacement expander, scroll expander, screw expander, twin-screw expander, vane expander, piston expander, other volumetric expander, and/or any other expander suitable for an ORC operation or cycle. For example and as illustrated, the gas expander **291** may be a turbine expander. As gas flows through the turbine expander, a rotor **293** connected to the turbine expander may begin to turn, spin, or rotate. The rotor **293** may include an end with windings. The end with windings may correspond to a stator **294** including windings and a magnetic field. As the rotor **293** spins within the stator **294**, electricity may be generated. Other generators may be utilized, as will be understood by those skilled in the art. The generator **293** may produce DC power, AC power, single phase power, or three phase power.

FIG. 3A and FIG. 3B are block diagrams illustrating other novel implementations of a geothermal power generation enabled well to provide electrical power to one or more of in-field equipment, equipment at other wells, energy storage devices, and the grid power structure, according to one or more embodiment of the disclosure. FIGS. 3A and 3B may represent a side-view perspective block diagram of a well and the components at the well. In an example, wellhead fluid may flow from underground **304**. The wellhead fluid flow **308** may include hydrocarbons or a mixture of hydrocarbons and other fluids, e.g., water, chemicals, fluids left-over from fracturing operations, other residuals, and/or other fluids as will be understood by those skilled in the art. The wellhead fluid may flow from a wellbore.

As the wellhead fluid flows from the wellhead **306**, the wellhead fluid may flow to the high-pressure heat exchanger, through a bypass pipe, and/or a combination thereof based

on various factors or characteristics, e.g., wellhead fluid temperature and/or pressure and/or working fluid temperature. For example, if the wellhead fluid flow **308** is above a vaporous phase change threshold for a working fluid flow **310**, then valve **332** may open, at least partially, to allow the wellhead fluid flow to the high pressure heat exchanger **312**. In such examples, the wellhead fluid may continue to flow through the primary or bypass wellhead fluid pipe. As such, valve **330** may remain open, whether completely or at a certain percentage. From the high-pressure heat exchanger **312**, the wellhead fluid may flow back to the primary or bypass wellhead fluid pipe, to a condenser **316** or other cooling apparatus, and/or a combination thereof. If the wellhead fluid is at a temperature to provide cooling to the working fluid flow **310**, then valve **336** may open to allow wellhead fluid to flow therethrough. In such examples, the valve **334** may close to prevent wellhead fluid from flowing back. If the wellhead fluid is not at a temperature to allow for cooling of the working fluid flow **310**, then valve **336** may close or remained closed and valve **334** may open or remain open. From the condenser **316** or the primary or bypass wellhead fluid pipe, the wellhead fluid may flow to in-field equipment **316**, storage tanks, and/or other processing equipment at the well. The valves described above may be controlled via controller **320**.

In another embodiment, rather than basing the opening and closing of valve **332** and/or valve **336** on wellhead fluid flow **308** temperature, the valve **332** and/or valve **336** may be opened or closed based on the temperature of the working fluid flow **310**. For example, prior to activating the wellhead **306** (e.g., allowing wellhead fluid to flow or pumping wellhead fluid from the wellhead **306**), valve **332** may be open, fully or partially. As the wellhead fluid flows through the high-pressure heat exchanger **312**, the temperature of the working fluid flow **310** may be measured. Based on the working fluid flow **310** temperature, taken at continuously or at periodic intervals, and after a specified period of time, if the working fluid flow **310** does not reach a vaporous phase change temperature, then valve **332** may be closed. Further, such operations may be performed in conjunction with measuring wellhead fluid flow **308** and opening or closing valve **332** based on such measurements.

The wellhead fluid flowing through the high-pressure heat exchanger **312** may be at a temperature to facilitate heat transfer to a working fluid flow **310**. The working fluid may further flow, as a vaporous state working fluid flow to an ORC expander/generator **314**. The vaporous state working fluid may cause the ORC expander/generator **314** to generate electrical power to be utilized at equipment at the well (e.g., in-field equipment), energy storage device, or a grid power structure (via a transformer and power lines). The working fluid may then flow to a condenser **316** or other cooling apparatus. The condenser **316** or other cooling apparatus may facilitate cooling of the working fluid flow **310** via the wellhead fluid flow, air, another liquid, and/or other types of heat sinks or heat exchangers. The liquid state working fluid may then flow back to the high-pressure heat exchanger **312**.

In another embodiment, the high-pressure heat exchanger **312** may connect to an ORC unit/module **340** or one or more ORC units or modules. The number of ORC units/modules may scale based on power to be utilized by in-field equipment, the amount or potential capacity of electricity generation at the well, and/or other factors. After production of hydrocarbons begins, additional ORC units/modules may be added at the well or existing ORC units/modules may be removed from the well.

FIG. 4A and FIG. 4B are simplified diagrams illustrating a control system for managing the geothermal power production at a well, according to one or more embodiment of the disclosure. A master controller **402** may manage the operations of geothermal power generation at a wellhead during hydrocarbon production. The master controller **402** may be one or more controllers, a supervisory controller, programmable logic controller (PLC), a computing device (such as a laptop, desktop computing device, and/or a server), an edge server, a cloud based computing device, and/or other suitable devices. The master controller **402** may be located at or near the well. The master controller **402** may be located remote from the well. The master controller **402**, as noted, may be more than one controller. In such cases, the master controller **402** may be located near or at various wells and/or at other off-site locations. The master controller **402** may include a processor **404**, or one or more processors, and memory **406**. The memory **406** may include instructions. In an example, the memory **406** may be a non-transitory machine-readable storage medium. As used herein, a “non-transitory machine-readable storage medium” may be any electronic, magnetic, optical, or other physical storage apparatus to contain or store information such as executable instructions, data, and the like. For example, any machine-readable storage medium described herein may be any of random access memory (RAM), volatile memory, non-volatile memory, flash memory, a storage drive (e.g., hard drive), a solid state drive, any type of storage disc, and the like, or a combination thereof. As noted, the memory **406** may store or include instructions executable by the processor **404**. As used herein, a “processor” may include, for example one processor or multiple processors included in a single device or distributed across multiple computing devices. The processor may be at least one of a central processing unit (CPU), a semiconductor-based microprocessor, a graphics processing unit (GPU), a field-programmable gate array (FPGA) to retrieve and execute instructions, a real time processor (RTP), other electronic circuitry suitable for the retrieval and execution instructions stored on a machine-readable storage medium, or a combination thereof.

As used herein, “signal communication” refers to electric communication such as hard wiring two components together or wireless communication for remote monitoring and control/operation, as understood by those skilled in the art. For example, wireless communication may be Wi-Fi®, Bluetooth®, ZigBee, cellular wireless communication, satellite communication, or forms of near field communications. In addition, signal communication may include one or more intermediate controllers or relays disposed between elements that are in signal communication with one another.

The master controller **402** may include instructions **408** to measure temperature at various points or locations of the system (e.g., as illustrated in, for example, FIG. 2E). For example, temperature may be measured at a wellhead fluid temperature sensor **1 416**, heat exchanger temperature sensors **418**, wellhead fluid temperature sensor **2 420**, condenser temperature sensors **422**, and/or at various other points in the system **400**. Other characteristics may be measured as well, such as flow, density, pressure, composition, or other characteristics related to the wellhead fluid and/or working fluid.

Utilizing the characteristics noted above, the master controller **402** may control various aspects of the system **400**. For example, the master controller **402** may include flow control adjustment instructions **412**. The system **400** may include one or more valves placed in various locations (For example, but not limited to, FIGS. 2A through 2H). Valves

of the system **400** may include a wellhead fluid valve **1 426**, heat exchanger valves **428**, wellhead fluid valve **2 430**, and condenser valves **432**. As noted other valves may be included in the system and controlled by the master controller **402**. The valves may operate to adjust flow based on a number of factors. Such factors may include temperature of a wellhead fluid, flow rate of the wellhead fluid, temperature of the working fluid, flow rate of the working fluid, pressure of the wellhead fluid at various points in the system, pressure of the working fluid at various points in the system, and/or some combination thereof.

In an example, the system **400** may include a user interface **436**, e.g., such as a monitor, display, computing device, smartphones, tablets, and other similar devices as will be understood by those skilled in the art. A user may view data, enter thresholds or limits, monitor status of the equipment, and perform other various tasks in relation to the equipment at the well. For example, a specific flow rate may be set for hydrocarbon production. As a wellhead begins producing hydrocarbons (e.g., wellhead fluids begin flowing from a wellhead), the master controller **402** may monitor flow rate and compare the flow rate to the threshold either set by a user or pre-set in the master controller **402**. If the master controller **402** determines that the heat exchanger valves **428** (e.g., via flow control adjustment instructions **412**) should be open or are open and that the flow of wellhead fluid is higher or lower than the threshold, the master controller **402** may adjust the appropriate valves, e.g., wellhead fluid valve **426** and/or heat exchanger valves **428**. The valve associated with the primary or bypass wellhead fluid pipe, e.g., wellhead fluid valve **426**, may open or close by varying degrees based on such determinations.

In another example, the master controller **402** may include instructions **410** to control a pump for the ORC unit, e.g., working fluid pump **424**. If heat exchanger valves **428** and condenser valves **432** are closed, the master controller **402** may transmit a signal to shut down or cease operation of the working fluid pump **424**, if the working fluid pump **424** is operating. The master controller **402** may further transmit a signal, based on the heat exchanger valves **428** being open, to initiate or start operations of the working fluid pump **424**. The working fluid pump **424** may be a fixed pressure pump or a variable frequency pump. The master controller **402** may further include instructions **414** to monitor the power output from an ORC unit or from expanders/generators **434** (e.g., expander/generator A **434A**, expander/generator B **434B**, and/or up to expander/generator N **434N**). If the system **400** utilizes ORC units, the master controller **402** may determine the electrical power generated or output based on an output from, for example, ORC unit controller A **438A**, ORC unit controller B **438B**, and/or up to ORC unit controller N **438N**. If the power output drops to an un-economical or unsustainable level or electrical power generation ceases completely while the heat exchanger valves **428** are open, the master controller **402** may transmit signals to close the heat exchanger valves **428**. In another example, the master controller **402** may monitor electrical power output from other wells. The master controller **402** may monitor or meter the amount of electrical power being utilized at each of the wells and/or the amount of electrical power being generated at each of the wells. If an excess of electrical power exists, the master controller **402** may transmit signals causing the excess energy at any particular well to be stored in energy storage devices, transmitted to the grid, and/or transmitted to another well. If a deficit of electrical power exists, the master controller **402** may transmit a signal causing other wells to transmit electrical power

to the well experiencing an electrical power deficit. In another example, the metered electrical power may be utilized for commercial trade, to determine a cost of the electricity generated, and/or for use in determining emissions or emission reductions through use of an alternate energy source (e.g., geothermal power).

In another example, the master controller **402** may include instructions to maximize energy output from an ORC unit. In such examples, the ORC unit may be connected to a plurality of high-pressure heat exchangers. Further, each of the high-pressure heat exchangers may connect to one or more wellheads. As a wellhead produces a wellhead fluid, the pressure and temperature of the wellhead fluid may vary, over time, as well as based on the location of the wellhead. The master controller **402** may determine the temperature of the wellhead fluid at each high-pressure heat exchanger and/or the temperature of the working fluid in each high-pressure heat exchanger. Based on these determinations, the master controller **402** may open/close valves associated with one or more particular high-pressure heat exchangers to ensure the most efficient heat transfer. Further, the master controller **402** may determine the amount of electrical power output from the ORC unit. Based on a power rating of the ORC unit (e.g., the maximum power output the ORC unit is able to produce) and/or the amount of electrical power output from the ORC unit, the master controller **402** may adjust valves associated with the one or more particular high-pressure heat exchangers to thereby increase electrical power output. Additional ORC units may be utilized and electrical power output for each may be optimized or efficiently generated. The master controller **402** may determine, for each ORC unit, the optimal amount or efficient amount of heated working fluid flowing from each high-pressure heat exchanger to ensure the highest amount of electrical power possible is generated per ORC unit or that each ORC unit meets a preselected electrical power output threshold. In such examples, each ORC unit may be connected to each high-pressure heat exchanger and the master controller **402** may determine which set of valves to open/close based on such an optimization or electrical power output threshold.

In another example, the master controller **402** may include failover instructions or instructions to be executed to effectively reduce or prevent risk. The failover instructions may execute in the event of ORC unit and/or high-pressure heat exchanger failure or if an ORC unit and/or high-pressure heat exchanger experiences an issue requiring maintenance. For example, the ORC unit and/or high-pressure heat exchanger may have various sensors or meters. Such sensors or meters, when providing measurement to the master controller **402**, may indicate a failure in the ORC unit and/or high-pressure heat exchanger. In another example, the master controller **402** may include pre-determined parameters that indicate failures. If the master controller **402** receives such indications, the master controller **402** may open, if not already open, the wellhead fluid valve **1 426** and wellhead fluid valve **2 430**. After the wellhead fluid valve **1 426** and wellhead fluid valve **2 430** are opened, the master controller **402** may close the heat exchanger valves **428**, the condenser valves **432**, or any other valve associated with the flow of fluid to the ORC unit and/or high-pressure heat exchanger. In such examples, the master controller **402** may prevent further use of the ORC unit and/or high-pressure heat exchanger until the issue or failure indicated is resolved. Such a resolution may be indicated by a user via the user interface **436** or based on measurements from sensors and/or meters.

21

In another example, the master controller **402** may, as noted, determine an amount of electrical power output by an ORC unit. The master controller **402** may additionally determine different characteristics of the electrical power output. For example, the master controller **402** may monitor the output voltage and frequency. Further, the master controller **402** may include pre-set or predetermined thresholds, limits, or parameters in relation to the monitored characteristics of the electrical power output. Further still, the master controller **402** may connect to a breaker or switchgear. In the event that the master controller **402** detects that an ORC unit exceeds any of the thresholds, limits, and/or parameters, the master controller **402** may transmit a signal to the breaker or switchgear to break the circuit (e.g., the flow of electricity from the ORC unit to a source) and may shut down the ORC unit (e.g., closing valves preventing further flow to the ORC unit, as described above).

FIG. 5 is a flow diagram of geothermal power generation in which, when a wellhead fluid is at or above a vaporous phase change temperature threshold, heat exchanger valves may be opened to allow wellhead fluid to flow therethrough, thereby facilitating heating of a working fluid for use in a ORC unit, according to one or more embodiment of the disclosure. The method is detailed with reference to the master controller **402** and system **400** of FIGS. 4A and 4B. Unless otherwise specified, the actions of method **500** may be completed within the master controller **402**. Specifically, method **500** may be included in one or more programs, protocols, or instructions loaded into the memory of the master controller **402** and executed on the processor or one or more processors of the master controller **402**. The order in which the operations are described is not intended to be construed as a limitation, and any number of the described blocks may be combined in any order and/or in parallel to implement the methods.

At block **502**, the master controller **402** may determine whether the wellhead is active. Such a determination may be made based on sensors located at or near the wellhead, e.g., a pressure sensor indicating a pressure or a flow meter indicating a flow of a wellhead fluid or hydrocarbon stream from the wellhead. In other examples, a user may indicate, via the user interface **436**, that the wellhead is active. If the wellhead is not active, the master controller **402** may wait for a specified period of time and make such a determination after the period of time. In an example, the master controller **402** may continuously check for wellhead activity.

At block **504**, in response to wellhead activity or during hydrocarbon production, the master controller **402** may determine a wellhead fluid temperature. The wellhead fluid temperature may be measured by a wellhead fluid temperature sensor **1 416** disposed at or near the wellhead. The wellhead fluid temperature sensor **1 416** may be disposed on or in a pipe. In an example, various other temperature sensors may be disposed at other points in the system **400**, e.g., heat exchanger temperature sensor **418**, wellhead fluid temperature sensor **2 420**, condenser temperature sensor **422**, and/or other temperature sensors. The temperature measurements provided by such sensors may be utilized by the master controller **402** to determine which valves to open or close.

At block **506**, the master controller **402** may determine whether the wellhead fluid is at or above a vaporous phase change temperature threshold of a working fluid. In such examples, the vaporous phase change temperature may be based on the working fluid of the ORC unit. For example, for pentafluoropropane the vaporous phase change temperature or boiling point may be 15.14 degrees Celsius. In another

22

example, or factors may be taken into account when determining whether to open heat exchanger valves **428**. For example, whether the pressure is within operating range of a high-pressure heat exchanger, whether the flow rate at a primary or bypass pipeline is sufficient to prevent impedance of hydrocarbon production, whether power generation costs are offset by power generation needs, among other factors.

At block **508**, if the wellhead fluid is at or above the vaporous phase change temperature, the master controller **402** may transmit a signal to heat exchanger valves **428** to open to a specified degree. In an example, the heat exchanger valves **428** may be may be fully opened or partially opened. The degree to which the heat exchanger valves **428** opens may depend on the temperature of the wellhead fluid, the flow rate of the wellhead fluid, and/or the pressure of the wellhead fluid.

At block **510**, the master controller **402** may close wellhead fluid valves (e.g., wellhead fluid valve **1 426**) to divert a portion of the flow of wellhead fluids to the high-pressure heat exchanger. The wellhead fluid valves (e.g., wellhead fluid valve **1 426**) may close partially or completely, depending on various factors, such as heat exchanger flow capacity, current flow rate, current pressure, current temperature, among other factors. Once the wellhead fluid valves (e.g., wellhead fluid valve **1 426**) are closed, at block **512**, a working fluid pump **424** of the ORC unit may begin pumping the working fluid through the ORC loop. At block **514**, the master controller **402** may determine whether electricity is being generated. If not, the master controller **402** may check if the wellhead is still active and, if the wellhead is still active, the master controller **402** may adjust the valves (e.g., wellhead fluid valve **1 426** and heat exchanger valves **428**) as appropriate (e.g., increasing flow through the heat exchanger to facilitate an increase in heat transfer).

At block **516**, if the wellhead fluid is lower than the vaporous phase change temperature, the master controller **402** may open or check if the wellhead fluid valves (e.g., wellhead fluid valve **1 426**) are open. Further, the wellhead fluid valves may already be open to a degree and, at block **516**, may open further or fully open, depending on desired wellhead fluid flow. In an example, the wellhead fluid valve (e.g., wellhead fluid valve **1 426**) may be used, with or without a separate choke valve, to choke or partially choke the wellhead fluid flow. Further, once the wellhead fluid valves are open, at block **518**, the master controller **402** may close the heat exchanger valves **428** fully or partially in some cases.

FIG. 6 is another flow diagram of geothermal power generation in which, when a wellhead fluid is at or above a vaporous phase change temperature, heat exchanger valves may be opened to allow wellhead fluid to flow therethrough, thereby facilitating heating of a working fluid for use in a ORC unit, according to one or more embodiment of the disclosure. The method is detailed with reference to the master controller **402** and system **400** of FIGS. 4A and 4B. Unless otherwise specified, the actions of method **600** may be completed within the master controller **402**. Specifically, method **600** may be included in one or more programs, protocols, or instructions loaded into the memory of the master controller **402** and executed on the processor or one or more processors of the master controller **402**. The order in which the operations are described is not intended to be construed as a limitation, and any number of the described blocks may be combined in any order and/or in parallel to implement the methods.

Blocks **602** through **610** correspond to blocks **502** through **610**, as described above. Once it has been determined that

23

the heat exchanger valves **428** should be open and after the heat exchanger valves **428** open, wellhead fluid may flow through the heat exchanger and/or the primary or bypass wellhead fluid pipe to a choke valve. The choke valve may reduce the pressure of the wellhead fluid and, thus, reduce the temperature of the wellhead fluid. At block **612**, the master controller **402** may determine the reduced pressure wellhead fluid temperature at or near a condenser or heat sink valve based on a measurement from the condenser temperature sensors **422**. The master controller **402** may, at block **614**, determine whether the wellhead fluid is at cool enough temperatures to facilitate cooling of a working fluid flow. The working fluid may have a condensation point or a temperature at which the working fluid changes phase from a vapor to a liquid. Such a temperature may be utilized as the threshold for such determinations.

At block **616**, if the temperature is cool enough, the master controller **402** may open the condenser valves **432**, allowing wellhead fluid to flow through the condenser or other cooling apparatus. At block **617** the master controller **402** may transmit a signal to the working fluid pump **424** to start or begin pumping working fluid through an ORC loop. In another example, at block **619**, if the temperature of the working fluid is not cool enough to facilitate cooling of the working fluid to any degree, the master controller **402** may close the condenser valves **432**. In another example, the master controller **402** may determine the temperature of the reduced pressure wellhead fluid flow and whether the temperature of the reduced pressure wellhead fluid flow, in conjunction with a primary or secondary cooler, may cool the working fluid to a point. The master controller **402**, in such examples, may consider the temperature of the working fluid entering the condenser and the temperature of the reduced pressure wellhead fluid flow at or near the condenser valve **432**.

As noted and described above, the master controller **402** may, at block **618**, determine whether electric power is generated. In another example, if the wellhead temperature is not high enough to produce geothermal power, the master controller **402** may, at block **620**, open the wellhead fluid valves. At block **622**, the master controller **402** may close, if the heat exchanger valves **428** are open, the heat exchanger valves **428**. Finally, at block **624**, the master controller **402** may close condenser valves **432**.

FIG. 7A is a flow diagram of geothermal power generation in which, when a working fluid or ORC fluid is at or above a vaporous phase change temperature threshold, heat exchanger valves may remain open to allow wellhead fluid to flow therethrough, thereby facilitating heating of the working fluid or ORC fluid for use in an ORC unit, according to one or more embodiment of the disclosure. The method is detailed with reference to the master controller **402** and system **400** of FIGS. 4A and 4B. Unless otherwise specified, the actions of method **700** may be completed within the master controller **402**. Specifically, method **700** may be included in one or more programs, protocols, or instructions loaded into the memory of the master controller **402** and executed on the processor or one or more processors of the master controller **402**. The order in which the operations are described is not intended to be construed as a limitation, and any number of the described blocks may be combined in any order and/or in parallel to implement the methods.

At block **702**, the master controller **702** may determine whether the wellhead is active. Such a determination may be made based on sensors located at or near the wellhead, e.g., a pressure sensor indicating a pressure or a flow meter

24

indicating a flow of a wellhead fluid or hydrocarbon stream from the wellhead. In other examples, a user may indicate, via the user interface **436**, that the wellhead is active. If the wellhead is not active, the master controller **402** may wait for a specified period of time and make such a determination after the specified period of time. In an example, the master controller **402** may continuously check for wellhead activity.

At block **704**, in response to wellhead activity or during hydrocarbon production, the master controller **402** may open heat exchanger valves **428**. At block **706**, the master controller **402** may close wellhead fluid valve **1 426**, at least partially. At block **708**, when the heat exchanger valves **428** is open and wellhead fluid valve **1 426** is fully or partially closed, working fluid or ORC fluid may be pumped through the ORC unit.

At block **710**, the master controller **402** may measure the temperature of the working fluid or ORC fluid. At block **712**, the master controller **402** may determine whether the wellhead fluid is at or above a vaporous phase change temperature threshold. In such examples, the vaporous phase change may include when the working fluid or ORC fluid changes from a liquid to a vapor or gas. For example, for pentafluoropropane the vaporous phase change temperature or boiling point may be 15.14 degrees Celsius. In another example, other factors may be taken into account when determining whether to maintain an open percentage of the heat exchanger valves **428**. For example, whether the pressure is within operating range of a high-pressure heat exchanger, whether the flow rate at a primary or bypass pipeline is sufficient to prevent impedance of hydrocarbon production, whether power generation costs are offset by power generation needs, among other factors.

At block **712**, if the working fluid or ORC fluid is at or above the vaporous phase change temperature, the master controller **402** may determine whether electricity is generated at the ORC unit. If electricity is not generated, the master controller **402** may check, at block **702**, whether the wellhead is active and perform the operations of method **700** again.

If the working fluid or ORC fluid, at block **712** is not at a vaporous phase change temperature, then, at block **714**, the master controller **402** may first determine whether a first specified period of time has lapsed. The first period of time may be period of time of sufficient length to determine whether or not the working fluid or ORC fluid may reach a vaporous phase change state. Such a first specified period of time may be about an hour or more, two hours, three hours, four hours, or some other length of time during wellhead activity.

If the first specified period of time has not lapsed, at block **716**, the master controller **402** may wait a second specified period of time before measuring the temperature of the working fluid or ORC fluid. The second specified period of time may be less than the first specified period of time.

If the first specified period of time has lapsed, then the master controller **402** may have determined that, based on the temperature of the working fluid or ORC fluid, that the wellhead fluid may not reach temperatures sufficient to cause a vaporous phase change of the working fluid or ORC fluid. As such, at block **718**, the master controller may close the open wellhead fluid valves and, at block **720**, close the heat exchanger valves **428**.

FIG. 7B is another flow diagram of geothermal power generation in which, when a working fluid or ORC fluid is at or above a vaporous phase change temperature threshold, heat exchanger valves may remain open to allow wellhead fluid to flow therethrough, thereby facilitating heating of the

25

working fluid or ORC fluid for use in an ORC unit, according to one or more embodiment of the disclosure. The method is detailed with reference to the master controller 402 and system 400 of FIGS. 4A and 4B. Unless otherwise specified, the actions of method 701 may be completed within the master controller 402. Specifically, method 701 may be included in one or more programs, protocols, or instructions loaded into the memory of the master controller 402 and executed on the processor or one or more processors of the master controller 402. The order in which the operations are described is not intended to be construed as a limitation, and any number of the described blocks may be combined in any order and/or in parallel to implement the methods.

Blocks for FIG. 7B may perform the same functions as described for blocks of FIG. 7A. As such, those blocks include the same numbers, such as block 702 through block 722. In addition to those operations, at block 724 the master controller 402 may check whether the heat exchanger valves may be open. If they are not the master controller 402 may open the heat exchanger valves and close, at least partially, the wellhead fluid valves. If the heat exchanger valves are open, the master controller 402 may not change or adjust any of the valves open percentage, but rather continue to or start pumping the working fluid or ORC unit through the ORC unit.

In addition, after the master controller 402 closes the heat exchanger valves at block 720, the master controller 402 may determine, at block 726, the wellhead fluid temperature at the heat exchanger. At block 728, the master controller may determine whether the wellhead fluid is at a vaporous phase change temperature of the working fluid or ORC fluid. If the wellhead fluid temperature is less than such a value, the master controller 402 may wait and measure the temperature again after a period of time. If the wellhead fluid temperature is greater than or equal to such a value, the master controller 402 may perform the operations of method 701 starting at block 724 again.

FIG. 8 is a flow diagram of geothermal power generation in which a working fluid flow is determined based on a preselected electrical power output threshold, according to one or more embodiment of the disclosure. The method is detailed with reference to the master controller 402 and system 400 of FIGS. 4A and 4B. Unless otherwise specified, the actions of method 800 may be completed within the master controller 402. Specifically, method 800 may be included in one or more programs, protocols, or instructions loaded into the memory of the master controller 402 and executed on the processor or one or more processors of the master controller 402. The order in which the operations are described is not intended to be construed as a limitation, and any number of the described blocks may be combined in any order and/or in parallel to implement the methods.

At block 802, each of one or more heat exchangers may be connected to one or more wellhead fluid lines. Each of the one or more wellhead fluid lines may correspond to a wellhead. At block 804, an ORC unit may be connected to the one or more heat exchangers. In another example, the system 400 may include one or more ORC units and each of the one or more ORC units may connect to one or more heat exchangers or two or more heat exchangers.

At block 806, heat exchanger valves positioned between the one or more heat exchangers and the one or more wellhead fluid lines may be opened. Once opened, the heat exchanger valves may allow for continuous diversion of the flow of wellhead fluid through the heat exchanger. The flow of wellhead fluid through the heat exchanger may facilitate

26

transfer of heat from the flow of wellhead fluid to a flow of working fluid or intermediate working fluid.

At block 808, ORC unit valves may be opened. The ORC unit valves may initially be fully opened or partially opened. The ORC unit valves, when open, may allow for working fluid from each of the heat exchangers to flow into the ORC unit. Each working fluid flow may be combined and may pass through the ORC unit.

At block 810, the master controller 402 may determine one or more operating conditions of the ORC unit and/or the system 400. The one or more operating conditions may include the flow rate and/or pressure of working fluid flowing through each of the one or more heat exchangers, the flow rate and/or pressure of wellhead fluid flowing through each of the one or more heat exchangers or at any other point downstream of the wellhead, the temperature of the working fluid in each of the one or more heat exchangers, the temperature of wellhead fluid at each of the one or more heat exchangers, the temperature of the combined working fluid flow at the ORC unit, the electrical power output from the ORC unit, and/or the open position of each of the valves included in the system 400.

At block 812, based on the determined operating conditions, the master controller 402 may determine an optimal or efficient working fluid flow of the ORC unit. The optimal or efficient working fluid flow may depend on the temperature of the combined working fluid flowing to the ORC unit. The other operating conditions, described above, may be utilized to determine the optimal or efficient working fluid flow. The optimal or efficient working fluid flow may comprise the combined flow of working fluid flowing into the ORC unit to thereby produce a maximum amount of electrical power possible or to enable the ORC unit to meet a preselected electrical power output threshold. The optimal or efficient working fluid flow may be at a temperature sufficient to produce such an amount of electrical power (e.g., a temperature greater than or equal to the boiling point of the working fluid within the ORC unit). The optimal or efficient working fluid flow may be indicated by the master controller 402 as one or more open positions for each of the ORC unit valves and/or heat exchanger valves.

At block 814, the master controller 402 may, based on a determined optimal or efficient working fluid flow, determine whether to adjust the one or more ORC unit valves. Other valves within the system 400 may be adjusted based on the optimal or efficient working fluid flow, such as one or more heat exchanger valves and/or one or more wellhead fluid valves. If it is determined that the ORC unit valves or any other valves are to be adjusted, the master controller 402, at block 816, may transmit a signal to the valve to be adjusted indicating a new open position or closed position for the valve to adjust to. After the signal is transmitted, the valve may automatically adjust to the position indicated. If the valve is not to be adjusted or after the valves have been adjusted, the master controller 402 may determine operating conditions again. In an example, the master controller 402 may wait for a period of time, allowing the system to adjust to the new temperatures and flow rates or to reach equilibrium, prior to determining the operating conditions.

FIG. 9A, FIG. 9B, FIG. 9C, FIG. 9D, and FIG. 9E are block diagrams illustrating novel implementations of one or more geothermal power generation enabled wells to provide electrical power to one or more of in-field equipment, equipment at one of the other wells, energy storage devices, and the grid power structure, according to one or more embodiment of the disclosure. As illustrated in FIG. 9A, various wells 902A, 902B, up to 902N may be positioned in

proximity to one another. For example, one well (e.g., well A 902A) may be located at a distance of over about 1 mile, about 5 miles, about 10 miles, about 25 miles, or more to another well (e.g., well B 902B or well N 902N). Each well 902A, 902B, 902N may include various in-field equipment 910A, 910B, 910N. Each well 902A, 902B, 902N may include one or more wellheads 904A, 904B, 904N. Each wellhead 904A, 904B, 904N may connect to a high-pressure heat exchanger 906A, 906B, 906N and the high-pressure heat exchanger 906A, 906B, 906N may connect to ORC equipment 908A, 908B, 908N. The ORC equipment 908A, 908B, 908N may generate electrical power. The electrical power may be provided to the in-field equipment 910A, 910B, 910N. A surplus of electrical power may be provided to an energy storage device 920A, 920B, 920N. The ORC equipment 908A, 908B, 908N may also transmit electrical power energy to other wells, energy storage devices, or to a grid power structure 912.

As noted and as illustrated in FIG. 9B, the high-pressure heat exchanger N 906N may connect to one or more wellheads 904A, 904B, 904N. For example, a well N 902 may include 1, 2, 3, or more wellheads (e.g., wellhead A 904A, wellhead B 904B, and/or up to wellhead N 904N). All of the wellheads 904A, 904B, 904N may connect to one high pressure heat exchanger N 906N. In another example, a well N 902N may include one or more high-pressure heat exchangers, as illustrated in FIG. 9D and FIG. 9E. In such examples, the wellheads 904A, 904B, 904N may connect to one or more of the high-pressure heat exchangers.

As described and as illustrated in FIG. 9C, the well N 902N may include one or more wellheads (e.g., wellhead A 904A, wellhead B 904B, and/or up to wellhead N 904N), one or more sets of ORC equipment (e.g., ORC equipment A 908A, ORC equipment B 908B, and/or up to ORC equipment N 908N), and one or more sets of in-field equipment (e.g., in-field equipment A 910A, in-field equipment B 910B, and/or up to in-field equipment N 910N). As noted, a well N 902N may include one high-pressure heat exchanger, as illustrated in FIG. 9D and FIG. 9E. The well 902 may further include additional high-pressure heat exchangers 906.

As illustrated in FIGS. 9D and 9E, a well N 902N may include one or more wellheads (e.g., wellhead A 904A, wellhead B 904B, and/or up to wellhead N 904N). Each of the one or more wellheads 904A, 904B, 904N may correspond to a high-pressure heat exchanger (e.g., heat exchanger A 906A, heat exchanger B 906B, and/or up to heat exchanger N 906N). In other examples, two or more high-pressure heat exchangers may correspond to a particular wellhead, while in other examples, two or more wellheads may correspond to a high-pressure heat exchanger. Further, a well N 902N may include one ORC equipment A 908A or unit, as illustrated in FIG. 9D. A well N 902N may include two ORC equipment (e.g., ORC equipment A 908A and ORC equipment B 908B) or units or, in some cases, more. In such examples, each ORC equipment 908A, 908B may connect to each of the high-pressure heat exchangers 906A, 906B, 906N. As wellhead fluid flows from a wellhead 904A, 904B, 904N, the temperature and pressure may vary based on a number of factors (e.g., production, type of fluids, distance from the high-pressure heat exchanger, among other factors).

As such, a working fluid of a particular high-pressure heat exchanger 906A, 906B, 906N may be heated to a degree sufficient, insufficient, or more than sufficient to cause the working fluid of the ORC equipment 908A, 908B to exhibit a vaporous phase change. Since the temperature of the

wellhead fluid varies, a controller (e.g., controller 916A, 916B, up to 916N or master controller 918, as illustrated in FIG. 9F) may determine the temperature of working fluid at each high-pressure heat exchanger 906 and determine the most efficient and/or optimal combination to be utilized for generating the most electrical power or for generating an amount of electrical power to meet a preselected electrical power output threshold at the ORC equipment 908A, 908B. The electrical power output by the ORC equipment 908A, 908B may be measured by, for example, an electrical power meter or other device suitable to determine electrical power output. The controller may also determine the temperature of the combination of working fluid or intermediate working fluid entering the ORC equipment 908A, 908B, via, for example, one or more temperature sensors positioned at or near an inlet of the ORC equipment 908A, 908B. The inlet may allow working fluid or intermediate working fluid to flow into the ORC equipment 908A, 908B. The controller may determine the most efficient and/or optimal combination or amount of working fluid or intermediate working fluid from the one or more high-pressure heat exchangers 906A, 906B, 906N based on a variety of factors noted above. For example, the most efficient and/or optimal combination or amount may be based on wellhead fluid temperature from each of the one or more wellheads 904A, 904B, 904N, the working fluid or intermediate working fluid temperature at each of the high-pressure heat exchangers 906A, 906B, 906N, and/or electrical power output (e.g., a measurement indicative of the electrical power generated) by the ORC equipment 908A, 908B. Other factors may include flow rate and pressure of each of the high-pressure heat exchangers 906A, 906B, 906N, current open positions high-pressure heat exchanger valves, and/or current open positions of other valves included at the well N 902N.

For example, if high-pressure heat exchanger A 906A includes a working fluid at a temperature slightly less than a temperature to cause vaporous phase change, then valves providing working fluid or intermediate working fluid from the high-pressure heat exchanger A 906A to the ORC equipment 908A, 908B may be closed. In another example, if high-pressure heat exchanger B 906B is providing working fluid at a temperature well above a temperature to cause vaporous phase change, then valves providing working fluid or intermediate working fluid from high-pressure heat exchanger B 906B to ORC equipment A 908A and/or to ORC equipment B 908B may be adjusted to positions such that a greater portion of the working fluid or intermediate working fluid is transported to ORC equipment A 908A and/or to ORC equipment B 908B.

In yet another example, all valves for allowing flow of working fluid or intermediate working fluid to the ORC equipment A 908A and/or ORC equipment B 908B may be, at least, in a partially open position. The temperature of the wellhead fluid and/or working fluid of each heat exchanger 906A, 906B, 906N may be determined or measured. Further, the electrical power output of the ORC equipment A 908A and/or ORC equipment B 908B may be determined. The positions of each valve for allowing flow of working fluid or intermediate working fluid to the ORC equipment A 908A and/or ORC equipment B 908B may be adjusted to different partially open positions, fully opened positions, or fully closed positions. Such valve adjustments may be based on maximization of the resultant temperature and heat delivered to the ORC equipment A 908A and/or ORC equipment B 908B once the combined working fluid flows into the ORC equipment 908 A 908A and/or ORC equipment B 908B. The valve adjustments may be based on, rather than or in

addition to other factors, the maximization of the electrical power output from the ORC equipment A 908A and/or ORC equipment B 908B. Valve adjustments may further be based on wellhead fluid temperature and/or some combination of the factors described herein.

As noted and described above and as illustrated in FIG. 9F, a controller (e.g., controller A 916A, controller B 916B, and/or up to controller N 916N) may be included at one or more of the wells 902A, 902B, 902N. The controller 916A, 916B, 916N may control and monitor various aspects of the well 902A, 902B, 902N. The controller 916A, 916B, 916N of each well 902A, 902B, 902N may connect to a master controller 918, the master controller 918 may control the operations of the ORC equipment 908A, 908B, 908N, as well as other equipment (e.g., valves and/or pumps) at each of the wells 902A, 902B, 902N. As described above and illustrated in FIG. 9G, the master controller 918 may connect to one or more wellheads 904A, 904B, 904N. The ORC enabled wells 902A, 902B, 902N may provide power to one or more of in-field equipment at the ORC enabled well 902A, 902B, 902N, to an energy storage device, and/or a grid structure device (see 912).

This application is a continuation of U.S. application Ser. No. 17/305,297, filed Jul. 2, 2021, titled "SYSTEMS FOR GENERATING GEOTHERMAL POWER IN AN ORGANIC RANKINE CYCLE OPERATION DURING HYDROCARBON PRODUCTION BASED ON WORKING FLUID TEMPERATURE", which claims priority to and the benefit of U.S. Provisional Application No. 63/200,908, filed Apr. 2, 2021, titled "SYSTEMS AND METHODS FOR GENERATING GEOTHERMAL POWER DURING HYDROCARBON PRODUCTION," the disclosures of which are incorporated herein by reference in their entirety.

In the drawings and specification, several embodiments of systems and methods to provide geothermal power in the vicinity of a wellhead during hydrocarbon production have been disclosed, and although specific terms are employed, the terms are used in a descriptive sense only and not for purposes of limitation. Embodiments of systems and methods have been described in considerable detail with specific reference to the illustrated embodiments. However, it will be apparent that various modifications and changes can be made within the spirit and scope of the embodiments of systems and methods as described in the foregoing specification, and such modifications and changes are to be considered equivalents and part of this disclosure.

What is claimed is:

1. A system for generating geothermal power in the vicinity of a wellhead during hydrocarbon production, thereby to supply electrical power to one or more of in-field equipment, a grid power structure, or energy storage devices, the system comprising:

- a first temperature sensor to provide a first temperature, the first temperature defined by a temperature of a flow of working fluid;
- a heat exchanger valve positioned to divert flow of wellhead fluid from one or more wellheads;
- a controller configured to signal the heat exchanger valve to divert the flow of the wellhead fluid based on when the first temperature reaches a vaporous phase-change temperature threshold;
- a high-pressure heat exchanger including a first fluid path to accept and output the flow of wellhead fluid from the heat exchanger valve and a second fluid path to accept and output the flow of the working fluid, the high-pressure heat exchanger positioned to indirectly transfer heat from the flow of wellhead fluid to the flow of

the working fluid, thereby to cause the working fluid to change phases from a liquid to a vapor; and

- a power generation unit including a generator, a gas expander, and a partial loop for the flow of the working fluid, the partial loop defined by a fluid path through a condenser, the gas expander, and a pump, the partial loop further defining a complete loop when connected to the second fluid path of the high-pressure heat exchanger, the flow of the working fluid, as a vapor, positioned to cause the generator to generate electrical power via rotation of the gas expander, the condenser positioned to cool the flow of the working fluid so that the cooled flow of the working fluid changes phase from the vapor to the liquid, the pump positioned to transport the liquid state working fluid from the condenser for heating.

2. The system of claim 1, further comprising a first wellhead fluid valve to adjust flow of wellhead fluid from the one or more wellheads based on the diversion of the flow of wellhead fluid to the heat exchanger valve.

3. The system of claim 1, further comprising another one or more power generation units connected to the high-pressure heat exchanger, and wherein each of the one or more power generation units and the another one or more power generation units includes a respective working fluid reservoir to store a corresponding working fluid.

4. The system of claim 1, wherein the first fluid path of the high-pressure heat exchanger is configured to withstand corrosion caused by the wellhead fluid via an anti-corrosive coating applied on the first fluid path or an injection point for anti-corrosive chemical additive injections into the first fluid path.

5. A system for generating power in thermal power operation in the vicinity of a wellhead during hydrocarbon production, thereby to supply electrical power to one or more of in-field equipment, a grid power structure, or energy storage devices, the system comprising:

- a first heat exchanger (a) including a first fluid path to accept and output a flow of wellhead fluid and a second fluid path to accept and output a flow of a first working fluid and (b) positioned to indirectly transfer heat from the flow of wellhead fluid to the flow of the first working fluid;
- a first temperature sensor to provide a first temperature of the flow of the first working fluid output from the second fluid path;
- a first valve positioned to divert the flow of the wellhead fluid from one or more wellheads to the first fluid path;
- a controller configured to signal the first valve to divert the flow of the wellhead fluid based on (a) when the first temperature reaches a vaporous phase-change temperature threshold and (b) production of wellhead fluid from the one or more wellheads;
- a second valve to adjust the flow of wellhead fluid from the one or more wellheads downstream based on (a) diversion of the flow of wellhead fluid to the first valve, (b) when the first temperature reaches the vaporous phase-change temperature threshold, and (c) production of wellhead fluid from the one or more wellheads; and
- a thermal power generator including a second heat exchanger, the second heat exchanger (a) including a third fluid path to accept and output the flow of first working fluid from the first heat exchanger and a fourth fluid path to accept and output the flow of a second working fluid and (b) positioned to indirectly transfer heat from the flow of the first working fluid to the flow

31

of the second working fluid and cause the second working fluid to change phases from a liquid to a vapor, thereby to cause the thermal power generator to generate electrical power.

6. The system of claim 5, further comprising:

a second temperature sensor to provide to the controller a second temperature of the flow of the second working fluid output from the second heat exchanger, and wherein the controller is configured to signal the first valve to divert the flow of the wellhead fluid from the one or more wellheads to the first fluid path based on the second temperature.

7. The system of claim 5, further comprising:

a third temperature sensor to provide to the controller a third temperature of the flow of the wellhead fluid accepted at the first heat exchanger, and

wherein the controller is configured to signal the first valve to divert the flow of the wellhead fluid from the one or more wellheads to the first fluid path based on the third temperature.

8. The system of claim 5, further comprising a flow meter (a) positioned downstream of the first heat exchanger and the first valve and (b) configured to measure the flow of wellhead fluid downstream of the first heat exchanger and the first valve, and

wherein the flow of wellhead fluid downstream of the first heat exchanger and the second valve indicates the production of wellhead fluid from the one or more wellheads.

9. A system for generating geothermal power in the vicinity of a wellhead during hydrocarbon production, thereby to supply electrical power to one or more of in-field equipment, a grid power structure, or energy storage devices, the system comprising:

a first temperature sensor to provide a first temperature, the first temperature defined by a temperature of a flow of working fluid;

a heat exchanger valve to divert flow of wellhead fluid from one or more wellheads based on when the first temperature reaches a vaporous phase-change temperature threshold within a selected period of time;

a high-pressure heat exchanger including a first fluid path to accept and output the flow of wellhead fluid from the heat exchanger valve and a second fluid path to accept and output the flow of the working fluid, the high-pressure heat exchanger positioned to indirectly transfer heat from the flow of wellhead fluid to the flow of the working fluid, thereby to cause the working fluid to change phases from a liquid to a vapor; and

a power generation unit including a generator, a gas expander, and a partial loop for the flow of the working fluid, the partial loop defined by a fluid path through a condenser, the gas expander, and a pump, the partial loop further defining a complete loop when connected to the second fluid path of the high-pressure heat exchanger, the flow of the working fluid, as a vapor, positioned to cause the generator to generate electrical power via rotation of the gas expander as defined by a power generation operation, the condenser positioned to cool the flow of the working fluid so that the cooled flow of the working fluid change phases from the vapor to the liquid, and the pump positioned to transport the liquid state working fluid from the condenser for heating.

10. A system for generating power in thermal power operation in the vicinity of a wellhead during hydrocarbon production, thereby to supply electrical power to one or

32

more of in-field equipment, a grid power structure, or energy storage devices, the system comprising:

a first heat exchanger (a) including a first fluid path to accept and output a flow of wellhead fluid and a second fluid path to accept and output a flow of a first working fluid and (b) positioned to indirectly transfer heat from the flow of wellhead fluid to the flow of the first working fluid;

a first temperature sensor to provide a first temperature of the flow of the first working fluid output from the second fluid path;

a first valve to divert the flow of the wellhead fluid from one or more wellheads to the first fluid path based on (a) when the first temperature reaches a vaporous phase-change temperature threshold and (b) production of wellhead fluid from the one or more wellheads;

a second fluid valve to adjust the flow of wellhead fluid from the one or more wellheads downstream based on (a) diversion of the flow of wellhead fluid to the first valve, (b) when the first temperature reaches the vaporous phase-change temperature threshold, and (c) production of wellhead fluid from the one or more wellheads;

a thermal power generator including a second heat exchanger, the second heat exchanger (a) including a third fluid path to accept and output the flow of first working fluid from the first heat exchanger and a fourth fluid path to accept and output the flow of a second working fluid and (b) positioned to indirectly transfer heat from the flow of the first working fluid to the flow of the second working fluid and cause the second working fluid to change phases from a liquid to a vapor so as to cause the thermal power generator to generate electrical power; and

a second temperature sensor to provide a second temperature of the flow of the second working fluid output from the second heat exchanger, the first valve positioned to divert the flow of the wellhead fluid from one or more wellheads to the first fluid path based on the second temperature.

11. A system for generating power in thermal power operation in the vicinity of a wellhead during hydrocarbon production, thereby to supply electrical power to one or more of in-field equipment, a grid power structure, or energy storage devices, the system comprising:

a high-pressure intermediate heat exchanger (a) including a first fluid path to accept and output a flow of wellhead fluid and a second fluid path to accept and output a flow of a first working fluid and (b) positioned to indirectly transfer heat from the flow of wellhead fluid to the flow of the first working fluid;

a first temperature sensor to provide a first temperature of the flow of the first working fluid output from the second fluid path;

a first valve to divert the flow of the wellhead fluid from one or more wellheads to the first fluid path based on (a) when the first temperature reaches a vaporous phase-change temperature threshold and (b) production of wellhead fluid from the one or more wellheads;

a second valve to adjust the flow of wellhead fluid from the one or more wellheads downstream based on (a) diversion of the flow of wellhead fluid to the first valve, (b) when the first temperature reaches the vaporous phase-change temperature threshold, and (c) production of wellhead fluid from the one or more wellheads;

a thermal power generator including a heat exchanger, the heat exchanger (a) including a third fluid path to accept

33

and output the flow of first working fluid from the high-pressure intermediate heat exchanger and a fourth fluid path to accept and output the flow of a second working fluid and (b) to indirectly transfer heat from the flow of the first working fluid to the flow of the second working fluid and cause the second working fluid to change phases from a liquid to a vapor so as to cause the thermal power generator to generate electrical power; and

- a flow meter (a) positioned downstream of the high-pressure intermediate heat exchanger and the first valve and (b) configured to measure the flow of wellhead fluid downstream of the high-pressure intermediate heat exchanger and the first valve.

12. A system for generating geothermal power in the vicinity of a wellhead during hydrocarbon production, thereby to supply electrical power to one or more of in-field equipment, a grid power structure, or energy storage devices, the system comprising:

- a first temperature sensor to provide a first temperature, the first temperature defined by a temperature of a flow of working fluid;
- a heat exchanger valve to divert flow of wellhead fluid from one or more wellheads based on (a) when the first temperature reaches a vaporous phase-change tempera-

34

ture threshold and (b) a pressure of the flow of wellhead fluid being below a pressure threshold;

- a high-pressure heat exchanger including a first fluid path to accept and output the flow of wellhead fluid from the heat exchanger valve and a second fluid path to accept and output the flow of the working fluid, the high-pressure heat exchanger positioned to indirectly transfer heat from the flow of wellhead fluid to the flow of the working fluid, thereby to cause the working fluid to change phases from a liquid to a vapor; and
- a power generation unit including a generator, a gas expander, and a partial loop for the flow of the working fluid, the partial loop defined by a fluid path through a condenser, the gas expander, and a pump, the partial loop further defining a complete loop when connected to the second fluid path of the high-pressure heat exchanger, the flow of the working fluid, as a vapor, positioned to cause the generator to generate electrical power via rotation of the gas expander as defined by a power generation operation, the condenser positioned to cool the flow of the working fluid so that the cooled flow of the working fluid changes phase from the vapor to the liquid, and the pump positioned to transport the liquid state working fluid from the condenser for heating.

* * * * *